# MAT 2000: Introduction to Statistics

Worksheets for Grace College from Zybooks

# MAT 2000: Introduction to Statistics

Worksheets for Grace College from Zybooks

Compiled by Dr. Ryan Johnson
Grace College

October 15, 2025

©2022 Zyante Inc. (zyBooks.com)

# Preface

This is a workbook using copyright material from Zybooks.com for Grace College compiled by Dr. Ryan Johnson.

# Contents

**Preface** v

**1 Preliminaries** 1

1.1 Preliminary: Introduction to the six-step method . . . . . 2
1.2 Preliminary: Exploring Data . . . . . 4
1.3 Preliminary: Exploring random processes . . . . . 5
1.4 Data and formulas . . . . . 6

**2 Significance** 7

2.1 Introduction to chance models . . . . . 8
2.2 Measuring the strength of evidence . . . . . 13
2.3 Alternative measure of strength of evidence . . . . . 18
2.4 What impacts strength of evidence? . . . . . 22
2.5 Inference on a single proportion: Theory-based approach . . . 25
2.6 Data and Formulas . . . . . 29

**3 Generalization** 31

3.1 Sampling from a finite population . . . . . 32
3.2 Inference for a single quantitative variable . . . . . 40
3.3 Theory-based Inference for a Population Mean . . . . . 45
3.4 Other Statistics . . . . . 49
3.5 Tools, data, and formulas . . . . . 53

**4 Estimation** 55

4.1 Statistical inference: Confidence intervals . . . . . 56
4.2 2SD and theory-based confidence intervals for a single proportion . . . . . 60
4.3 2SD and theory-based confidence intervals for a single mean . . 64
4.4 Factors that affect the width of a confidence interval . . . . . 66
4.5 Factors that affect the width of a confidence interval (continued) 68
4.6 Tools, data, and formulas . . . . . 72

**5 Causation** 75

5.1 Association and confounding . . . . . 76
5.2 Observational studies versus experiments . . . . . 79

**6 Comparing Two Proportions** **85**

6.1 Comparing two groups: Categorical response . . . . . 86
6.2 Comparing two proportions: Simulation-based approach . . . . 90
6.3 Comparing two proportions: Theory-based approach . . . . . 97
6.4 Formulas . . . . . .101

**7 Comparing Two Means** **103**

7.1 Comparing two groups: Quantitative response . . . . . .104
7.2 Comparing two means: Simulation-based approach. . . . . .107
7.3 Comparing two means: Theory-based approach . . . . . .114
7.4 Data and Formulas . . . . . .118

**8 Paired Data** **121**

8.1 Paired designs . . . . . .122
8.2 Simulation-based approach for analyzing paired data . . . . .125
8.3 Theory-based approach for analyzing paired data . . . . . .130
8.4 Data and Formulas . . . . . .133

**9 Comparing More Than Two Proportions** **137**

9.1 Comparing multiple proportions: Simulation-based approach . .138
9.2 Comparing multiple proportions: Theory-based approach (Part A) . . . . . .142
9.3 Comparing multiple proportions: Theory-based approach (Part B) . . . . . .147
9.4 Chi-square goodness-of-fit test . . . . . .150
9.5 Data and Formulas . . . . . .155

**10 Comparing More Than Two Means** **159**

10.1 Comparing multiple means: Simulation-based approach . . . .160
10.2 Comparing multiple means: Theory-based approach . . . . .165
10.3 Data and Formulas . . . . . .170

**11 Two Quantitative Variables** **175**

11.1 Two quantitative variables: Scatterplot and correlation . . . .176
11.2 Inference for correlation coefficient . . . . . .181
11.3 Least squares regression . . . . . .186

**Back Matter**

# Chapter 1

# Preliminaries

## 1.1 Preliminary: Introduction to the six-step method

**Statistics** is a discipline that guides researchers in collecting, exploring, and drawing conclusions from data. This course will demonstrate the statistical investigation process for a variety of fields of application.

### The six steps of a statistical investigation

The **six steps of a statistical investigation** is a framework for learning from data. The six steps are a guide for collecting and interpreting data in keeping with the scientific method. When research questions are investigated with the six-step method, conclusions are supported by systematic study rather than individual experiences. The six steps are described in the table below.

| | |
|---|---|
| **1. Ask a research question.** | Confirm the question can be addressed by collecting data. Typical questions involve comparing groups, asking how two characteristics are related, or surveying people's opinions. |
| **2. Design a study and collect data.** | Plan how to select the people or objects to be studied, decide how to gather the relevant data, and carry out the data collection in a careful, systematic manner. |
| **3. Explore the data.** | Look for patterns that relate to the research question or raise additional questions. |
| **4. Draw inferences beyond the data.** | Determine whether any findings in the data reflect a genuine tendency and estimate the strength of that tendency. |
| **5. Formulate conclusions** | Consider whether inferences made in step 4 can be generalized to processes or groups beyond those represented in the study. Identify whether a cause-and-effect conclusion is warranted. |
| **6. Look back and ahead.** | Point out limitations of the study and suggest new studies to build on the findings. |

**Exercise 1.1.1 Asking a research question.** State a research question you would be interested in answering.

### What affects the choice to become an organ donor?

Researchers Johnson and Goldstein[1] asked 161 volunteer participants to imagine moving to a new state and applying for a driver's license. One part of the application asked whether the participants would agree to be an organ donor. Participants were presented with one of three different forms of the question:

- You will not be a donor unless you opt in.
- You will be a donor unless you opt out.
- Choose whether or not to be a donor (neutral).

[1] papers.ssrn.com/sol3/papers.cfm?abstract_id=1324774

## Basic terminology

Some basic terminology of data is required.

- **Data** are the values measured or categories recorded on individual objects of interest.
- The **observational unit** is the individual object on which data are recorded.
- A **variable** is a characteristic being recorded about the observational units.
    - A **quantitative variable** has numeric values for which ordinary arithmetic (addition, multiplication) makes sense.
    - A **categorical variable** places observational units into categories.
- A **distribution** is the pattern of outcomes of a variable.

**Key Idea 1.1.1** The best way to determine if a variable is quantitative or categorical is to ask the question, "Can I take the average of this? And would that make sense?" For example, I could ask a group of people to tell me what ZIP code they currently live in, and I could technically take the average of those numbers, but it wouldn't make sense to do so. That is because ZIP codes are categorical.

**Exercise 1.1.2 Study design elements.** For the research question you stated at the beginning of the section, identify the observational units and variable(s) you would use, and describe how you would collect the data. For each variable, indicate whether the variable would be quantitative or categorical.

## 1.2 Preliminary: Exploring Data

In this section, step 3 of the Statistical Investigation Process is further explored. The goal is to describe, measure, and, if possible, explain the patterns of variation (the *distribution*) in the values of the variable.

### Graphs for quantitative variables

A **dotplot** is a graph with one dot representing the variable outcome for each observational unit. Dotplots, like the graph of the performance lengths in the animation above, are good for smaller datasets. Another common graph for a quantitative variable is a histogram. A **histogram** separates the quantitative variable into bins and displays the number of observations in each bin.

### Describing the distribution

Four important features are discussed when describing the distribution of a quantitative variable:

- The **shape** of a distribution is described by the number of peaks and whether one side of the distribution tends to stretch out longer than the other.
- The **center** of a distribution is the middle or typical value.
- The **variability** of a distribution is the spread in the observations.
- **Unusual observations** are observations that do not follow the overall pattern of the distribution.

## 1.3 Preliminary: Exploring random processes

### Random processes

A **random process** is a repeatable process where the result for an individual attempt is not known in advance but there is a predictable pattern to the results in the long run. Playing *Let's Make a Deal* is an example of a random process: The result for the next round of the game is uncertain, but if the game is played many, many times, a predictable pattern is expected.

### Long-run proportion

A **simulation** is an artificial re-creation of a random process.

### Interpretation of probability

The **probability** of an outcome is the long-run proportion of times an outcome from a random process occurs if the process is repeated indefinitely under identical conditions. A probability can be written as a decimal (0.3333), a fraction (1/3), or a percentage (33.33%).

**Checkpoint 1.3.1 Switch or stay.** How do you think the probability of winning with the switch strategy will compare to the probability of winning with the stay strategy? Briefly explain your reasoning.

**Key Idea 1.3.2** The sections in this chapter may have felt unrelated and brief, but they each contributed important components of drawing conclusions from data, including the six-step investigation process, variation in data, and exploring randomness and probability.

# 1.4 Data and formulas

## Data

Organ Donation

| | |
|---|---|
| **Section** | Preliminary: Introduction to the six-step method |
| **Data** | OrganDonation.csv[1] |
| **Reference** | Johnson and Goldstein (Science 2003)[2] |

Super Bowl National Anthem Durations

| | |
|---|---|
| **Section** | Preliminary: Exploring data |
| **Data** | NationalAnthemTimes.csv[3] |
| **Reference** | Grewal (Covers 2019)[4] |

## Preliminary formulas

- $x_i$ is the value of the $i$th observation for the variable of interest.
- $n$ is the sample size.
- $\overline{x} = \frac{\text{sum of observations in the sample}}{n} = \frac{\sum x_i}{n} = \frac{x_1+\ldots+x_n}{n}$ is the sample mean, which is a measure of center for a sample. The sample mean is sometimes called the average.
- $s = \sqrt{\frac{\sum(x_i-\overline{x})^2}{n-1}} = \sqrt{\frac{(x_1-\overline{x})^2+\ldots+(x_n-\overline{x})^2}{n-1}}$ is the sample standard deviation, which is a measure of the typical deviation of the data values from the mean for a sample.

[1] static-resources.zybooks.com/ISI/Ch09/OrganDonation.csv
[2] papers.ssrn.com/sol3/papers.cfm?abstract_id=1324774
[3] static-resources.zybooks.com/ISI/Ch01/NationalAnthemTimes.csv
[4] www.covers.com/nfl/super-bowl/props/national-anthem

# Chapter 2

# Significance

## 2.1 Introduction to chance models

### Can dogs understand human cues?

Dogs have been domesticated for about 14,000 years. In that time, have they been able to develop an understanding of human gestures such as pointing or glancing? How about similar nonhuman cues? Researchers Udell, Giglio, and Wynne[1] tested a small number of dogs in order to answer these questions.

In this exploration, we will first see whether dogs can understand human gestures as well as nonhuman gestures. To test this, the researchers positioned the dogs about 2.5 meters from the experimenter. Two cups were placed, one on each side of the experimenter. The experimenter would perform some sort of gesture (pointing, bowing, looking) toward one of the cups or there would be some other nonhuman gesture (a mechanical arm pointing, a doll pointing, or a stuffed animal looking) toward one of the cups. The researchers would then see whether the dog would go to the cup that was indicated. There were six dogs tested. We will look at one of the dogs in two of his sets of trials. This dog, a four-year-old mixed breed, was named Harley. Each trial involved one gesture and one pair of cups, with a total of 10 trials in a set. We will start out by looking at one set of trials where the experimenter bowed toward one of the cups to see whether Harley would go to that cup.

### Step 1: State the research question

**Exercise 2.1.1** Based on the description of the study, state the research question.

### Step 2: Design a study and collect data

Harley was tested 10 times and 9 of those times he chose the correct cup.

**Exercise 2.1.2** What are the observational units?

**Exercise 2.1.3** Identify the variable in the study. What are the possible outcomes of this variable? Is this variable quantitative or categorical?

**Definition 2.1.1** The set of observational units on which we collect data is called the **sample**. The number of observational units in the sample is the **sample size**. A **statistic** is a number summarizing the results in the sample. ◊

### Step 3: Explore the data

With categorical data, we typically report the number of "successes" or the proportion of successes as the statistic.

**Exercise 2.1.4** What is the number of observational units (sample size) in this study?

**Exercise 2.1.5** Determine the observed statistic and produce a simple bar graph of the data (have one bar for the proportion of times Harley picked the correct cup and another for the proportion of times he picked the wrong cup).

**Exercise 2.1.6** If the research conjecture is that Harley can understand what the experimenter means when they bow toward an object, is the statistic in the direction suggested by the research conjecture?

[1] htt//doi.org/10.1037/0735-7036.122.1.84

**Exercise 2.1.7** Could Harley have gotten 9 out of 10 correct even if he really didn't understand the human gesture and so was randomly guessing between the two cups?

**Exercise 2.1.8** Do you think it is likely Harley would have gotten 9 out of 10 correct if he was just guessing randomly each time?

## Step 4: Draw inferences beyond the data

There are two possibilities for why Harley chose the correct cup 9 out of 10 times:

- He is merely picking a cup at random and in these 10 trials happened to guess correctly in 9 of them. That is, he got more than half correct just by random chance alone.
- He is doing something other than merely guessing and perhaps understands what the experimenters mean when they bow towards the cup.

The unknown long-run proportion (i.e., probability) that Harley will choose the correct cup is called a **parameter**.

**Definition 2.1.2** For a random process, a **parameter** is a long-run numerical property of the process. ◊

We don't know the value of the parameter, but the two possibilities listed above suggest two different possibilities.

**Exercise 2.1.9** What is the value of the parameter if Harley is picking a cup at random? Give a specific value.

**Exercise 2.1.10** What is the possible range of values (greater than or less than some value) for the parameter if Harley is not just guessing and instead understands the experimenter?

We will show you how statisticians use simulation to make a statement about the strength of evidence for these two possible statements about the parameter's value.

## The chance model

Statisticians often use chance models to generate data from random processes to help them investigate the process. In particular, they can see whether the observed statistic is consistent with the values of the statistic simulated by the chance model. If we determine that Harley's results are not consistent with the results from the chance model, we will consider this to be evidence against the chance model and in favor of the research conjecture, that he understands the bowing gesture. In this case, we would say Harley's results are statistically significant, meaning unlikely to have occurred by chance alone.

**Definition 2.1.3** A result is **statistically significant** if it is unlikely to occur just by random chance. If our observed result appears to be consistent with the chance model, we say that the chance model is **plausible** or believable. ◊

We can't perform the actual study more times in order to assess the second possibility, but we can simulate the behavior of Harley's choices if we were to assume the first possibility (that he is simply guessing every time).

**Exercise 2.1.11** Explain how you could use a coin toss to represent Harley's choices if he is guessing between the two cups each time. How many times do you have to flip the coin to represent one set of Harley's attempts? What does heads represent?

**Exercise 2.1.12** If Harley was guessing randomly each time, on average, how many out of the 10 times would you expect him to choose the correct cup?

**Exercise 2.1.13** Simulate one repetition of Harley guessing randomly by flipping a coin 10 times (why 10?) and letting heads represent selecting the correct cup ("success") and tails represent selecting the incorrect cup ("failure"). Count the number of heads in your 10 flips. Combine your results with the rest of the class to create a *dotplot* of the distribution for the number of heads out of 10 flips of a coin.

**(a)** Where does 9 heads fall in the distribution? Would you consider it an unusual outcome or a fairly typical outcome for the number of heads in 10 flips?

**(b)** Based on your answer to Part a, p. 10 do you think it is plausible (believable) that Harley was just guessing which cup to choose?

## Using an Applet to Simulate Flipping a Coin Many Times

To really assess the typical values for the number of heads in 10 coin tosses (number of correct picks by Harley assuming he is guessing at random), we need to simulate many more outcomes of the chance model. Open the One Proportion applet[2].

Notice that the probability of heads has been set to be 0.50, representing Harley guessing between the two cups. Set the number of tosses to 10 and press the **Draw Samples** button. What was the resulting number of heads?

Notice that the number of heads in this set of 10 tosses is then displayed by a dot on the graph. Uncheck the **Animate** box and press the **Draw Samples** button 9 more times. This will demonstrate how the number of heads varies randomly across each set of 10 tosses. Nine more dots have been added to your dotplot. Is a pattern starting to emerge?

Now change the **Number of repetitions** from 1 to 990 and press **Draw Samples**. The applet will now show the results for the number of heads in 1,000 different sets of 10 coin tosses. So each dot represents the number of times Harley chooses the correct cup out of 10 attempts assuming he is just guessing.

Remember why we conducted this simulatito assess whether Harley's result (9 correct in 10 attempts) would be unlikely to occur by chance alone if he were just guessing between the pair of cups for each attempt.

**Exercise 2.1.14** Locate the result of getting 9 heads in the dotplot created by the applet. Would you consider this an unlikely result in the tail of the distribution of the number of heads?

**Exercise 2.1.15** Based on the results of 1,000 simulated sets of 10 coin flips each, would you conclude that Harley would be very unlikely to have picked the correct cup 9 times in 10 attempts if he was randomly guessing between the two cups each time? Explain how your answer relates to the applet's dotplot.

**Exercise 2.1.16** Do the results of this study appear to be statistically significant?

**Exercise 2.1.17** Do the results of this study suggest that Harley just guessing is a plausible explanation for Harley picking the correct cup 9 out of 10 times?

[2] `static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1`

## Summarizing Your Understanding

**Exercise 2.1.18** To make sure that you understand the coin-flipping chance model, what parts of the real study correspond to the physical (coin-flipping) simulation. Specifically, what do a coin flip, heads, tails, chance of heads, and one repetition represent.

## 3S Strategy

We will call the process of simulating could-have-been statistics under a specific chance model the **3S strategy**. After forming our research conjecture and collecting the sample data, we will use the 3S strategy to weigh the evidence against the chance model. This 3S strategy will serve as the foundation for addressing the question of statistical significance in Step 4 of the statistical investigation method.

**Key Idea 2.1.4 3S Strategy for measuring strength of evidence..**

1. **Statistic** Compute the statistic from the observed sample data.
2. **Simulate** Identify a "by-chance-alone" explanation for the data. Repeatedly simulate values of the statistic that could have happened when the chance model is true.
3. **Strength of evidence** Consider whether the value of the observed statistic from the research study is unlikely to occur if the chance model is true. If we decide the observed statistic is unlikely to occur by chance alone, then we can conclude that the observed data provide strong evidence against the plausibility of the chance model. If not, then we consider the chance model to be a plausible (believable) explanation for the observed data; in other words what we observed could plausibly have happened just by random chance.

Let's review how we have already applied the 3S strategy to this study.

**Exercise 2.1.19** Statistic: What is the statistic in this study?

**Exercise 2.1.20** Simulate: Fill in the blanks to describe the simulation. We flipped a coin _____ times and kept track of how many times it came up heads. We then repeated this process _____ more times, each time keeping track of how many heads were obtained in each of the _____ flips.

**Exercise 2.1.21** Strength of evidence: Fill in the blanks to summarize how we are assessing the strength of evidence for this study. Because we rarely obtained a value of _____ heads when flipping the coin _____ times, this means that it is _____ (believable/unlikely) that Harley is just guessing, because if Harley was just guessing he _____ (rarely/often) would get a value like _____ correct out of _____ attempts.

## Step 5: Formulate conclusions

**Exercise 2.1.22** Based on this analysis, are you convinced that Harley can understand human cues? Why or why not?

### Step 6: Look back and ahead

**Exercise 2.1.23** A single study will not provide all of the information needed to fully understand a broad, complex research question. Thinking back to the original research question, what additional studies would you suggest conducting next?

### Exploring further

One important step in a statistical investigation is to consider other models and whether the results can be confirmed in other settings.

**Exercise 2.1.24 Mechangical Arm Study.** In a different study, the researchers used a mechanical arm (roughly the size of a human arm) to point at one of the two cups. The researchers tested this to see whether dogs understood nonhuman gestures. In 10 trials, Harley chose the correct cup 6 times.

**(a)** Using the dotplot you obtained when you simulated 1,000 sets of 10 coin flips assuming Harley was just guessing, locate the result of getting 6 heads. Would you consider this an unlikely result in the tail of the distribution?

**(b)** Based on the results of 1,000 simulated sets of 10 coin flips each, would you conclude that Harley would be very unlikely to have picked the correct cup 6 times in 10 attempts if he was randomly guessing between the two cups each time? Explain how your answer relates to the applet's dotplot.

**(c)** Is this study's result statistically significant?

**(d)** Do the results of this study suggest that Harley just guessing is a plausible explanation for Harley picking the correct cup 6 out of 10 times?

**(e)** Does this study prove that Harley cannot understand the mechanical arm?

**Exercise 2.1.25** Compare the analyses between the two studies. How does the unusualness of the observed statistic compare between the two studies? Does this make sense based on the value of the observed statistic in the two studies? Does this make sense based on how the two studies were designed? Explain. (Hint: Why might the results differ for human and mechanical arms? Why would this matter?)

## 2.2 Measuring the strength of evidence

### Tasting water

People spend a lot of money on bottled water. But do they really prefer bottled water to ordinary tap water? Researchers at Longwood University (Lunsford and Dowling Fink, 2010[1]) investigated this question by presenting people who came to a booth at a local festival with four cups of water. Three cups contained different brands of bottled water, and one cup was filled with tap water. Each subject (person) was asked which of the four cups of water they most preferred. Researchers kept track of how many people chose bottled water in order to see whether bottled water was chosen significantly more often than would be expected by random chance.

### Step 1: Ask a research question:

**Exercise 2.2.1** What is the research question that the researchers hoped to answer?

### Step 2: Design a study and collect data:

**Exercise 2.2.2** Identify the observational units in this study.

**Exercise 2.2.3** Identify the variable. Is the variable quantitative or categorical?

**Definition 2.2.1** A **binary variable** is a categorical variable with only two outcomes. Often we convert categorical variables with more than two outcomes (e.g., four brands of water) into binary variables (e.g., tap water or not). In this case we also define one outcome to be a "success" and one to be a "failure."

◊

**Exercise 2.2.4** Rewrite the variable as a binary variable.

**Exercise 2.2.5** Describe the parameter of interest (in words). (Hint: The parameter is the long-run proportion of …?)

**Exercise 2.2.6** One possibility here is that subjects have an equal preference among all four waters and so are essentially selecting one of the four cups at random. In this case what is the long-run proportion (i.e., probability) that a subject in this study would select bottled water?

**Exercise 2.2.7** Another possibility is that the subjects are more likely to prefer bottled water than the tap water. In this case what can you say about the long-run proportion that a subject in this study would select bottled water? (Hint: You are not to specify a particular value this time; instead indicate a direction from a particular value.)

**Definition 2.2.2**

- **Null hypothesis** typically states the "by-random-chance-alone" explanation, which is used to create the chance model, also called the null model. A null hypothesis is expressed symbolically as $H_0$.
- **Alternative hypothesis** typically states the "an-effect-exists" explanation that contradicts the null hypothesis. An alternative hypothesis is expressed symbolically as $H_a$ .

◊

[1] doi.org/10.1080/10691898.2010.11889484

**Exercise 2.2.8** Your answers to Exercise 2.2.6, p. 13 and Exercise 2.2.7, p. 13 should be the null and alternative hypotheses for this study. Which is which?

### Step 3: Explore the data:

**Exercise 2.2.9** Researchers found that 24 of the 27 subjects selected bottled water. Calculate the value of the relevant statistic.

### Use of symbols

Mathematical symbols can be used to represent quantities and simplify your writing. The Greek letter $\pi$ (pronounced "pie") is used to represent a parameter that is a long-run proportion, such as the probability that a (future) subject in this study would choose bottled water as most preferred. The symbol $\hat{p}$ (pronounced "p-hat") is used to represent a statistic that is the proportion of "successes" in a sample, such as the proportion of subjects in this study who did choose bottled water as the most preferred. The symbol $n$ is used to represent the sample size.

**Exercise 2.2.10** What is the value of $\hat{p}$ in this study?

**Exercise 2.2.11** What is the value of n in this study?

**Exercise 2.2.12** Hypotheses are always conjectures about the unknown parameter $\pi$. You can also use H0 and HA as short-hand notation for the null and alternative hypotheses, respectively. A colon ":" is used to represent the word "is". Restate the null and alternative hypotheses using $\pi$.

### Step 4: Draw inferences:

**Exercise 2.2.13** Is the sample proportion who selected bottled water in this study more than the probability specified in the null hypothesis?

**Exercise 2.2.14** Is it possible that this proportion could turn out to be this large even if the null hypothesis was true (i.e., even if people did not really prefer bottled water and were essentially selecting at random from among the four cups)?

As we did with Buzz and Doris previously, we will use simulation to investigate how surprising the observed sample result (24 of 27 selecting bottled water) would be if in fact subjects did not prefer bottled water and so each had a 3/4 probability of selecting bottled water. (Note also that our null model assumes the same probability for all subjects.)

**Key Idea 2.2.3 Think About It.** Can we use a coin to represent the chance model specified by the null hypothesis like before? If not, can you suggest a different random device we could use? What needs to be different about our simulation this time?

**Exercise 2.2.15** Explain why we cannot use a simple coin toss to simulate the subjects' choices, as we did with the Buzz/Doris study.

**Exercise 2.2.16** We could do the simulation using a set of four playing cards: one black and three red. Explain how the simulation would work in this case.

**Exercise 2.2.17 Simulating with Spinners.** Another option would be to use a spinner like you would use when playing a child's board game.

**(a)** Explain how the simulation would work if you were using a spinner. In particular: What does each region represent?

**(b)** How many spins of the spinner will you need to do in order to simulate one repetition of the experiment when there is equal preference between the four waters (null hypothesis is true)?

**Exercise 2.2.18 Simulating with an applet.** We will now use the One Proportion applet[2] to conduct this simulation analysis. Notice that the applet will show us what it would be like if we were simulating with spinners.

**(a)** First enter the **probability of heads/probability of success** value specified in the null hypothesis.

**(b)** Enter the appropriate **sample size** (number of subjects in this study) into the tool.

**(c)** Enter 1 for the number of samples, and press Draw Samples. Report the number of "successes" in this simulated sample.

**(d)** Now, select the radio button for "Proportion of successes." Report the proportion of successes in this simulated sample. Use your answer to "c" to verify how this value is calculated.

**(e)** Leaving the "Proportion of successes" radio button selected but unchecking the "Animate" box, click on Draw Samples four more times. Do you get the same results each time?

**(f)** Now enter 995 for the number of samples and click on Draw Samples, bringing the number of simulated samples to 1,000. Comment on the center, variability, and shape of the resulting distribution of sample proportions.

This distribution of simulated sample proportions is called the **null distribution**, because it is based on assuming the null hypothesis to be true.

**Exercise 2.2.19** Recall that the observed value of the sample proportion who selected bottled water in this study was $\hat{p} = 24/27 \approx 0.8889$. Looking at the null distribution you have simulated, is this a very unlikely result when the null hypothesis is true? In other words, is this value far in the tail of the null distribution?

You might very well find that Exercise 2.2.19, p. 15 is a bit of a tough call. The value 0.8889 is not far in the tail of the distribution, but it's also not near the middle of the distribution. To help make a judgment about strength of evidence in this case, we can count how many (and what proportion) of the simulated sample proportions are as extreme or more extreme than the observed value.

**Exercise 2.2.20** Use the applet to count how many (and what proportion) of the simulated sample proportions are as or more extreme than the observed value. Make sure that the inequality symbol is selected (to match the alternative hypothesis). Then enter 0.8889 (the observed sample proportion who chose bottled water) in the box to the left of the Count button. Then click on the Count button. Record the number and proportion of simulated sample proportions that are as extreme or more extreme than the observed value.

**Definition 2.2.4** The **p-value** is estimated as the proportion of simulated statistics in the null distribution that are at least as extreme (in the direction of the alternative hypothesis) as the value of the statistic actually observed in the research study. ◊

[2] static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1

How do we evaluate this p-value as a judgment about strength of evidence provided by the sample data against the null hypothesis? One answer is: The smaller the p-value, the stronger the evidence against the null hypothesis and in favor of the alternative hypothesis. But how small is small enough to regard as convincing? There is no definitive answer, but here are some guidelines:

**Table 2.2.5 Guidelines for evaluating strength of evidence from a p-value.**

| Range | Strength of evidence |
|---|---|
| $0.10 <$ p-value | Little to no evidence against the null hypothesis. |
| $0.05 <$ p-value $\leq 0.10$ | Moderate evidence against the null hypothesis. |
| $0.01 <$ p-value $\leq 0.05$ | Strong evidence against the null hypothesis. |
| p-value $\leq 0.01$ | Very strong evidence against the null hypothesis. |

**Exercise 2.2.21** Is the approximate p-value from your simulation analysis (your answer to Exercise 2.2.20, p. 15) small enough to provide much evidence against the null hypothesis that subjects prefer bottled water equally to the tap water? If so, how strong is this evidence? Explain.

## Step 5: Formulate conclusions:

**Exercise 2.2.22** When computing p-values, "more extreme" is always measured in the direction of the alternative hypothesis. Use this fact to explain why you counted 24 in Exercise 2.2.20, p. 15.

**Exercise 2.2.23** Do you consider the observed sample result to be statistically significant? Recall that this means that the observed result is unlikely to have occurred by chance alone.

## Step 6: Look back and ahead:

**Exercise 2.2.24** How broadly are you willing to generalize your conclusions? Would you be willing to generalize your conclusions to water drinkers beyond the subjects in this study? How broadly? Explain your reasoning.

**Exercise 2.2.25** Suggest a new research question that you might investigate next, building on what you learned in this study.

## Exploring further

Instead of focusing on the subjects who chose bottled water, you could instead analyze the data based on the subjects who chose tap water. Because 24 of 27 subjects chose bottled water, we know 3 of the 27 subjects chose tap water. Now let the parameter of interest (denoted by $\pi$) be the probability that a subject will select tap water as most preferred.

**Exercise 2.2.26 Tap Water Investigation.** Conduct a simulation analysis to assess the strength of evidence provided by the sample Data.

**(a)** The research conjecture is that subjects tend to select tap water (more or less) often than bottled water.

**(b)** State the null hypothesis in words and in terms of the (newly defined) parameter $\pi$.

**(c)** State the alternative hypothesis in words and in terms of the (new) parameter $\pi$.

**(d)** Calculate the observed value of the relevant statistic.

**(e)** Before you use the One Proportion applet to analyze these data, indicate what values you will input for Probability of success, Sample size, and Number of samples.

**(f)** Use the applet to produce the null distribution of simulated sample proportions. What is the center, variability, and shape of the null distribution?

**(g)** In order to approximate the p-value, you will count how many of the simulated pro- portions are _____ or _____ (larger/smaller) and then divide by _____.

**(h)** Estimate the p-value from your simulation results.

**(i)** Interpret this p-value. (Hint: This is the probability of what, assuming what?)

**(j)** Evaluate this p-value: How much evidence do the sample data provide against the null hypothesis?

**Exercise 2.2.27** Does your analysis based on the number who chose tap water produce similar conclusions to your previous analysis based on the number who chose bottled water? Explain.

You should have found that it does not matter whether you focus on the number/ proportion that chose tap water or the number/proportion that chose bottled water. In other words, it does not matter which category you define to be a "success" for the preferred water variable. Your findings should be very similar provided that you make the appropriate adjustments in your analysis:

- Using 0.25 instead of 0.75 as the null value of the parameter
- Changing the alternative hypothesis to "$\pi < 0.25$" rather than "$\pi > 0.75$"
- Calculating the p-value as the proportion of samples with $\hat{p} \leq 0.1111$ rather than $\hat{p} \geq 0.8889$

## 2.3 Alternative measure of strength of evidence

A study in Psychonomic Bulletin and Review[1] (Lea, Thomas, Lamkin, and Bell, 2007) presented evidence that "people use facial prototypes when they encounter different names." Participants were given two faces and asked to identify which one was Tim and which one was Bob. The researchers wrote that their participants "overwhelmingly agreed" on which face belonged to Tim and which face belonged to Bob but did not provide the exact results of their study.

### STEP 1: Ask a research question

We will gather data from your class to investigate the research question of whether students have a tendency to associate certain facial features with a name.

### STEP 2: Design a study and collect data

Each student in your class will be shown the same two pictures of men's faces used in the research study. You will be asked to assign the name Bob to one photo and the name Tim to the other. Once everyone has made their guess, we will count the proportion of people who chose Tim as the face on the left.

**Exercise 2.3.1** Identify the observational units in this study.

**Exercise 2.3.2** Identify the variable. Is the variable categorical or quantitative?

The parameter of interest here is the probability that a student in your class would assign the same name to the face on the left.

**Exercise 2.3.3** State the null and alternative hypotheses to be tested when the data are collected. Express these both in words and symbols. (Hint: Think about the parameter and the research question of interest here.)



Melissa A. Lea/Robin D. Thomas/Nathan A. Lamkin/Aaron Bell ©Psychonomic Bulletin & Review 2007, 14 (5), 901–907.

**Figure 2.3.1** Consider these two photos:

**Exercise 2.3.4** Do you think the face on the left is Bob or Tim?

[1] doi.org/10.3758/BF03194119

## STEP 4: Explore the data

**Exercise 2.3.5** Collect the responses (data) for all the students in your class. How many students put Tim as the name on the left? How many students participated in this study (sample size)? What proportion put Tim's name on the left?

When we conduct analyses with binary variables, we often call one of the outcomes a "success" and the other a "failure" and then focus the analysis on the "success" outcome. It is arbitrary which outcome is defined to be a success, but you need to make sure you do so consistently throughout the analysis. In this case we'll call "Tim on left" a success because that's what previous studies have found to be a popular choice.

## STEP 4: Draw inferences

You will use the One Proportion[2] applet to investigate how surprising the observed class statistic would be if students were just randomly selecting which name to put with which face.

**Exercise 2.3.6** Before you use the applet, indicate what you will enter for the following values:

**(a)** Probability of success:

**(b)** Sample size:

**(c)** Number of repetitions:

**Exercise 2.3.7** Conduct this simulation analysis. Make sure the Proportion of heads button is selected in the applet and not Number of heads.

**(a)** Indicate how to calculate the approximate p-value (count the number of simulated statistics that equal _____ or _____).

**(b)** Report the approximate p-value.

**(c)** Use the p-value to evaluate the strength of evidence provided by the sample data against the null hypothesis, in favor of the alternative that students really do tend to assign the name Tim (as the researchers predicted) to the face on the left.

The p-value is the most common way to evaluate strength of evidence against the null hypothesis, but now we will explore a common alternative way to evaluate strength of evidence. The goal of any measure of strength of evidence is to use a number to assess whether the observed statistic falls in the tail of the null distribution (and is therefore surprising when the null hypothesis is true) or among the typical values we see when the null hypothesis is true.

**Exercise 2.3.8** Check the **Summary Stats** box in the applet.

**(a)** Report the mean (average) value of the simulated statistics.

**(b)** Explain why it makes sense that this mean is close to 0.50.

**(c)** Report the standard deviation (SD) of the simulated statistics.

**(d)** Report (again) the observed class value of the statistic. (What proportion of students in your class put Tim's name on the left?) $\hat{p} =$

[2] static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1

0.72

**(e)** Calculate how many standard deviations the observed class value of the statistic is from the hypothesized mean of the null distribution, 0.50. In other words, subtract the 0.50 from the observed value and then divide by the standard deviation. In still other words, calculate:

$$\frac{\text{observed statistic}(\hat{p}) - 0.50}{\text{SD of null distribution}}.$$

(0.72-0.496)/0.099 = 2.26

Your calculation in Part 2.3.8.e, p. 20 is called "standardizing the statistic." It is telling us how far above the mean the observed statistic is in terms of the "how many standard deviations."

**Definition 2.3.2** To **standardize** a statistic, compute the distance of the statistic from the (hypothesized) mean of the null distribution and divide by the standard deviation of the null distribution.

$$\text{standardized statistic} = z = \frac{\text{statistic} - \text{mean of null distribution}}{\text{standard deviation of null distribution}}$$

◊

Once you calculate this value, you interpret it as "how many standard deviations the observed statistic falls from the hypothesized parameter value."

The next question is how to evaluate strength of evidence against the null hypothesis based on a standardized value. Here are some guidelines:

Standardizing gives us a quick, informal way to evaluate the strength of evidence against the null hypothesis. For standardized statistics:

**Table 2.3.3 Guidelines for evaluating strength of evidence from standardized values of statistics**

| | |
|---|---|
| between -1.5 and 1.5 | **little or no evidence** against the null hypothesis |
| below -1.5 or above 1.5 | **moderate** evidence against the null hypothesis |
| below -2 or above 2 | **strong** evidence against the null hypothesis |
| below -3 or above 3 | **very strong** evidence against the null hypothesis |

The diagram in the figure below illustrates the basis for using a standardized statistic to assess strength of evidence against the null hypothesis for a mound-shaped, symmetric distribution.

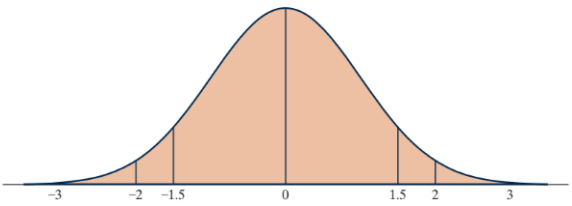



**Figure 2.3.4** Positions of standardized statistics for a bell-shaped distribution.

The figure can be summarized by the following key idea.

**Key Idea 2.3.5** Observations that fall more than 2 or 3 standard deviations from the mean can be considered in the tail of the distribution.

## STEP 5: Formulate conclusions.

**Exercise 2.3.9** Let's examine the strength of evidence against the null.

**(a)** Based on the value of the standardized statistic, z, in Part 2.3.8.e, p. 20 and the guidelines shown above, how much evidence do the class data provide against the null hypothesis?

**(b)** How closely does your evaluation of strength of evidence based on the standardized statistic compare to the strength of evidence based on the p-value in Part 2.3.7.c, p. 19?

Now, let's step back a bit further and think about the scope of inference. We have found that in most classes, the observed data provide strong evidence that students do better than random guessing which face is Tim's and which is Bob's. In that case, do you think that most students at your school would agree on which face is Tim's? Do you think this means that most people can agree on which face belongs to Tim? Furthermore, does this mean that all people do ascribe to the same facial prototyping?

## STEP 6: Look back and ahead.

**Exercise 2.3.10** Based on the limitations of this study, suggest a new research question that you would investigate next.

## Exploring Further

**Exercise 2.3.11** In Exercise 5, p. 19 you recorded the proportion of students in your class who put Tim's name with the photo on the left. Imagine that the proportion was actually larger than that (e.g., if your class was 60%, imagine it was 70%).

**(a)** How would this have affected the p-value?

**(b)** How would this have affected the absolute value of the standardized statistic?

**(c)** How would this have affected the strength of evidence against the null hypothesis?

**Exercise 2.3.12** Suppose that less than half of the students in your class had put Tim's name on the left, so your class result was in the opposite direction of the research conjecture and the alternative hypothesis.

**(a)** What can you say about the standardized value of the statistic in this case? Explain. (Hint: You cannot give a value for the standardized statistic, but you can say something specific about its value.)

**(b)** What can you say about the strength of evidence against the null hypothesis and in favor of the alternative hypothesis in this case?

## 2.4 What impacts strength of evidence?

### Does Uniform Color Matter?

In this exploration, we are going to explore three factors that influence the strength of evidence in a test of significance: (1) the difference between the observed sample statistic and the value of the parameter used in the null hypothesis; (2) the sample size; and (3) one-sided tests versus two-sided tests. To do this we will look at a study conducted by Hill and Barton (Nature, 2005)[1] to investigate whether Olympic athletes in certain uniform colors have an advantage over their competitors. They noticed that competitors in the combat sports of boxing, tae kwon do, Greco-Roman wrestling, and freestyle wrestling are randomly assigned red or blue uniforms. For each match in the 2004 Olympics, they recorded the uniform color of the winner.

**Exercise 2.4.1** What are the observational units? What is the variable measured? Is the variable categorical or quantitative?

The observational units in this study are the matches, and the variable is whether the match was won by someone wearing red or someone wearing blue—a categorical variable. Let's suppose that going into this study, the researchers wanted to see whether the color red had an advantage over blue. In other words, competitors that wear red uniforms will win a majority of the time.

**Exercise 2.4.2** State the null and the alternative hypotheses in words.

**Exercise 2.4.3** We will let $\pi$ represent the probability that a competitor wearing a red uniform wins. Using this, restate the hypotheses using symbols.

**Exercise 2.4.4** Researchers Hill and Barton used data collected on the results of 457 matches and found that the competitor wearing red won 248 times, whereas the competitor wearing blue won 209 times. We will carry out a simulation to assess whether or not the observed data provide evidence in support of the research conjecture. This simulation will employ the **3S strategy**: Determine the **s**tatistic, **s**imulate could-have-been outcomes of the statistic under the null model, and assess the **s**trength of evidence against the null model by estimating the p-value or the standardized statistic.

**(a)** What is the statistic we will use? Calculate the observed value of the statistic in this study.

**(b)** Describe how you could use a coin to develop a null distribution to test our hypothesis.

**(c)** Use the One Proportion applet[2] to test our hypothesis. Based on your simulation, find the p-value and write a conclusion. Also write down the mean and standard deviation from your null distribution when the proportion of successes is used for the variable on the horizontal axis. You will need this later.

### One-Sided vs. Two-Sided Tests

One factor that influences strength of evidence is whether we conduct a two-sided or a one-sided test. Up until now we have only done one-sided tests. In a one-sided test the alternative hypothesis is either $>$ or $<$. In a two-sided test,

[1] doi.org/10.1038/435293a

[2] static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1

the alternative hypothesis is $\neq$. So, why would you do a two-sided test and what are the implications?

Suppose the researchers did not necessarily think that red would win more often, but they also didn't necessarily think that blue would win more often. They were just interested in whether one color would win more often than the other. A two-sided alternative hypothesis (red wins at a rate other than 50% of the time, or at a rate not equal to 50%) allows the researchers to be less sure of the anticipated value of the parameter than a one-sided test.

**Exercise 2.4.5** If we let $\pi$ equal the probability that a competitor wearing a red uniform wins, state the hypotheses for this study in symbols using a two-sided alternative.

**Exercise 2.4.6** Return to the One Proportion[3] applet to approximate the p-value for our original overall proportion of red winning 248 times out of 457 matches, but now select the "Two-sided" check box to find the "two-sided p-value."

**(a)** Describe how the portion of the null distribution that is shaded red is different than our first test done in #3c.

**(b)** Describe how the p-value is different from the p-value that was obtained in our original test done in #3c.

**(c)** To find the two-sided p-value, the applet is looking to see how often 0.543 or larger occurs and how often the comparable value on the other side of the null distribution (or smaller) occurs. To find this value, first compute how far 0.543 is from the center of the null distribution and then go that same distance to the left (less than) the center of the null distribution. What is the comparable value?

**(d)** Complete the following sentence: The two-sided p-value of _____ is the probability of obtaining _____ or larger plus the probability of obtaining _____ or smaller if the _____ is true.

**(e)** You should have seen that when the alternative hypothesis is two-sided, the p-value is computed by looking at how extreme the observed data is in both tails on the null distribution. This makes the p-value about twice as large. Because of this, explain how switching from a one-sided to a two-sided test influences the strength of evidence against the null.

**Key Idea 2.4.1** Because the p-value for a two-sided test is about twice as large as that for a one-sided test, they provide less evidence against the null hypothesis. However, two-sided tests are used more often in scientific practice.

## Difference Between Statistic and Null Hypothesis Parameter Value

**Exercise 2.4.7** A second factor that influences the strength of evidence against the null is how far apart the observed sample statistic and the value of the parameter specified under the null hypothesis are. For this study the null value was 0.50 and the observed sample statistic was about 0.543 (or 54.3% of the competitors wearing red won their matches). Suppose a larger proportion of competitors wearing red won their matches. In fact, suppose 57% of the 457 matches were won by a competitor wearing red.

[3] static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1

**(a)** Go back to the One Proportion applet and approximate the (one-sided) p-value for this situation where again we are testing to see whether the overall probability of winning is more than 0.50.

**(b)** Is your p-value larger or smaller than your original one? Explain why this makes sense.

**(c)** Write a sentence explaining the relationship between the distance between the observed sample statistic and the value of the parameter specified under the null hypothesis to the strength of evidence against the null hypothesis.

**Key Idea 2.4.2** The farther away the observed statistic is from the average value of the null distribution, the more evidence there is against the null hypothesis.

## Sample Size

**Exercise 2.4.8** The third factor we will look at that influences strength of evidence against the null hypothesis is the sample size. As we said earlier, the data for this study came from four combat sports in the 2004 Olympics. One of those sports was boxing. The researchers found that out of the 272 boxing matches, 150 of them were won by competitors wearing red. This proportion of 150/272  0.551 is similar to the overall proportion of times the competitor wearing red won. Let's see what the smaller sample size does to the strength of evidence. Use the One Proportion[4] applet to test the same hypotheses as we originally did, but with just the boxing matches as our sample.

**(a)** Compare the null distribution you generate in this case to that generated in #3. In particular, how do the center (mean) and variability (standard deviation) compare?

**(b)** What is your new p-value? Is it larger or smaller than your original p-value from #3c? Explain why this makes sense.

**(c)** Write a sentence explaining the relationship between sample size and the strength of evidence against the null hypothesis.

**Key Idea 2.4.3** As the sample size increases (and the value of the observed sample statistic stays the same) the strength of evidence against the null hypothesis increases.

[4] static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1

## 2.5 Inference on a single proportion: Theory-based approach

### Eye Dominance

Just like handedness where people prefer to use one hand over another, eye dominance, sometimes called eyedness, is the tendency to prefer to see using one eye over the other. Interestingly though the side of the dominant eye does not always match that of the dominant hand. Let's investigate whether people are equally likely to have left-eye or right-eye dominance by collecting some data from you and your classmates.

**Exploration 2.5.1** To figure out which of your eyes is the dominant eye, carry out the following "dominant eye test":

- Extend both your arms in front of you and create a triangular opening using your thumbs and pointer fingers.
- With both eyes open, center your triangular opening on a distant object such as a clock or a projector.
- Close your left eye. If the object stays centered in your triangular opening, your right eye (as that is the one that's open) is your dominant eye. If the object is no longer in the triangular opening, your left eye is your dominant eye.
- Double check this by closing your right eye. If the object stays centered in your triangular opening, your left eye (as that is the one that's open) is your dominant eye. If the object is no longer in the triangular opening, your right eye is your dominant eye.

**Exercise 2.5.2** Record whether you have left-eye or right-eye dominance.

Before we combine your data with the data from your classmates, let's think about what we want to test here. Conventional wisdom says that more often people are right-handed than left, so for now let's use our research hypothesis to be that more often people tend to have right-eye dominance than left-eye dominance.

**Exercise 2.5.3** In this study with your class:

**(a)** What are the observational units in this study?

**(b)** What is the variable that is recorded?

**(c)** Describe the parameter of interest in words. (Use the symbol $\pi$ to represent this parameter.)

**(d)** If right-eye and left-eye dominance are equally prevalent, what would you expect the numerical value of the parameter to be? Is this the null hypothesis or the alternative hypothesis?

**(e)** If people are more likely to be right-eye dominant than left, what can you say about the numerical value of the parameter? Is this the null hypothesis or the alternative hypothesis?

**Exercise 2.5.4** Including yourself and your classmates, how many people participated in this study? How many have right-eye dominance? Calculate the sample proportion who are right-eye dominant.

**Exercise 2.5.5** To have a larger sample size to analyze, combine your class results with the results from some of the author's classes, in which 70 of 115 students had right-eye dominance. Now what are the sample size and the sample proportion that are right-eye dominant?

Use the One Proportion[1] applet to test the hypotheses from Part 2.5.3.d, p. 25 and Part 2.5.3.e, p. 25.

**Exercise 2.5.6**

**(a)** Describe the shape of the null distribution of sample proportions. Does this shape look familiar? Where is the null distribution centered? Does this make sense? Check the Summary Stats box and report the mean and standard deviation as reported by the applet.

**(b)** Approximate the p-value and summarize the strength of evidence that the sample data provide regarding the research hypothesis.

**(c)** Determine the standardized statistic, z, and summarize the strength of evidence. Confirm that the strength of evidence obtained using the standardized statistic is similar to that obtained using the p-value.

## Theory-Based Approach (One-Proportion z-test)

In Part 2.5.6.a, p. 26, you probably described the shape of the null distribution using words such as bell-shaped, symmetric, or maybe even normal. You have seen many null distributions in this chapter that have had this same basic shape. You should have also noticed that the null distributions have all been centered at the hypothesized value of the long-run proportion used in the null hypotheses. You probably could have predicted that your null distribution was going to be somewhat bell-shaped and centered at 0.50. You probably would have a harder time predicting your null distribution's variability (standard deviation), but this too can be predicted in advance, as we will see shortly. We can use mathematical models known as normal distributions (bell-shaped curves) to approximate many of the null distributions we have generated so far in this text. When rules and theories are used to predict what the value of the standardized statistic and p-value would be if someone carried out a simulation, we call the approach a *theory-based approach*. The normal distribution provides a second way, in addition to simulation, to approximate a p-value.

**Exercise 2.5.7** Check the box next to Normal Approximation in the applet. Does the region shaded in blue seem to be a good description (model) of what we actually got in the simulation?

## Validity Conditions for Theory-Based Approach

The normal approximation to the null distribution is valid whenever the sample size is reasonably large. One convention is to consider the sample size to be large enough whenever there are at least 10 observations in each category.

**Exercise 2.5.8** According to this convention, is the sample size large enough in this study to use the normal approximation and theory-based inference? Justify your answer.

**Validity Conditions 2.5.1** The normal approximation can be thought of as a prediction of what would occur if a simulation-based analysis was carried out. Many times this prediction is valid, but not always. It is only valid when the

[1] `static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1`

condition (at least 10 successes and at least 10 failures) is met.

## Formulas

The normal approximation will also give you values of the standardized statistic and p-value based on its mathematical predictions. As you learned previously, the standardized score is calculated as

$$\text{standardized statistic} = z = \frac{\text{statistic} - \text{mean of null distribution}}{\text{standard deviation of null distribution}}$$

The mean of the null distribution is the hypothesized value of the long-run proportion ($\pi$). The standard deviation can be obtained in two ways:

First, find the standard deviation of the null distribution by simulating.

Second, predict the value of the standard deviation by substituting into this formula:

$$\text{standard deviation of null distribution of single proportion} = \sqrt{\frac{\pi(1-\pi)}{n}}$$

**Exercise 2.5.9** Use the formula to determine the (theoretical; predicted) standard deviation of the sample proportion. Then compare this to the SD from your simulated sample proportions, as recorded in Part 2.5.6.a, p. 26. Are they similar?

The predicted value of the standard deviation (using the formula) will be very close to the simulated standard deviation of the null distribution. The validity condition mentioned earlier says the shape will be approximately normal when the sample size is large enough where, a "large enough" sample size means at least 10 successes and at least 10 failures. This mathematical prediction is often called the "central limit theorem."

**Theorem 2.5.2 Central Limit Theorem.** *If the sample size (n) is large enough, the distribution of sample proportions will be bell-shaped (or normal), centered at the long-run proportion ($\pi$), with a standard deviation of* $\sqrt{\frac{\pi(1-\pi)}{n}}$ .

**Exercise 2.5.10** Use the predicted value of the standard deviation from Exercise 9, p. 27 to calculate the standardized statistic (z) by hand and confirm that your answer is very close to what you found in Part 2.5.6.c, p. 26 when using simulation.

In the applet, see that the predicted value of the standardized statistic, z, is given immediately below the button for "Normal approximation" in parentheses and should match your answer to Exercise 10, p. 27.

**Exercise 2.5.11** The theory-based (normal approximation) p-value is also now displayed. Compare this p-value to the one you got from simulation Part 2.5.6.b, p. 26. Are they similar?

**Exercise 2.5.12** Why are the standard deviation (Exercise 9, p. 27), standardized statistic (Exercise 10, p. 27) and p-value (Exercise 11, p. 27) similar when using the theory-based (one-proportion z-test; normal approximation) to what you got in your simulation? When would they be different?

## Exploring Further

### Follow-up Analysis #1

There are several research papers (see, for example, Ehrenstein et al., 2005[2]) on eye preference that say that the long-run proportion of right-eyed people is two-thirds.

**Exercise 2.5.13** Use the theory-based approach to test the claim that there is a two-thirds chance that a person will be right-eye dominant, using the sample data (your class combined with author's class) and a two-sided alternative. Report the null and alternative hypothesis, standardized statistic, and p-value. Summarize your conclusion and explain the reasoning process by which it follows from your analysis.

### Follow-up Analysis #2

In a small class of 14 students, nine students turned out to be right-eye dominant.

**Exercise 2.5.14** Use simulation to generate a two-sided p-value evaluating the strength of evidence that the long-run proportion of right-eyed students is different than 50% based on this small class's data alone.

**Exercise 2.5.15** Why can't you use the normal approximation in this case?

**Exercise 2.5.16** Use the normal approximation anyway. Compare and comment on the p-values obtained from the two methods.

[2] doi.org/10.1007/s00417-005-1128-7

# 2.6 Data and Formulas

## Tools

**Table 2.6.1 Table 2121: One-proportion relevant features by section**

| Section | Feature |
|---|---|
| Introduction to chance models | Input chance model parameters and build repetitions. |
| Measuring strength of evidence | Calculate p-value. |
| Alternative measures of strength of evidence | Calculate standardized statistic. |
| What impacts strength of evidence? | Specify the sidedness of the alternative hypothesis. |
| Inference on a single proportion: theory-based approach | Overlay a normal distribution and calculate theory-bas |
| Tools, data, and formulas | Specify statistic as "proportion of successes" or "numb |

## Data

**Table 2.6.2 References**

| Section | Data |
|---|---|
| Example: Introduction to chance models | $\pi$=0.5, n=16, $\hat{p}$=0.9375 |
| Supplemental Exploration: Introduction to chance models | $\pi$=0.5, n=10, $\hat{p}$=0.9 |
| Example: Measuring strength of evidence | $\pi$=0.333, n=12, $\hat{p}$=0.833 |
| Supplemental Exploration: Measuring strength of evidence | $\pi$=0.75, n=27, $\hat{p}$=0.8889 |
| Example: Alternative measures of strength of evidence | $\pi$=0.15, n=361, $\hat{p}$=0.197 |
| Supplemental Exploration: Alternative measures of strength of evidence | $\pi$=0.5, n=34, $\hat{p}$=0.705 |
| Example: What impacts strength of evidence? | $\pi$=0.5, n=32, $\hat{p}$=0.719 |
| Supplemental Exploration: What impacts strength of evidence? | $\pi$=0.5, n=32, $\hat{p}$=0.719 |
| Example: Inference on a single proportion: Theory-based approach | $\pi$=0.5, n=248, $\hat{p}$=0.543 $\pi$= |
| Supplemental Exploration: Inference on a single proportion: Theory-based approach | $\pi$=0.5, n=116, $\hat{p}$=0.612 |
| Investigation: Tire story falls flat | $\pi$=0.25, n=28, $\hat{p}$=0.5 |

## Formulas for one proportion

- $\pi$ is the underlying process probability of success or population proportion of success.
- $n$ is the sample size.
- $\hat{p} = \frac{\text{number of successes in the sample}}{n}$ is the sample proportion.
- $SD = \sqrt{\pi(1-\pi)/n}$ is the theory-based standard deviation of the null distribution.

[1] apps.dtic.mil/sti/citations/AD0646241
[2] doi.org/10.1037/0735-7036.122.1.84
[3] www.maa.org/programs/faculty-and-departments/classroom-capsules-and-notes/winning-at-rock-paper-scissors
[4] doi.org/10.1080/10691898.2010.11889484
[5] doi.org/10.1136/bmj.37956.520567.44
[6] doi.org/10.3758/BF03194119
[7] doi.org/10.1126/science.1110589
[8] doi.org/10.1038/435293a
[9] doi.org/10.1016/S1499-4046(06)60335-7
[10] doi.org/10.1007/s00417-005-1128-7
[11] doi.org/10.1007/s00417-005-1128-7

- standardized statistic $= \frac{\text{statistic} - \text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
    - $z = \frac{\hat{p} - \pi}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one proportion.
    - $z = \frac{\hat{p} - \pi}{SD} = \frac{\hat{p} - \pi}{\sqrt{\pi(1-\pi)/n}}$ is the theory-based standardized statistic for one proportion.

- $z = \frac{\hat{p} - \pi}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one proportion.

- $z = \frac{\hat{p} - \pi}{SD} = \frac{\hat{p} - \pi}{\sqrt{\pi(1-\pi)/n}}$ is the theory-based standardized statistic for one proportion.

# Chapter 3

# Generalization

## 3.1 Sampling from a finite population

### Literary Computing

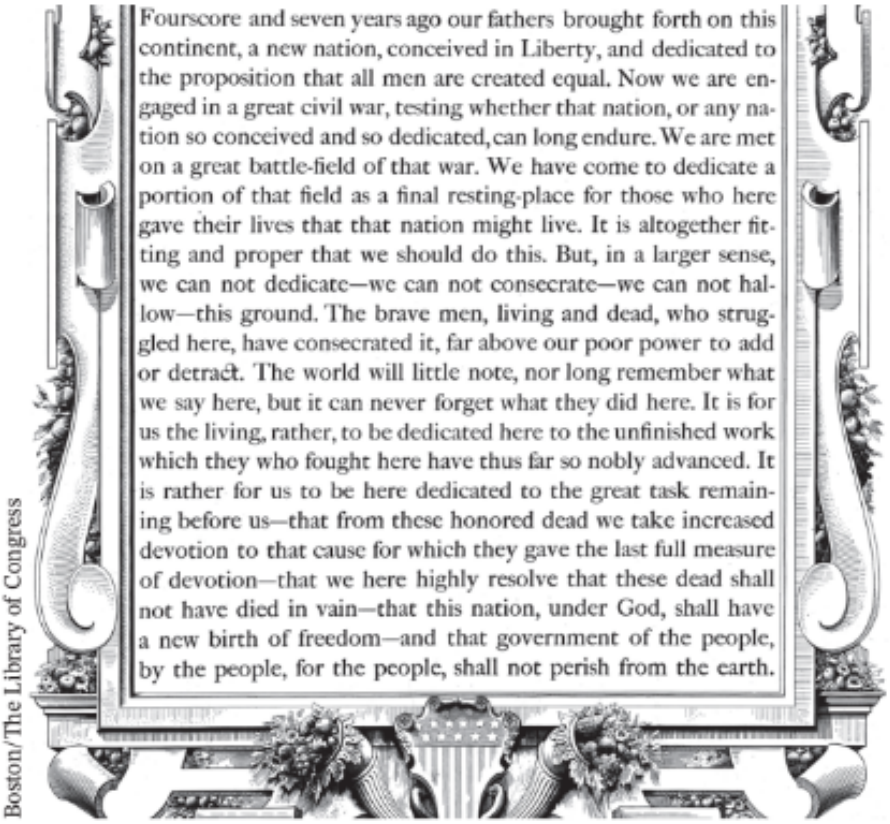



**Figure 3.1.1**

**Exercise 3.1.1** Select a representative set of 10 words from the passage above by circling them with your pen or pencil.

The authorship of literary works is often a topic for debate. For example, researchers have tried to determine whether some of the works attributed to Shakespeare were actually written by Bacon or Marlow. The field of "literary computing" examines ways of numerically analyzing authors' works, looking at variables such as sentence length and rates of occurrence of specific words.

**Exercise 3.1.2** Complete the following data table by recording each word from your sample and then indicate the length of the word (number of letters) and whether or not the word is "short" (3 or fewer letters).

| Word | Length (no. of letters) | Short (3 or fewer letters)? (Y or N) |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |
| 8 | | |
| 9 | | |
| 10 | | |

**Exercise 3.1.3** Identify the observational units and the variables you have recorded on these observational units (Keep in mind that observational units do not have to be people!)

**Definition 3.1.2** The **population** is the entire collection of observational units of interest, and the **sample** is a subset of the population on which we record data. ◊

The passage is, of course, Lincoln's Gettysburg Address, given November 19, 1863, on the battlefield near Gettysburg, Pennsylvania. We are considering this passage a **population** of 268 words, and the 10 words you selected are a **sample** from this population. As before, we can use $\hat{p}$ to refer to a sample proportion and $\pi$ to refer to the parameter, this time the proportion of short words in the population.




**Exercise 3.1.4** Do you think the sample proportion of short words you found will be a good estimate of the population proportion, $\pi = 0.41$ ? Why or why not?

**Exercise 3.1.5** Suggest a method for deciding whether the sample proportion of short words is likely to be close to the population proportion of short words. (Hint: Whereas any one sample may not produce a statistic that exactly equals the population parameter, what would we like to be true in general?)

**Exercise 3.1.6** Combine your results with your classmates' by producing a dotplot of the distribution of the proportion of short words in your samples.

**Exercise 3.1.7** What label should be along the horizontal axis of this graph? How many dots do you have? In other words, clarify what each dot represents. (Hint: If you wanted to add another dot to the graph, what would you need to do?)

**Exercise 3.1.8** Based on the class dotplot, does this sampling method tend to produce sample proportions close to the population proportion, 0.41? Justify your answer.

**Definition 3.1.3** A sampling method is **biased** if, when using that sampling method, statistics from different samples *consistently* overestimate or consistently underestimate the population parameter of interest. ◊

Note that bias is a property of a sampling *method*, not a property of an individual sample. Also note that the sampling method must *consistently* produce nonrepresentative results in the same direction in order to be considered biased

**Exercise 3.1.9** Does asking you to quickly select 10 representative words appear to be a biased or unbiased sampling method? If biased, what is the direction of the bias (tendency to overestimate or underestimate the proportion of short words)?

**Exercise 3.1.10** Explain why we might have expected this sampling method (asking you to quickly pick 10 representative words) to be biased for this variable.

**Exercise 3.1.11** Do you think that if we'd asked each of you to take 20 words instead of 10 words it would have helped with this issue? Explain.

**Exercise 3.1.12** Suggest another technique for selecting 10 words from this population in order for the sampling method to be unbiased and likely to produce samples that are representative of the population proportion of short words.

## Taking a Simple Random Sample

**Key Idea 3.1.4** A **simple random** sample ensures that every sample of size $n$ is equally likely to be the sample selected from the population. In particular, each observational unit has the same chance of being selected as every other observational unit.

The key to obtaining a representative sample is using some type of *random* mechanism to select the observational units from the population, rather than relying on **convenience samples** or any type of human judgment.

**Definition 3.1.5** A **convenience sample** is a nonrandom sample of a population. ◊

Instead of having you choose "random" words using your own judgment, we will now ask you to take a simple random sample of words and evaluate your results. The first step is to obtain a **sampling frame**—a complete list of every member of the population where each member of the population can be assigned a number. Below is a copy of the Gettysburg Address that includes numbers in front of every word. For example, the 43rd word is "nation". *Note*: In this sampling frame, "four score" is listed as two words.

**Exercise 3.1.13**

**(a)** Go to the **Random Numbers**[1] applet (or random.org[2]).

- Specify that you want 5 *Numbers per replication* in the range from 1 to 268.
- Press *Generate* to view the 5 random numbers.
- *Enter* the random numbers in the table below:

**Generate Random Number**

Number of replications: 1

Numbers per replication: 5

Number range: From: 1

To: 268

Randomly generated five ID values from 1 to 268:

| | | | | |
|---|---|---|---|---|
| | | | | |

**(b)** Using your randomly generated numbers, look up the corresponding word from the sampling frame. Fill in the ***data table*** below.

| Word | Length (no. of letters) | Short (3 or fewer letters)? (Y or N) |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Four | 17 | conceived | 33 | are | 49 | and | 65 | We | 81 | those |
| 2 | score | 18 | in | 34 | engaged | 50 | so | 66 | have | 82 | who |
| 3 | and | 19 | liberty | 35 | in | 51 | dedicated | 67 | come | 83 | here |
| 4 | seven | 20 | and | 36 | a | 52 | can | 68 | to | 84 | gave |
| 5 | years | 21 | dedicated | 37 | great | 53 | long | 69 | dedicate | 85 | their |
| 6 | ago | 22 | to | 38 | civil | 54 | endure | 70 | a | 86 | lives |
| 7 | our | 23 | the | 39 | war | 55 | We | 71 | portion | 87 | that |
| 8 | fathers | 24 | proposition | 40 | testing | 56 | are | 72 | of | 88 | that |
| 9 | brought | 25 | that | 41 | whether | 57 | met | 73 | that | 89 | nation |
| 10 | forth | 26 | all | 42 | that | 58 | on | 74 | field | 90 | might |
| 11 | upon | 27 | men | 43 | nation | 59 | a | 75 | as | 91 | live |
| 12 | this | 28 | are | 44 | or | 60 | great | 76 | a | 92 | It |
| 13 | continent | 29 | created | 45 | any | 61 | battlefield | 77 | final | 93 | Is |
| 14 | a | 30 | equal | 46 | nation | 62 | of | 78 | resting | 94 | altogether |
| 15 | new | 31 | Now | 47 | so | 63 | that | 79 | place | 95 | fitting |
| 16 | nation | 32 | we | 48 | conceived | 64 | war | 80 | for | 96 | and |

*(continued)*

**Figure 3.1.6** Gettysburg Address Sampling Frame

[1] `static-content.zybooks.com/applets/2021/randomnumbers/GenRandom01.htm?hideExtras=1`

[2] `www.random.org/`

*(continued)*

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 97 | proper | 126 | struggled | 155 | never | 184 | so | 213 | that | 242 | under |
| 98 | that | 127 | here | 156 | forget | 185 | nobly | 214 | cause | 243 | God |
| 99 | we | 128 | have | 157 | what | 186 | advanced | 215 | for | 244 | shall |
| 100 | should | 129 | consecrated | 158 | they | 187 | It | 216 | which | 245 | have |
| 101 | do | 130 | It | 159 | did | 188 | is | 217 | they | 246 | a |
| 102 | this | 131 | far | 160 | here | 189 | rather | 218 | gave | 247 | new |
| 103 | But | 132 | above | 161 | It | 190 | for | 219 | the | 248 | birth |
| 104 | in | 133 | our | 162 | is | 191 | us | 220 | last | 249 | of |
| 105 | a | 134 | poor | 163 | for | 192 | to | 221 | full | 250 | freedom |
| 106 | larger | 135 | power | 164 | us | 193 | be | 222 | measure | 251 | and |
| 107 | sense | 136 | to | 165 | the | 194 | here | 223 | of | 252 | that |
| 108 | we | 137 | add | 166 | living | 195 | dedicated | 224 | devotion | 253 | government |
| 109 | cannot | 138 | or | 167 | rather | 196 | to | 225 | that | 254 | of |
| 110 | dedicate | 139 | detract | 168 | to | 197 | the | 226 | we | 255 | the |
| 111 | we | 140 | The | 169 | be | 198 | great | 227 | here | 256 | people |
| 112 | cannot | 141 | world | 170 | dedicated | 199 | task | 228 | highly | 257 | by |
| 113 | consecrate | 142 | will | 171 | here | 200 | remaining | 229 | resolve | 258 | the |
| 114 | we | 143 | little | 172 | to | 201 | before | 230 | that | 259 | people |
| 115 | cannot | 144 | note | 173 | the | 202 | us | 231 | these | 260 | for |
| 116 | hallow | 145 | nor | 174 | unfinished | 203 | that | 232 | dead | 261 | the |
| 117 | this | 146 | long | 175 | work | 204 | from | 233 | shall | 262 | people |
| 118 | ground | 147 | remember | 176 | which | 205 | these | 234 | not | 263 | shall |
| 119 | The | 148 | what | 177 | they | 206 | honored | 235 | have | 264 | not |
| 120 | brave | 149 | we | 178 | who | 207 | dead | 236 | died | 265 | perish |
| 121 | men | 150 | say | 179 | fought | 208 | we | 237 | in | 266 | from |
| 122 | living | 151 | here | 180 | here | 209 | take | 238 | vain | 267 | the |
| 123 | and | 152 | but | 181 | have | 210 | increased | 239 | that | 268 | earth |
| 124 | dead | 153 | it | 182 | thus | 211 | devotion | 240 | this | | |
| 125 | who | 154 | can | 183 | far | 212 | to | 241 | nation | | |

**Figure 3.1.7**

**Exercise 3.1.14** Let's examine your sample of five words, as well as those of your classmates.

**(a)** Calculate the proportion of short words in your random sample.

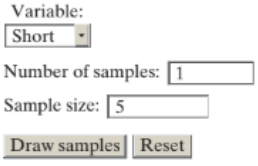
**(b)** Again, produce a dotplot of the distribution of proportions of short words for your sample and those of your classmates. Make sure you label the horizontal axis appropriately.

**(c)** Comment on how this distribution compares to your class' dotplot of sample proportions from Exercise 6, p. 34 which used nonrandom sampling.

**(d)** Does this second method appear to be an unbiased sampling method? How are you deciding?

Let's use technology to take even more random samples.

- Open the Sampling Words[3] applet.
- Use the *Variable* pull down menu, to choose "Short". Confirm the population proportion is 0.41.
- Check the *Show Sampling Options* box. Specify 5 in the *Sample Size* box and press the *Draw Samples* button.




You will see the five selected words appear in the box below. You will also see five blue dots within the population distribution representing the five words you have sampled (and whether they are short). These five yes/no outcomes for your sample are also displayed in a dotplot in the middle panel. The blue triangle indicates the proportion of short words in this sample. This value also appears on the Sampling Distribution graph to the right. Press *Draw Samples* again.

[3] `static-content.zybooks.com/applets/2021/sampling/OneSample.html?population=gettysburg`

**Exercise 3.1.15** Did you get the same sample this time? Did you get the same sample proportion?

**Exercise 3.1.16** Suppose you take 1,000 different random samples and examined the distribution of the 1,000 sample proportions. Where do you think this distribution will be centered? What shape do you think it will have? Predict the largest and smallest values.

**Exercise 3.1.17** Change the Number of samples from 1 to 9998, for a total of 10,000 samples and press Draw Samples. Describe the behavior of the distribution of sample proportions and contrast it with your prediction, as well as to the class distribution of nonrandom samples using 10 words.

**Definition 3.1.8** A **sampling distribution** is the distribution of a statistic for all possible samples of size n randomly selected from the same population. By taking lots of samples, you can approximate this sampling distribution. ◊

The following table outlines the simulation process you are using to approximate the sampling distribution of sample proportions of short words from the Gettysburg Address.

**Table 3.1.9 Parallels between population distributions and sampling distributions**

| | |
|---|---|
| Population | All 268 words in the Gettysburg Address |
| Parameter | Proportion of words in the population that are short ($\pi = 0.41$) |
| Sample | 5 randomly selected words |
| Statistic | Proportion of sampled words that are short ( $\hat{p}$ varies from sample to sample) |

One big change between the random samples and the previous convenience samples is the center of the sampling distribution should now be close to $\pi$, the population proportion. In other words, simple random sampling is an unbiased sampling method because there is no tendency to over- or underestimate the parameter.

**Key Idea 3.1.10** The mean of the sampling distribution of sample proportions for all possible simple random samples from a population with success proportion $\pi$ will be equal to $\pi$.

As we saw, this sampling method (taking 5 randomly selected words) is better (unbiased) even though we selected fewer words.

## Sampling Distribution of Sample Proportions

Based on what you learned about sample proportions previously, you should not be too surprised that the distribution of sample proportions you produced in Exercise 17, p. 37 is somewhat symmetric (because $\pi$ is close to 0.50), with a mean close to 0.41 (because the sampling method is unbiased). In fact, our previous formula for the standard deviation of sample proportions, $\sqrt{\frac{\pi(1-\pi)}{n}}$, still works as well!

**Exercise 3.1.18** Verify that the standard deviation of your 10,000 sample proportions is close to the standard deviation of sample proportions defined above. (Hint: What values are you using for $\pi$ and n?)

**Key Idea 3.1.11** When sampling from a large population, the standard deviation of sample proportions is estimated by the same formula,

$$\text{standard deviation of sample proportions} = \sqrt{\frac{\pi(1-\pi)}{n}},$$

where $\pi$ represents the proportion of successes in the population and n represents the sample size. The population size is considered large enough when it is more than 20 times that sample size.

Notice one new aspect here compared to earlier: When we are sampling from a finite population rather than from an infinite random process, we want the population to be large in order to assume $\pi$ represents a constant probability of success. In this case, $268 > 200 (= 20 \times 10)$, and the formula should apply. In which case, we can use the same theory-based methods we used before.

**Exercise 3.1.19** How would our prediction of the behavior (shape, mean, largest value/smallest value) of the distribution of sample proportions change if the population size had been 2,680?

**Exercise 3.1.20** Name two things that do impact the standard deviation of the distribution of sample proportions.

**Exercise 3.1.21** In the applet, select the radio button labeled Fixed. Then change the Sample size from 5 to 10 and the Number of samples from 9,998 to 10,000. Press Draw Samples and describe the behavior of the new sampling distribution. Does the standard deviation change as you predicted? Is the sampling method still unbiased? How are you deciding?

**Exercise 3.1.22** Repeat Exercise 21, p. 38 for a sample size of 20.

Keep in mind that our population size (268) is not all that large here and you may start to see some discrepancies with the standard deviation formula with samples of 20 or more (Why?). But also notice that the shape of the distribution of sample proportions is looking more and more like a normal distribution as we increase the sample size, just as you saw previously.

**Key Idea 3.1.12** The **Central Limit Theorem for sampling from a large finite population** says that the distribution of sample proportions from repeated random samples will be approximately normal if there at least 10 successes and at least 10 failures in each sample.

In the following figure, we have overlaid the normal distribution predicted by the Central Limit Theorem and determined the proportion of samples and the theoretical probability for a sample proportion of 0.55 and higher.

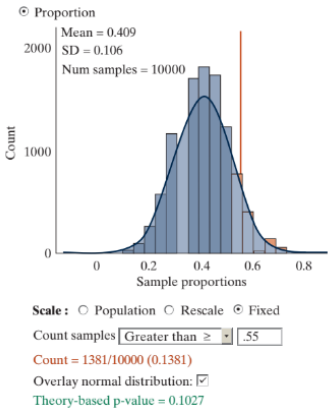



**Figure 3.1.13** Figure 321: Distribution of sample proportions compared to predicted distribution

**Exercise 3.1.23** Does the normal approximation appear to be valid here? Which value (0.1381 or 0.1027) do you trust more? Explain your reasoning.

## Other Considerations

Another key property of our sampling method is that we had a list of every word in the population. This list is referred to as a **sampling frame**. An incomplete sampling frame (e.g., only the first paragraph of the speech) would also produce a biased sampling method if the observational units not in the sampling frame were systematically different from the rest of the population.

**Key Idea 3.1.14** If a sampling frame does not include every member of the population (e.g., last paragraph missing, nouns not listed), then we cannot claim to represent those segments in the population with our sample. An incomplete sampling frame can also lead to biased sampling.

You also need to be sure there are not any other **nonsampling concerns**. These represent problems that arise even with a carefully selected sample. For example, suppose we were not able to accurately measure the lengths of the words. Or, different individuals used different definitions of "short."

**Key Idea 3.1.15** In addition to carefully selecting the sample, you need to guard against other possible sources of bias as well. For example, when talking to people, their suspect memories and truthfulness can impact people's answers. Other examples of **nonsampling concerns** include the effects that the wording of a question can have on people's responses (e.g., how extreme the phrasing is), the effects that the interviewer can have by virtue of their demeanor, sex, race, and other characteristics, using an uncalibrated scale, and so on.

See Example 2.1 for additional discussion of nonsampling concerns.

## Exploring Further

**Exercise 3.1.24** As part of the General Social Survey in 2018, a random sample of U.S. adults were asked whether at least one of their grandparents was born outside the United States. In the sample of 2,214 respondents, 881 said "yes."

**(a)** Define the parameter of interest in words. Do you know its value (if so, what is it)?

**(b)** Describe the statistic in words. Do you know its value (if so, what is it)?

**(c)** Do you consider this sample of 2,214 U.S. adults to be generalizable to the roughly 253,881,929 U.S. adults in 2018? Explain why or why not.

**(d)** Suppose we assume the population proportion is 0.50. If we were to predict the behavior of the distribution of sample proportions, what do you predict for: Mean? Standard deviation? Shape?

**(e)** Based on the observed sample proportion, is it plausible that the observed sample came from a population with $\pi = 0.50$? Be clear how you are deciding.

**(f)** Is it plausible that the observed sample came from a population with $\pi = 0.40$? Explain how you are deciding.

**(g)** Do you have any reason to suspect the respondents could be giving misleading information to this question? What are the implications of your answer?

## 3.2 Inference for a single quantitative variable

### Sampling words (cont.)

Let's reconsider the *population* of words in the Gettysburg Address[1] you first looked at in a previous exploration. Suppose instead of looking at the proportion of short words (3 or fewer letters) we look at the *number of letters per word*.

**Exercise 3.2.1** Is number of letters per word a quantitative or categorical variable? Why?

**Exercise 3.2.2** Suggest a statistic (a single number) you could calculate from your sample that summarizes the number of letters per word variable.

**Exercise 3.2.3** What parameter are you hoping to estimate with your statistic? (Hint: Be clear how this relates to the population.)

**Exercise 3.2.4** How could we explore whether the sampling methods used previously are unbiased in estimating the parameter you identified in #3? What does it mean to be unbiased in this context?

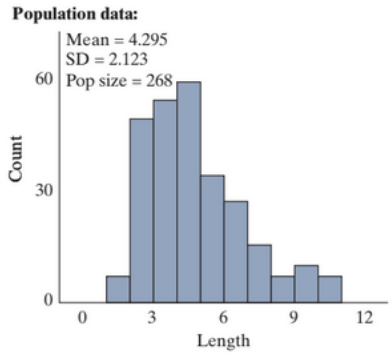



**Figure 3.2.1** Population distribution for lengths of words (number of letters per word) in the Gettysburg address

**Definition 3.2.2** A distribution of data is **skewed** if it is not symmetric and, instead, the bulk of values tend to fall on one side of the distribution with a "longer tail" on the other. Right-skewed distributions have their tail on the right, and left-skewed distributions have their tail on the left. ◊

With skewed data, we might prefer to measure the "center" of the distribution with the median rather than the mean.

**Definition 3.2.3** The **median** is the middle data value when the data are sorted in order from smallest to largest. ◊

**Exercise 3.2.5** The mean is found to be 4.295 letters, do you think the median is larger or smaller than 4.295? Briefly explain your reasoning.

**Exercise 3.2.6** Summarize the behavior of this population. Recall that when we have a quantitative variable, we should talk about shape, center (using the mean and/or the median), variability (using the standard deviation), and outliers, as well as making sure that we are making comments relevant to the context of the study.

**Key Idea 3.2.4** To distinguish a quantitative variable from a categorical variable, we will use different symbols to refer to the statistics and parameters. In

[1] www.loc.gov/exhibits/gettysburg-address/

particular,

**Table 3.2.5**

| **Statistics** | **Parameters** |
|---|---|
| $\bar{x}$ = sample mean | $\mu$ = population mean |
| $s$ = sample standard deviation | $\sigma$ = population standard deviation |

Note that we are focusing on the mean and the standard deviation.

**Exercise 3.2.7** Identify the values of $\mu$ and $\sigma$ for the length of words in the Gettysburg Address.

To evaluate our sampling method for a quantitative variable, we want to see whether the means from different samples center around the population mean.

To evaluate our sampling method for a quantitative variable, we want to see whether the means from different samples center around the population mean.

**Exercise 3.2.8** What term do we use to describe a sampling method where the distribution of the statistic centers around the population mean?

**Exercise 3.2.9** Open the Sampling Words applet[2]. Make sure that the Variable is set to "Length." Check the Show Sampling Options box, change the Sample size to 5, and press Draw Samples.

Notice that the graph on the left in the applet and in Figure 3.2.6, p. 42 is the population, the graph in the middle is your sample, and the mean of this sample is placed in the graph on the right. Fill in your results for the middle and right most graphs in Figure 3.2.6, p. 42. Also indicate the value of the sample mean that you obtained.

[2] static-content.zybooks.com/applets/2021/sampling/OneSample.html?population=gettysburg&hideExtras=1

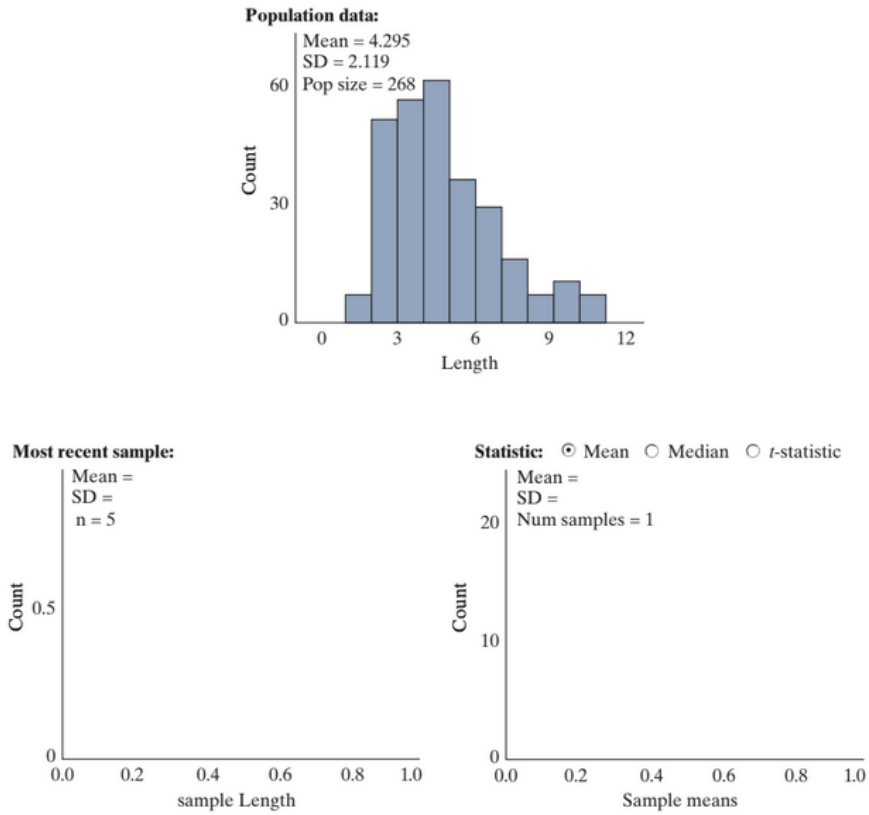
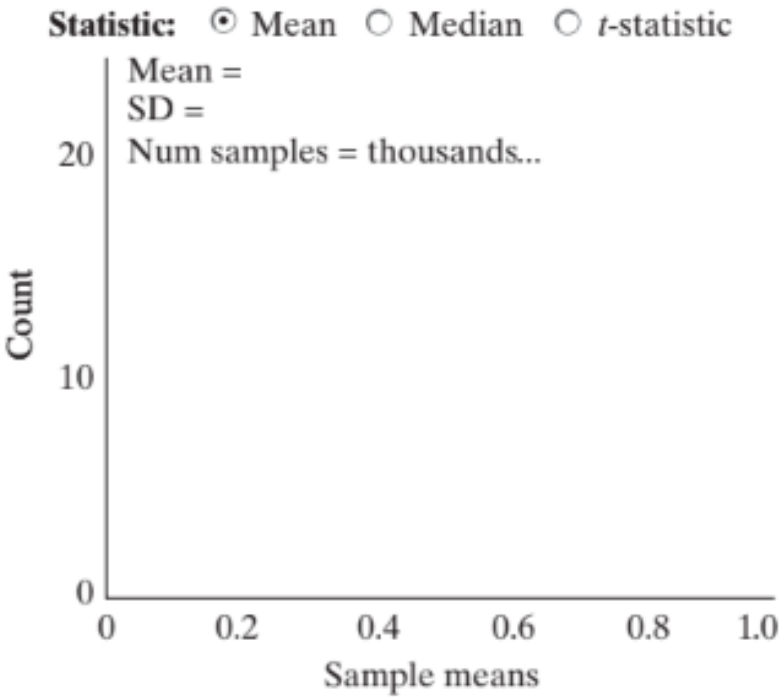



**Figure 3.2.6** Outputs in the Sampling Words applet

**Exercise 3.2.10** If we were to repeat the process in #9 thousands of times, predict the behavior of the distribution of sample means—the graph all the way to the right in the applet. (Shape? Center? Variability? Compared to the population?) Sketch your prediction, changing the horizontal axis scale if necessary.

**Exercise 3.2.11** Change the Number of samples from 1 to 9999 (for 10,000 samples) and press Draw Samples.

**(a)** Create a sketch of the distribution in the box below — including carefully labeling the horizontal axis and indicating what each dot on the graph represents. (Hint: How many dots do you have? What would you/the applet do to add one more dot to the graph?)




**(b)** Describe the shape of the distribution of sample means. How does it compare to the shape of the population?

**(c)** What is the mean of the sampling distribution of sample means? How does it compare to the mean of the population?

**(d)** What is the standard deviation of the sampling distribution of sample means? How does it compare to the standard deviation of the population?

**Exercise 3.2.12** Does simple random sampling for quantitative data appear to be an unbiased sampling method? How are you deciding?

**Exercise 3.2.13** Suppose we wanted less sample-to-sample variation in the sample means. What would you change about the sampling process?

**Key Idea 3.2.7** When taking random samples from a large population (more than 20 times the size of the sample), the distribution of sample means will

1. Center around the population mean ($\mu$)
2. Have a standard deviation smaller than the population, $SD(\bar{X}) = \dfrac{\text{population standard deviation}}{\sqrt{\text{sample size}}} = \dfrac{\sigma}{\sqrt{n}}$
3. Be approximately normal when the sample size is large (by the **Central Limit Theorem** or if the population distribution is normally distributed)

Like with proportions, we see that the sample mean ($\bar{x}$) is unbiased for estimating the population mean, but now we have a different formula for predicting the standard deviation of the statistic.

**Exercise 3.2.14** Use your applet output to verify that the standard deviation of sample means is approximately equal to $\sigma/\sqrt{n}$

The Central Limit Theorem tells us that we can also assume that the dis-

tribution of sample means is approximately normal when the sample size is large. How large is large? We cannot check the number of successes and failures because we are working with a quantitative variable. But in a similar manner, the sampling distribution will look normal as long as the population distribution is not too skewed. As we increase the sample size, the distribution of sample means will look more and more like a normal distribution. Generally, if the sample size is at least 20 we can assume that distribution of sample means is approximately normal. We also have a new case—the distribution of sample means will behave like a normal distribution for any sample size, if the population shape is itself approximately normal.

**Exercise 3.2.15** Next you will change the sample size from 5 to 10. Predict how this will impact the shape, center, and variability of the distribution of sample means.

**Exercise 3.2.16** Generate 10,000 random samples of 10 words. Sketch the resulting distribution of sample means, remembering to carefully label the horizontal axis. Were your predictions about how the increased sample size would impact the shape, center, and variability accurate?

**Exercise 3.2.17** Using what you have seen so far, which is least likely:

1. Randomly picking one word from the Gettysburg Address and finding it be 6 letters or more.

2. Randomly picking 5 words from the Gettysburg Address and finding the mean of the 5 words to be 6 letters or more in length.

3. Randomly picking 10 words from the Gettysburg Address and finding the mean of the 10 words to be 6 letters or more in length.

Explain your reasoning.

## 3.3 Theory-based Inference for a Population Mean

### STEP 1: Ask a research question.

How much do students at your school sleep on a typical night? Let's make the question more specific and ask about *last* night. Is the average less than the recommended eight hours? How can we estimate this population average?

**Exercise 3.3.1** Based on these questions, what is the population of interest? What is the parameter? What symbol do we use for this parameter?

### STEP 2: Design a study and collect data.

Let's investigate whether the mean amount of sleep last night for the population of all students at your school was less than 8 hours.

**Exercise 3.3.2** Express the null and alternative hypotheses for investigating this question.

To test these hypotheses we need to collect data.

**Exercise 3.3.3** Ideally, how should you obtain a sample from the population of all students at your school?

**Exercise 3.3.4** What variable would you measure on each student?

Whereas obtaining a simple random sample of students from your school is an ideal way to attempt to get a representative sample, there is no question that it would take a fair bit of work to make that happen. So much so, that we might not want to do the study anymore! This happens frequently in Statistics, and so researchers often opt to use a convenience sample instead. One choice of a convenience sample of students at your school is to use your class.

**Exercise 3.3.5** Identify at least one reason that your class is likely a good representation of all students at your school in terms of hours slept last night.

**Exercise 3.3.6** Identify at least one reason that your class is likely not a good representation of all students at your school in terms of hours slept last night.

**Exercise 3.3.7** Recall that convenience sampling may be _____ (biased/unbiased) whereas simple random sampling is _____ (biased/unbiased).

We will revisit the implications of using your class as a sample of students from your college later when we evaluate the study. For now, let's gather the data.

**Exercise 3.3.8** How much sleep (to the nearest quarter hour) did you get last night? Combine your answer with those of your classmates.

**Exercise 3.3.9** Did you report a number of hours or a number of minutes slept?

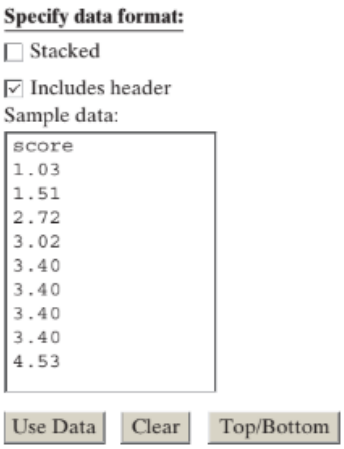
**Key Idea 3.3.1** When describing a quantitative variable, you should always be aware of the **measurement units** of the variable. Measurement units indicate information about the context and scaling of the variable (e.g., hours vs. days). Always include measurement units when describing your variable.

### STEP 3: Explore the data.

After you collect data, the next step is to explore the data. In the case of a quantitative variable (like we have here), exploring the data is a bit more involved than earlier. Recall from the Preliminaries that we can summarize

a quantitative variable using a dotplot and then describe the shape, center, variability, and unusual observations.

Use the Descriptive Statistics[1] applet to examine a dotplot of the sample data. To do this, press *Clear* to delete the existing data in the applet and then copy and paste your class data (with a one-word variable name) into the *Sample data* box and press *Use Data*.




**Exercise 3.3.10** Summarize the shape, center (mean with appropriate symbol and measurement units), and variability (standard deviation with appropriate symbol and measurement units) for the sample of sleep times for your class. Also be sure to discuss any unusual observations or outliers in the data.

**Exercise 3.3.11** Do these data provide any preliminary evidence that students at your school tend to sleep less than 8 hours a night?

## STEP 4: Draw inference beyond the data.

Let's now explore what the distribution of this sample tells us about the strength of evidence against the null hypothesis.

**Exercise 3.3.12** Suppose we had taken many, many samples of the same size from the population at your school. If the null hypothesis is true, how do you predict the distribution of sample means will behave? Be sure to discuss any information you do not know.

## Simulation-based Approach: Applying the 3S Strategy to Hypotheses about a Population Mean

To evaluate the strength of evidence that your sample data provide against the null hypothesis we need to investigate how often a sample mean as small as the one observed in your sample would occur if you were selecting a random sample from a population in which the mean sleep time really is 8 hours (null hypothesis). In essence, we will apply the 3S strategy as we have done before. A previous section showed that the sample mean should fall within two standard deviations $\frac{2\sigma}{\sqrt{n}}$ of the population mean. Let's now look at using the sample standard deviation(s) instead of $\sigma$.

**Exercise 3.3.13** Calculate and interpret the equation for the standard deviation.

[1] `static-content.zybooks.com/applets/2021/descstats/Dotplot.htm`

**Exercise 3.3.14** Based on your answer to #13, does it seem surprising to get a sample mean like the one in your class survey if the population mean is 8? Justify your answer.

**Key Idea 3.3.2** We can calculate a *standardized statistic* based on the observed sample data by computing

$$\text{standardized statistic} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

(where $\mu_0$ is a symbol used to represent the hypothesized value of the population mean).

Now let's explore whether "2" is still a reasonable cut-off for declaring this standardized statistic to be large.

## Let's Investigate

Open the One Mean[2] applet. Notice that "1" has been selected (this is the first of three hypothetical populations of sleep times pre-loaded into the applet) and a histogram displays this population distribution. For now we are **assuming** that the population distribution is normally distributed with mean very close to 8 hours and standard deviation about 1.5 hours.

Check the *Show Sampling Options* box. Set the *Number of samples* to 10000 and set the *Sample size* to match your class survey. Press *Draw Samples*.

**Exercise 3.3.15** Is the distribution of sample means centered around 8? Is the distribution approximately normal? (Why?) Is the standard deviation close to $\sigma/\sqrt{n}$? (Show calculation details.)

The histogram should behave as expected, but this is the distribution of sample means. What we are interested in now is the distribution of standardized statistics.

Above the distribution of sample means, select the radio button for *t-statistic*—the name for the standardized statistic we are using here—because if you now check the *Overlay t Distribution* box, you should see that these t-statistics are well modeled by something called the t-distribution.

The shape of a t-distribution is still mound-shaped, but it is centered at zero. It looks a lot like a normal distribution, but it is a bit more spread out than a normal distribution (more observations in the "tails," fewer in the middle). These "heavier tails" are beneficial for us here, where we have sample to sample variation in the means plus additional uncertainty by using the sample standard deviation each time rather than one population standard deviation. For this reason, the theory-based p-value will be based on the t-distribution rather than a normal distribution. But if you instead check the *Overlay Normal Distribution* box, there is not much difference in the two distributions. In general, when the sample size is large (e.g., more than 20), there is not a very large difference between the two distributions.

**Exercise 3.3.16** Use the applet, the t-statistic you calculated for your class survey, and the Count Samples box to estimate a p-value using the t-distribution (in the direction of the alternative hypothesis).

**Exercise 3.3.17** Based on this p-value do your sample data provide strong evidence that the population mean sleep time at your school last night was less than 8 hours? Justify your answer, as if to a friend who has not studied

[2] `static-content.zybooks.com/applets/2021/sampling/OneSample.html?hideExtras=1`

statistics.

### STEP 5: Formulate conclusions.

**Exercise 3.3.18** Think again about how the sample was selected from the population. Do you feel comfortable generalizing the results of your analysis to the population of all students at your school? Explain.

**Exercise 3.3.19** Looking back: Did anything about the design and conclusions of this study concern you? Issues you may want to critique include: the match between the research question and the study design, how the observational units were selected, how the measurements were recorded, and whether what we observed is of practical value.

**Exercise 3.3.20** Looking ahead: What should the researchers' next steps be to fix the limitations or build on this knowledge?

### Exploring Further

**Exercise 3.3.21** Suppose you rejected the null hypothesis in this study, does that mean you have proven that the population mean is not equal to 8 hours? If not, why not. (Hint: How does this relate to your p-value?)

**Exercise 3.3.22** Suppose we do not reject the null hypothesis that $\mu = 7$ hours. Does this mean you have proven that $\mu = 7$ hours? If not, why not?

# 3.4 Other Statistics

## Backpack weights

Carrying a heavy backpack can be a source of "chronic, low-level trauma," and can cause long-lasting shoulder, neck, and back pain. The American Academy of Orthopedic Surgeons recommends that a backpack not weigh more than 10 to 15% of the wearer's body weight. As the problem of overheavy backpacks has received more attention recently among primary and secondary school students, a student group decided to investigate the issue with students at their own university (approximately 20,000 enrolled students). Twenty-five students were sampled from each of four locations across the campus, during the course of four days, and four different times of the day (early morning, lunch time, evening, late afternoon). Scales were used to measure backpack weight to the nearest pound, and students were asked to report their body weight on a survey. One researcher was the designated talker and introduced the survey to the participants (survey questions included major, year in school, whether have back problems, and gender). Another person in the group was in charge of reading the scale, and another in charge of gathering the surveys.

## STEP 1: Ask a research question

**Exercise 3.4.1** Based on the study description, phrase a research question that you would be interested in testing with these students' data. Be sure to identify the population and parameter of interest. What symbol could you use for this parameter?

## STEP 2: Design study and collect data.

These data have already been collected, so now let us think about the students' data collection process.

**Exercise 3.4.2** Was this a random sample? Is it likely to be representative of the population you identified in Question 1?

**Exercise 3.4.3** What other steps did the students take to minimize non-sampling concerns?

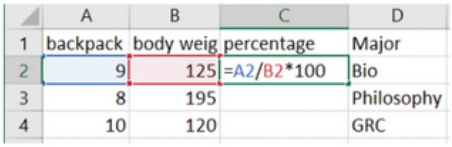
## STEP 3: Explore the data

The data in backpack.xlsx[1] contain several variables of interest, including backpack weight. But the recommendation was stated in terms of "percentage of person's weight," so we first convert this variable into the percentage of body weight for each individual in the dataset.

1. In the Excel file, create this variable: Insert a column after B. (You may have to press the Enable Editing button first.)
2. Name this column "percentage" in the square below the C.
3. In C2, type = a2/b2∗100, press Enter.
4. Now click on the cell, and then double click on the square in the lower right corner to fill down the column.

[1] `static-content2.zybooks.com/isi/data/chap2/Backpack.xlsx`

5. Copy and paste this column to your clipboard.



| | A | B | C | D |
|---|---|---|---|---|
| 1 | backpack | body weig | percentage | Major |
| 2 | 9 | 125 | =A2/B2*100 | Bio |
| 3 | 8 | 195 | | Philosophy |
| 4 | 10 | 120 | | GRC |

Once you have the column of ratios, open the One Mean applet[2]. Press *Clear* and then paste the percentages into the data window and press *Use Data.*

**Exercise 3.4.4** Describe the distribution of the percentages for this sample of 100 students. Remember to focus on shape, center, and variability, and any unusual observations or outliers.

The applet reports the mean and the standard deviation. Of course, another way to measure the "center" of the distribution is the **median**.

**Exercise 3.4.5** Predict how the median percentage will compare to the mean percentage, keeping in mind that the median will divide the dataset in half, 50 students below the median and 50 students above the median.

**Exercise 3.4.6** In the applet, select the Show Sampling Options button and then Median radio button. Was your conjecture correct? Explain to someone why the mean and median have this relationship for this dataset.

**Exercise 3.4.7** The largest percentage in this dataset is 18.1%. Suppose this had been incorrectly recorded as 181%! How will the mean change? How will the median change?

For this dataset, the difference between the mean and median is not large, but we still might consider the median a more appropriate statistic to focus on, as it would represent a typical percentage, rather than the mean which is being pulled to the right by a few large percentages. The problem is, all the theory you saw in previous sections only applies to means. So how can we get a sense for how much sample-to-sample variation there is in the sample medians?

## STEP 4: Draw inferences beyond the data.

Is it plausible that this sample came from a population with a population median of 10%?

**Exercise 3.4.8** State appropriate null and alternative hypotheses, in words, about the population median for this research question.

(Although inference for the median will follow the same logic as inference for the mean, we will not bother to define symbols for the sample or population medians.)

**Exercise 3.4.9** Outline a simulation strategy for this null hypothesis. Make sure you describe the population, parameter, sample, and statistic.

In a previous section, we created hypothetical populations from which to sample. We wanted the population to have similar characteristics as the observed sample, but also assuming the null hypothesis to be true. We could generate a population of 20,000 percentages, with a population median of 10, but what do we assume about the shape and the standard deviation of this population? Do they matter? Does the population size?

[2] `static-content.zybooks.com/applets/2021/sampling/OneSample.html?hideExtras=1`

One method for generating a hypothetical population is to make many copies of the observed sample. Before, we suggested that as long as the population is more than 20 times the size of the sample, then the actual population size does not matter.

**Exercise 3.4.10** In the applet, choose the $\times 40$ ratio button. This creates 40 copies of the observed sample and updates the "population distribution" in the first graph. How would you characterize the shape, center, and variability of this population compared to the actual sample?

**Exercise 3.4.11** In the Sample size box, use a value of 100 (the number of students actually measured), and press Draw Samples. If you scroll through the window showing the sample data, is it possible that you will see the same ID number more than once?

**Exercise 3.4.12** Now change the Number of samples to 999 (for 1,000 total). Choose Mean as the statistic and press Draw Samples. Compare the generated distribution of sample means (far right graph) to the distribution you would predict as you have in previous sections.

**Exercise 3.4.13** Now change the Statistic radio button to Median and produce a (well-labeled) representation of the distribution.

Notice that the distribution of medians does not follow a nice smooth bell-shaped curve like the distribution of means, but now we are able to see how the distribution behaves. This process, sampling from a large number of copies of the sample distribution, is equivalent to sampling from the original sample with replacement, a technique known as **bootstrapping**. The distributions you just generated are often called "bootstrap distributions."

**Definition 3.4.1 Bootstrapping** randomly samples n observations from the original sample of n observations with replacement to estimate the sample-to-sample variation in a statistic. This is equivalent to considering the "population" from which you are sampling to be a very large number of copies of the sample. ◊

**Key Idea 3.4.2** Bootstrapping allows you to estimate the sample to sample variation in any statistic.

## Null Distribution?

We cannot use the bootstrap distribution directly to estimate a p-value because it will be centered at the observed sample statistic rather than at the hypothesized value for the parameter. However, the bootstrap distribution does tell us about the sample-to-sample variation in the statistic. In fact, with quantitative data, we can simply shift the "population distribution" over to center at the hypothesized parameter value without changing the shape or variability of the distribution.

**Exercise 3.4.14** In the applet, use the Shift mean/median slider to the right so that the "population distribution" centers at (approximately) 10. Now specify the observed sample median (7.20) in the Count Samples box. Is it plausible that the observed sample of 100 percentages came from a population where the median percentage was 10? Explain your reasoning.

### STEP 5: Formula conclusions.

**Exercise 3.4.15** Write a paragraph summarizing your conclusions from this study. Be sure to discuss both what you learned from the sample and what you believe to be plausible about the population.

### STEP 6: Look back and ahead.

**Exercise 3.4.16** Return to your critique of the study design. If you were to carry out such a study on your own campus, discuss what you would do differently and why.

### Exploring Further

In the future, you can go directly to the link for Bootstrapping in the upper left corner of the applet.

**Exercise 3.4.17** Paste the original sample of 100 percentages into the sample data box and press Use Data. Check the Show Sampling Options box. Enter 10,000 for the number of samples and 100 for the sample size. Press Draw Samples. Select Median as the Statistic. How does the bootstrap distribution compare to what you generated in Exercise 3.4.13, p. 51?

# 3.5 Tools, data, and formulas

## Tools

**Step 3.5.1** In the tool samples are drawn without replacement. To carry out bootstrapping (sampling with replacement), a bootstrap distribution made of many copies of a set of sample data should be loaded into the tool as a .csv file and should then be used as the population distribution.

**Step 3.5.2** The tool bins data using the Freedman-Diaconis rule, which may produce different looking histograms when loading data from the sections.

**Table 3.5.3 Table 3101: One mean relevant features by section**

| Section | Feature |
|---|---|
| Sampling from a finite population | Draw samples from a finite population |
| Inference for a single quantitative variable | Input parameters |
| Theory-based Inference for a Population Mean | Calculate simulation-based and theory-based p-value and standa<br>with specified alternative hypothesis |
| Other statistics | Simulation-based analysis for medians<br>Specify statistic as "means" or "medians" |
| Tools, data, and formulas | Input hypothetical population by specifying shape, mean, and st<br>Input hypothetical population by uploading a .csv file |

## Data

**Table 3.5.4 References**

| Section | Data |
|---|---|
| Example: Sampling from a finite population | $\pi$=0.592, n=2181, $\hat{p}$=0.698 |
| Supplemental Exploration: Sampling from a finite population | $\pi$=0.41 |
| Example: Inference for a single quantitative variable | CollegeMidwestPopulation.csv[4] |
| Supplemental Exploration: Inference for a single quantitative variable | Gettysburg.csv[5] |
| Example: Theory-based Inference for a Population Mean | SnippetEstimate.csv[7] n=48, $\overline{x}$=13.71 |
| Supplemental Exploration: Theory-based Inference for a Population Mean | SleepTimes.csv[9] n=22, $\overline{x}$=6.705, $s$=1 |
| Example: Other statistics | SnippetEstimatePopulation.csv[11] Po |
| Supplemental Exploration: Other statistics | BackpackWeights.csv[12] n=100, $\overline{x}$=7.7 |
| Investigation: Faking cell phone calls | $\pi$=0.10, n=1858, $\hat{p}$=0.13 |

[1] www.electproject.org/2016g
[2] gss.norc.org/
[3] www.loc.gov/exhibits/gettysburg-address/
[4] static-resources.zybooks.com/ISI/Ch03/CollegeMidwestPopulation.csv
[5] static-resources.zybooks.com/ISI/Ch03/Gettysburg.csv
[6] www.loc.gov/exhibits/gettysburg-address/
[7] static-resources.zybooks.com/ISI/Ch03/SnippetEstimate.csv
[8] static-resources.zybooks.com/ISI/Ch03/SnippetEstimatePopulation.csv
[9] static-resources.zybooks.com/ISI/Ch03/SleepTimes.csv
[10] static-resources.zybooks.com/ISI/Ch03/SleepTimesPopulation.csv
[11] static-resources.zybooks.com/ISI/Ch03/SnippetEstimatePopulation.csv
[12] static-resources.zybooks.com/ISI/Ch03/BackpackWeights.csv
[13] static-resources.zybooks.com/ISI/Ch03/BackpackWeightsPopulation.csv
[14] www.causeweb.org/cause/filebrowser/download/356#page=17
[15] www.pewresearch.org/internet/dataset/may-2011-mobile/
[16] www.ibtimes.com/do-you-talk-cell-phone-avoid-interaction-others-302473

## Formulas for one mean

- $\mu$ is the population mean.
- $\sigma$ is the population standard deviation.
- $n$ is the sample size.
- $\overline{x}$ is the sample mean.
- $s$ is the sample standard deviation.
- $SD = \sigma/\sqrt{n}$ is the theory-based standard deviation of the null distribution.
- $SE = s/\sqrt{n}$ is the theory-based standard error, which approximates $SD$.
- standardized statistic $= \frac{\text{statistic} - \text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
    - $t = \frac{\overline{x}-\mu}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one mean.
    - $t = \frac{\overline{x}-\mu}{SE} = \frac{\overline{x}-\mu}{s/\sqrt{n}}$ is the theory-based standardized statistic for one mean.
- $t = \frac{\overline{x}-\mu}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one mean.
- $t = \frac{\overline{x}-\mu}{SE} = \frac{\overline{x}-\mu}{s/\sqrt{n}}$ is the theory-based standardized statistic for one mean.

# Chapter 4

# Estimation

## 4.1 Statistical inference: Confidence intervals

A worksheet for Grace College from Zybooks

### Kissing right?

Most people are right handed, and even the right eye is dominant for most people. Developmental biologists have suggested that late-stage human embryos tend to turn their heads to the right. In a study reported in Nature (2003)[1], German bio-psychologist Onur Güntürkün conjectured that this tendency to turn to the right manifests itself in other ways as well, so he studied kissing couples to see which side they leaned their heads to while kissing. He and his researchers observed kissing couples in public places such as airports, train stations, beaches, and parks. They were careful not to include couples who were holding objects such as luggage that might have affected which direction they turned. For each kissing couple observed, the researchers noted whether the couple leaned their heads to the right or to the left. They observed 124 couples, ages 13 to 70 years.

**Exercise 4.1.1** Identify the observational units in this study.

**Exercise 4.1.2** Identify the variable recorded in this study. Classify it as categorical or quantitative.

You will first use Güntürkün's data to test his conjecture that kissing couples tend to lean their heads to the right. Use the symbol $\pi$ to denote the proportion of all kissing couples in the countries where the data were collected that lean their heads to the right.

**Exercise 4.1.3** Is $\pi$ a parameter or a statistic? Explain how you are deciding.

**Exercise 4.1.4** Do we know the exact value of $\pi$ based on the observed data? Explain.

**Exercise 4.1.5** State the appropriate null and alternative hypotheses, both in words and in terms of the parameter $\pi$, for testing the conjecture that kissing couples tend to lean their heads to the right more often than by random chance.

Of the 124 couples observed, 80 leaned their heads to the right while kissing.

**Exercise 4.1.6** Calculate the sample proportion of the observed couples who leaned their heads to the right while kissing. Also indicate the symbol used to denote this value.

**Exercise 4.1.7** Conduct a simulation analysis (using the One Proportion applet[2]) to assess the strength of evidence that the sample data provide for Güntürkün's conjecture that kissing couples tend to lean their heads to the right more often than they would by random chance.

Report the approximate p-value and summarize your conclusion about this strength of evidence.

Your simulation analysis should convince you that the sample data provide very strong evidence to believe that kissing couples lean their heads to the right more than half the time in the long run. That leads to a natural follow-up question: How much more than half the time? In other words, we have strong evidence that the long-run proportion of kissing couples who lean to the right is greater than 1/2, but can we now estimate the value for that probability?

[1] doi.org/10.1038/421711a

[2] static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1

We will do this by testing many different (null) values for the probability that a couple leans to the right when kissing.

**Exercise 4.1.8** Now test whether the data provide evidence that the probability that a couple leans their heads to the right while kissing ($\pi$) is different from 0.60. Use the One Proportion applet[3] to determine the p-value for testing the null value of 0.60.

Report what you changed in the applet and report your p-value.

**Think About It 4.1.1** What kind of test are we conducting if we want to determine whether the population parameter is different from a hypothesized value: a one-sided test or a two-sided test?

Remember from previous sections that if we don't have a specific direction we are testing, greater than or less than, we need to use a two-sided test of significance. If you used a one-sided alternative in Exercise 4.1.8, p. 57, determine the two-sided p-value now by checking the *Two-sided box*.

We can specify a *level of significance* in order to decide whether the p-value is small. For example, we can say a p-value of 0.05 or less is strong evidence against the null hypothesis and in favor of the alternative hypothesis.

**Definition 4.1.2** The **significance level** is a value used as a criterion for deciding how small a p-value needs to be to provide convincing evidence to reject the null hypothesis. ◊

Thus, we can *reject* the null hypothesis when the p-value is less than or equal to 0.05. Otherwise, when the p-value is greater than 0.05, we do not have strong enough evidence against the null hypothesis and so we consider the null value to be *plausible* for the parameter.

**Key Idea 4.1.3** We will consider a value of the parameter to be plausible if the two-sided p-value for testing that parameter value is larger than the level of significance.

**Exercise 4.1.9** Is the p-value for testing the null value of 0.60 less than 0.05? Can the value 0.60 be rejected, or is the value 0.60 plausible for the probability that a couple leans their heads to the right while kissing?

The p-value you found in #9 should not have been smaller than 0.05. Hence, you do not reject the null hypothesis at the 0.05 level of significance and therefore you do not reject 0.60 as a plausible value for $\pi$. Thus, it is plausible (i.e., believable) that the probability a kissing couple leans their heads to the right is 0.60.

**Exercise 4.1.10** Does this mean that you've proven that exactly 60% of kissing couples lean right? Why or why not?

**Exercise 4.1.11** Use the applet to test the probability values given in the following table.

- Each time, change the *Probability of success* to match the value that you are testing (keeping the observed sample proportion that you count beyond the same).
- Everything else should stay the same; press *Draw Samples* and then *Count* to see the new two-sided p-value (with the *Two-sided* box checked). You can also check the *Show sliders* box and double click on the orange value to change the value of $\pi$ and press *Enter*.

[3] `static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1`

**Table 4.1.4**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Probability under $H_0$ | 0.54 | 0.55 | 0.56 | 0.57 | 0.58 | 0.59 | 0.60 |
| (Two-sided) p-value | | | | | | | |
| Reject or plausible? | | | | | | | |

**Table 4.1.5**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Probability under $H_0$ | 0.70 | 0.71 | 0.72 | 0.73 | 0.74 | 0.75 | 0.76 |
| (Two-sided) p-value | | | | | | | |
| Reject or plausible? | | | | | | | |

**Exercise 4.1.12** Using a 0.05 significance level and your results from question 11, provide a list of plausible values for $\pi$, the long-run proportion of kissing couples who lean their heads to the right.

This list of values represents an interval containing all values between two endpoints.

**Key Idea 4.1.6** The interval of plausible values that you found in question 12 is also called a **95% confidence interval for** $\pi$. Why is this called 95% confidence? This corresponds to the 5% (0.05) significance level that was used to decide whether there was enough evidence against a hypothesized value. Notice that 95% and 5% add up to give 100%. The 95% is called the **confidence level**, and is a measure of how confident we are about our interval estimate of the parameter. In this case, we are 95% confident that the population proportion of all couples who lean their heads to the right while kissing is in the interval that you found.

**Exercise 4.1.13** Does your 95% confidence interval from question 12 include the value of 0.50? Does it include the value 0.60? Explain how your answers relate to the significance tests and p-values that you calculated in Exercise 4.1.7, p. 56 and Exercise 4.1.8, p. 57.

**Exercise 4.1.14** Now suppose we were to use a significance level of 0.01 instead of 0.05 to decide whether or not to reject the corresponding null hypothesis for each listed value of $\pi$. How would you expect the interval of plausible values to change: wider, narrower, or no change? Explain your reasoning.

**Exercise 4.1.15** Implement the 0.01 significance level to determine plausible values for the population proportion of all kissing couples that lean their heads to the right. (Hint: Start with the null values listed in the table in question 11, although you might have to test more null values as well. You can also change the success probability on the slider box by clicking on the bar and then using the arrow keys to move the slider value left and right.) Report the interval of plausible values.

**Exercise 4.1.16** How confident are you about the interval of plausible values that you listed in question 15?

**Exercise 4.1.17** What is the primary difference between the 95% confidence interval reported in question 12 and the 99% confidence interval reported in question 15? Which interval is wider? Is this consistent with what you predicted in question 14?

**Exercise 4.1.18** How would you expect a 90% confidence interval to compare to the 95% and 99% confidence intervals? Explain.

**Key Idea 4.1.7** As the confidence level increases, the *width* of the confidence interval also increases.

You should have found very strong evidence, based on a small p-value, that a majority of kissing couples in the population lean their heads to the right. Moreover, your confidence intervals, which provide plausible values of the population proportion that lean to the right, contain only values above 0.50, confirming the conclusion made from the p-value.

In fact, confidence intervals and tests of significance generally give complementary information. Values that are not considered plausible by the (two-sided) test of significance should not be contained in the corresponding confidence interval (confidence level = 100% - level of significance). Because 0.50 is not inside the confidence interval, this matches the test. It is not "further evidence against the null" but confirms that your calculations agree. The confidence interval then gives us additional information in how large this majority is.

## 4.2 2SD and theory-based confidence intervals for a single proportion

### How mobile are we?

Did you move around a lot when you were younger? Suppose you wanted to estimate the proportion of students at your school that currently live in a different state from where they were born. How would you come up with an estimate? Do you think the students in your class are representative of students at your school on this issue? How would you estimate the proportion of adult Americans who currently live in a different state from where they were born?

The General Social Survey[1] (GSS) is a survey administered every two years to a random sample of adult Americans aged 18 and older. The survey asks a battery of questions focused on different social aspects of the population. In 2018, one of the questions was whether or not the adult lived in a different state from where they were born. Of the 2,348 surveyed that year, 36.2% replied that they do currently live in a different state from where they were born.

**Exercise 4.2.1** Identify the population and sample in this survey.

**Exercise 4.2.2** Is it reasonable to believe that the sample of 2,348 adult Americans is representative of the larger population? Explain why or why not.

**Exercise 4.2.3** Explain why 36.2% is a statistic and not a parameter. What symbol would you use to represent it?

**Exercise 4.2.4** Identify (in words) the parameter that the Gallup organization was interested in estimating.

**Exercise 4.2.5** Is it reasonable to conclude that exactly 36.2% of all adult Americans currently live in a different state from where they were born? Explain why or why not.

**Exercise 4.2.6** Although we expect $\pi$ to be close to 0.362, we realize there may be other plausible values for the population proportion as well. First consider the value of 0.382. Is this a plausible value for $\pi$?

Use the One Proportion applet[2] to simulate random samples of 2,348 people from such a population. (Hint: Keep in mind that 0.382 is what we are assuming for the population proportion and 0.362 is the observed sample proportion.)

What do you estimate for the two-sided p-value? Would you reject or fail to reject the null hypothesis at the 5% level of significance?

**Exercise 4.2.7** Also check the Summary Stats box and report the mean and standard deviation of this null distribution.

**Exercise 4.2.8** Now consider 0.50. Is this a plausible value for $\pi$? Repeat #6 and record the mean and standard deviation for this null distribution as well.

Clearly 0.50 is going to be "too far" from $\hat{p} = 0.362$ to be plausible. But how far is too far? We could use the plausible values method to produce a confidence interval for the proportion of all adult Americans who currently live in a different state from where they were born. But that approach is somewhat cumbersome and time-intensive, so we'll now learn some shortcut approaches.

[1] gss.norc.org/

[2] static-content.zybooks.com/applets/2021/oneprop/OneProp.htm?hideExtras=1

**Exercise 4.2.9** Reconsider our first guess of $\pi = 0.382$. How many standard deviations is 0.362 from 0.382? (Hint: Standardize the value by looking at the difference between 0.382 and 0.362 and divide by the standard deviation you found in Exercise 4.2.7, p. 60.)

You should notice that 0.382 and 0.362 are about 2 standard deviations apart and that the two-sided p-value is around 0.05, so this value (0.382) is close to the edge of values that can be considered plausible for $\pi$. Values between 0.362 and 0.382 are considered plausible and values larger than 0.382, or more than 2 standard deviations above 0.362, will not be plausible values for the population proportion.

**Key Idea 4.2.1** When a sampling distribution is bell-shaped, as your null distribution should be for this study, approximately 95% of the statistics in the sampling distribution will fall within 2 standard deviations of the mean. This implies that 95% of sample proportions will fall within 2 standard deviations of the parameter ($\pi$), which means that $\pi$ is within 2 standard deviations of the observed sample proportion for 95% of all samples.

We can then extend this idea to construct a 95% confidence interval.

**Key Idea 4.2.2** We can construct a 95% confidence interval of plausible values for a parameter by including all values that fall within 2 standard deviations of the sample statistic. This method is only valid when the sampling distribution follows a bell-shaped, symmetric distribution. We call this the **2SD method**. Thus we can present the 95% confidence interval for the long-run proportion (or population proportion), $\pi$, in symbols as

$$\hat{p} \pm 2 \times SD(\hat{p})$$

where $\hat{p}$ is the sample proportion and $SD(\hat{p})$ is the standard deviation of the sampling distribution of sample proportions. The value of $2 \times SD(\hat{p})$, which represents half the width of the confidence interval for 95% confidence, is called the **margin of error**.

**Think About It 4.2.3** So how do we find the standard deviation to use for the 2SD method?

**Exercise 4.2.10** How did the standard deviations you found in Exercise 4.2.7, p. 60 (with $\pi = 0.382$) and in Exercise 4.2.8, p. 60 (with $\pi = 0.50$) compare?

You should see that the standard deviation changes slightly when we change $\pi$, but not by much. We saw before that the variability in the sample proportions is in fact largest when $\pi = 0.50$. So one approach would be to carry out one simulation (with lots of trials) using $\pi = 0.50$ and use that value of the standard deviation to calculate the margin of error.

**Exercise 4.2.11 Question 11.** Determine a 95% confidence interval using the 2SD method:

**(a)** First calculate 2 × (standard deviation for your sampling distribution of sample proportions) using 0.5 in the simulation to estimate the SD. (This is the margin of error.)

**(b)** Use this SD to produce a 95% confidence interval for $\pi$. (Hint: Subtract the margin of error from $\hat{p}$ to determine the lower endpoint of the interval and then add the margin of error to $\hat{p}$ to determine the upper endpoint of the interval.)

**(c)** Interpret the confidence interval: You are 95% confident that what is between what two values?

One limitation to this method is that it only applies for 95% confidence. What if we wanted to be 90% or 99% confident instead? We can extend this 2SD method to a more general theory-based approach.

## Theory-based approach

As we saw previously, we don't always need to simulate a sampling distribution—not if we can accurately predict what would happen if we were to simulate. Instead, we can predict the standard deviation by the formula $\sqrt{\pi(1-\pi)/n}$. But when constructing a confidence interval, we don't have a hypothesized value of $\pi$, so to estimate this standard deviation, we will substitute the observed sample proportion.

**Definition 4.2.4** An estimate of the standard deviation of a statistic, based on sample data, is called the **standard error** (SE) of the statistic. In this case $\sqrt{\hat{p}(1-\hat{p})/n}$ is the standard error of a sample proportion $\hat{p}$. ♢

**Exercise 4.2.12** Calculate the standard error for this study. How does it compare to the standard deviations you found in Exercise 4.2.7, p. 60 and Exercise 4.2.8, p. 60?

So a more general formula for using the 2SD method to estimate a population proportion would be

$$\text{CI} = \hat{p} \pm 2 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

But then how do we change the confidence level? The 2SD method was justified by saying 95% of samples yield a sample proportion within 2 standard deviations of the population proportion. If we want to be more confident that the parameter is within our margin of error, we can create a larger margin of error by increasing the multiplier. In fact a multiplier of 2.576 gives us a 99% confidence level, whereas a multiplier of 1.645 gives us only 90% confidence.

**Exercise 4.2.13** In the Theory-Based Inference applet[3], specify the sample size (n) of 2,348 and the sample proportion of 0.362 and press **Calculate**. (The applet will fill in the count or specify the count of 850 and the applet will fill in the sample proportion.)

Check the box for Confidence interval, confirm the confidence level is 95% and press Calculate CI to generate a theory-based confidence interval. Report the 95% theory-based confidence interval.

**Exercise 4.2.14** Is this theory-based confidence interval similar to the one you obtained using the 2SD method?

**Validity Conditions 4.2.5** The theory-based approach for finding a confidence interval for $\pi$ (called a **one-sample z-interval**) is considered valid if there are at least 10 observational units ineach category of the categorical variable (i.e., at least 10 successes and at least 10 failures).

Because we have a large sample size here, the theory-based approach should produce very similar results to the plausible values method and the 2SD method. In such a case, the theory-based approach is often the most convenient, especially if our confidence level is not 95%.

**Exercise 4.2.15** Change the confidence level in the applet from 95% to 99% and press the Calculate CI button again. Report the 99% confidence interval given by the applet. How does it compare to the 95% interval? (Compare both

[3] `static-content.zybooks.com/applets/2021/tbia/TBIA.html?hideExtras=1`

the midpoint of the interval = (lower endpoint + upper endpoint)/2 and the margin of error = (upper endpoint − lower endpoint)/2.)

## Exploring further

**Exercise 4.2.16** Let's look back at your class data. Determine a theory-based confidence interval for the proportion of students at your school who currently live in a state different from where they were born. Do you think this interval is valid?

**Exercise 4.2.17** How does your interval compare to the interval you found from the GSS survey?

**Exercise 4.2.18** Based on characteristics of your sample and your school, does it make sense that the two intervals compare as you described in #17? Explain.

## 4.3 2SD and theory-based confidence intervals for a single mean

### Sleepless Nights (cont.)?

Recall that in a previous explanation you investigated whether, on average, students at your school get less than 8 hours of sleep on a typical night. Even though 8 hours is often cited as the recommended amount, many people believe that college students will average less than 8 hours. But how much less? In this exploration, you will use the same data to estimate the average number of hours of sleep students at your school get.

**Exercise 4.3.1** Let $\mu$ represent the average number of hours of sleep all the students at your school got the previous night. The question we asked in a previous exploration was, "How much sleep (to the nearest quarter hour) did you get last night?" If you still have the data from that exploration, give the summary statistics here. If not, collect a new set of data from your class and give those summary statistics here.

**(a)** Sample size, $n = ?$

**(b)** Sample mean, $\bar{x} = ?$

**(c)** Sample SD, $s =?$

Similar to what you did in a previous exploration, we could conduct a simulation to generate a sampling distribution of the sample mean number of hours of sleep, assuming different values for $\mu$ to determine which are plausible. (In the One Mean applet[1], you can use a slider below the population to quickly change the value of the population mean.) Recall that such an interval is called a confidence interval. But this would be rather tedious, so in this exploration we will consider some shortcuts to estimating a confidence interval for $\mu$

**Exercise 4.3.2** Based on your sample data, what is your best guess or estimate for the value of $\mu$, that is, the average number of hours of sleep students at your school got the previous night?

**Exercise 4.3.3** Suppose you repeated the survey by asking a different group of students at your school. Will the average for this new sample be exactly the same as the original sample average reported in question 2? Why or why not?

A natural guess or estimate for the population average number of hours of sleep students at your school got the previous night ($\mu$) is the observed value of the sample average sleep time. But, we know that if we were to conduct this study again we would most likely observe a different value for the sample average due to sampling variability—we need to account for this in our estimate for $\mu$. To do so, we create a confidence interval for $\mu$ by using the formula

$$\text{sample mean} \pm \text{multiplier} \times (\text{SD of sample mean})$$

To find a 95% confidence interval for $\mu$ we can use the 2SD method that you saw previously, but we need to know what to use for the SD of the sample mean.

**Exercise 4.3.4** In a previous exploration you simulated the distribution of sample means from the hypothetical population of sleep times (population 1). What did you find for the standard deviation of the sampling distribution?

[1] static-content.zybooks.com/applets/2021/sampling/OneSample.html?hideExtras=1

How did this value compare to the standard deviation of your sample?

**Key Idea 4.3.1** There will be less sample-to-sample variability in the sample mean than person-to-person variability in sleep times.

**Exercise 4.3.5** Previously, we saw that $s/\sqrt{n}$ could be used to approximate the standard deviation of the sample means. We often refer to such an estimate of the standard deviation of a statistic as the standard error. Report this value.

**Key Idea 4.3.2** An approximate 2SD method for a 95% confidence interval for a population mean $\mu$ is $\overline{x} \pm 2 \times s/\sqrt{n}$ . This method is valid when the sampling distribution follows a bell-shaped, symmetric distribution.

**Exercise 4.3.6** Use the value of s/√n you found in #5 and the 2SD method to approximate a 95% confidence interval for $\mu$ in the context of the sleep study, and then interpret this interval. (Hint: When interpreting the interval, be clear about what the interval estimates. You are estimating not the individual sleep times of students but how much sleep students at your school get on …)

**Exercise 4.3.7** Does the 95% confidence interval reported in #6 contain 8 (the population mean under the original null hypothesis)? Explain how you could have known this in advance based on your p-value from the previous sleep times exploration.

## Theory-based approach

Previously, you saw a theory-based approach to find the p-value when testing hypotheses about a single population mean. This approach uses a more precise value of the multiplier in the confidence interval formula based on the t-distribution.

**Exercise 4.3.8** Go to the Theory-Based Inference applet[2], and select the *One Mean* option from the *Scenario* pull-down menu. Then, you can either copy and paste your class data or just enter the sample statistics (as recorded in #1). Press *Calculate*, check the *Confidence Interval* box, and press *Calculate CI*.

Report the 95% confidence interval and compare it to the interval you found in #6. Are they about the same? Same midpoint? Same width? Could you have anticipated this? Explain your reasoning.

Recall that the theory-based approach uses a theoretical probability distribution called the t-distribution to predict the behavior of the standardized statistic (denoted by t) and that this approach required certain validity conditions to be met.

**Validity Conditions 4.3.3** The theory-based interval for a population mean (called a **one-sample *t*-interval**) requires that the quantitative variable should have a symmetric distribution or you should have at least 20 observations and the sample distribution should not be strongly skewed.

**Exercise 4.3.9** Are the validity conditions for using the 2SD or t-distribution based method to find a confidence interval met for your class data? Explain why or why not.

In summary, the formula for a theory-based confidence interval for a population mean ($\mu$) is given by

$$\overline{x} \pm (\text{multiplier}) \times \frac{s}{\sqrt{n}}$$

[2] static-content.zybooks.com/applets/2021/tbia/TBIA.html?hideExtras=1

## 4.4 Factors that affect the width of a confidence interval

### Holiday spending habits

An October 1-10, 2019, An October 1-10 2019 Gallup poll[1] asked 1,526 U.S. adults how much they planned to personally spend on Christmas gifts this year. The report cited an average of $942.

**Exercise 4.4.1** Identify the observational units and variable in this study. Classify the variable as quantitative or categorical.

**Exercise 4.4.2** Define the parameter of interest in this study. (Hint: Do you want to estimate a mean or a proportion? For what population?)

Because our variable of interest (dollar amount intending to spend) is quantitative, the parameter of interest here is the population mean for all adult Americans; call this $\mu$. To construct a confidence interval for $\mu$, we need to know the sample size, the sample mean, and the sample standard deviation.

**Exercise 4.4.3** Which two of these three quantities do you know? What symbols can be used to represent them?

**Exercise 4.4.4** What we are not told by Gallup is s, the sample standard deviation. Explain in your own words what s represents in this context.

**Exercise 4.4.5** Suppose s was equal to $150. Use the 2SD method (Hint: Reminder to consider the SD of the sample mean) to approximate and interpret a 95% confidence interval for $\mu$. (Hint: You are 95% confident that what is between what two values?)

**Exercise 4.4.6** Suppose s was equal to $300. Use the 2SD method to approximate the 95% confidence interval for $\mu$. How does this interval compare to the interval from #5 (examine both the midpoint and the width of the interval)? Explain why the changes you observe make intuitive sense.

### The impact of sample size

**Exercise 4.4.7** The poll involved 751 men and 775 women. Suppose we looked at only one of these subgroups. How do you expect the width of the intervals to compare to the interval based on the total sample (wider, narrower, or the same width as before)? Explain briefly.

**Exercise 4.4.8** Suppose the women reported planning to spend an average of $942 with standard deviation $150. Compare the margin of error of this interval (Hint: 2SD) to that in Question 5. Was your prediction correct?

**Exercise 4.4.9** In general, how does the width of a confidence interval change as the sample size increases? Why?

You should have seen that a smaller sample size produced a wider confidence interval. This makes sense because a smaller sample size conveys less information and therefore leaves more uncertainty about the actual value of the population mean. More directly, as we discussed before, increasing the sample size reduces the variability in the sampling distribution; thus the SD of the sampling distribution will be smaller, and so the confidence interval will also be narrower.

[1] news.gallup.com/poll/267914/americans-plan-spend-generously-christmas.aspx

You also saw previously that the confidence level will impact the width of the interval as well. So with quantitative data, there are three main factors that impact the width of the confidence interval, the sample standard deviation (variability within the sample), the sample size (variability between samples), and the confidence level.

**Key Idea 4.4.1** The width of a confidence interval for a population mean, $\mu$, decreases

- as the confidence level decreases (for a fixed sample size and fixed sample standard deviation), or
- as the sample size increases (for a fixed confidence level and fixed sample standard deviation), or
- as the sample standard deviation decreases (for a fixed confidence level and fixed sample size).

If you wanted to answer the question "Do U.S. adults spend significantly less than $950 on average for Christmas gifts?," a confidence interval could answer that question. If we look at the confidence interval that was constructed assuming the standard deviation of spending to be $150, we get a confidence interval entirely below $ 950. So, yes, on average U.S. adults are spending significantly less than $950 on Christmas gifts. While this may be a statistically significant finding giving us strong evidence against the average being $950, it doesn't mean that the average amount is far less than $950. The upper end of the confidence interval is $949.53, only 47 cents below $950. The margin of error of the interval is about $7 arguably less than would be spent on one additional gift.

**Key Idea 4.4.2** Especially with large sample sizes, even a fairly small difference between an observed and hypothesized value can be statistically significant. In such cases a confidence interval for the parameter can help to decide whether the difference is large.

The Gallup poll did give us a bit more information;

**Table 4.4.3**

| | $1000 or more | $500-999 | $250-499 | $100-249 | Under $100 | No opinion | Median | Mean (w |
|---|---|---|---|---|---|---|---|---|
| 2019Oct 1-10 | 37 | 21 | 12 | 15 | 3 | 10 | $500 | $942 |

**Exercise 4.4.10** Based on this information, which appears to be a more reasonable guess ($150 or $300) for the sample standard deviation? Explain your reasoning.

**Exercise 4.4.11** Based on this information, do you think the theory-based inference method is valid for these data? (Hint: Do you think the sample data are roughly symmetric? What is the sample size?)

The information provided in the table is enough to tell us that the sample distribution is skewed. We can consider the 2SD and theory-based approaches valid as long as the sample size is large. Our original sample size of 1,526 respondents should be sufficient to consider these methods valid. But with such a strong skewness in spending amounts, you may want to be cautious or consider first transforming your data.

We have calculated 90%, 95%, and 99% confidence intervals, and we have seen that using a higher confidence level produces a larger margin of error and therefore a wider confidence interval. But what does it mean to say that we have "95% confidence" that an interval contains the parameter of interest? This next exploration leads you to investigate this question.

# 4.5 Factors that affect the width of a confidence interval (continued)

## Reese's Pieces

Reese's Pieces candies come in three colors: orange, yellow, and brown. Suppose that you take a random sample of candies and want to estimate the long-run proportion of candies that is orange. Let's assume for now (although we would not know this when conducting the study) that this long-run proportion, symbolized by $\pi$, is equal to 0.50.

**Exercise 4.5.1** Suppose that you take a random sample of 100 Reese's Pieces candies and find the sample proportion of orange. Is there any guarantee that the sample proportion will equal 0.50?

**Exercise 4.5.2** Suppose we calculate a confidence interval from this sample proportion. Is there any guarantee that the interval will contain the value 0.50?

**Exercise 4.5.3** Suppose that you select another random sample of 100 Reese's Pieces candies. Is there any guarantee that the sample proportion will be the same as for the first sample? Will the confidence interval based on the new sample necessarily be the same as the confidence interval based on the first sample? How do you think they will differ?

**Exercise 4.5.4** To explore the behavior of confidence intervals arising from different random samples with a sample size of 100 from the same population with 50% orange, open the Simulating Confidence Intervals applet[1].

- Make sure that *Proportions* and *Wald* are selected from the pull-down menus.
- Set $\pi$ to be 0.50 and $n$ to be 100.
- Let *Intervals* be set to 1.
- Make sure that *Conf level* is set to 95%.
- Press *Sample*.
- Notice that this adds an interval (depicted by a horizontal line) to the graph in the middle with a dot corresponding to the observed sample proportion value to the graph labeled "Sample Statistics." Click on this interval. Doing this reveals the value of the midpoint ($\hat{p}$) (at the bottom) and the endpoints of the 95% confidence interval based on the sample proportion.

Record the following:

**(a)** Sample proportion.

**(b)** Endpoints of the 95% confidence interval.

**(c)** Is the sample proportion within the confidence interval? Explain why this does not surprise you.

**(d)** Is the specified long-run proportion ($\pi = 0.50$) within the confidence interval?

[1] static-content.zybooks.com/applets/2021/confsim/ConfSim.html?hideExtras=1

Notice that 0.50 is either in the interval reported in Question 4b or not. Suppose the interval turned out to be (0.473, 0.537). It's not like 0.50 is sometimes between 0.473 and 0.537 and sometimes not between those two values. Similarly, even if we didn't know the value of $\pi$, the long-run proportion is still some value, it doesn't change if you take a different sample. Therefore, it's technically incorrect to make a statement like "there is a 95% probability that $\pi$ is between 0.473 and 0.537." This doesn't fit our "long-run proportion" interpretation of probability. So what is happening 95% of the time?

**Exercise 4.5.5** Return to the Simulating Confidence Intervals applet[2].

- Change *Intervals* from 1 to 99 for a total of 100 random samples from a process where $\pi = 0.50$.
- Press *Sample*.
- Notice that this produces a total of 100 possible values of a sample proportion which are graphed in the graph labeled "Sample Proportions" and the 100 confidence intervals (with 95% confidence) that are based on these sample proportions.

Study the graph labeled "Sample Proportions" closely. At about what number is this graph of sample proportions centered? Were you expecting this? Why or why not?Notice that some of the dots on the "Sample Proportions" graph are colored red and some are colored green.

- Click on any one of the red dots. Doing this will reveal the value of the sample propor- tion and the corresponding 95% confidence interval.

Record the following:Sample proportion.Endpoints of the 95% confidence interval.Is the sample proportion within the confidence interval?Is the probability you set for the overall process ($\pi = 0.50$) within the confidence interval?

- Now click on any one of the green dots. Doing this will reveal the value of the sample proportion and the corresponding 95% confidence interval.

Record the following:Sample proportion.Endpoints of the 95% confidence interval.Is the sample proportion within the confidence interval?Is the long-run proportion of orange ($\pi = 0.50$) within the confidence interval?

- The graph in the middle displays the 100 different 95% confidence intervals, depicting in green the ones that succeed in capturing the actual value of the long-run proportion (which, you'll recall, we assumed to be $\pi = 0.50$) and in red the intervals that fail to capture the actual value of $\pi$.

In the bottom left corner, the applet maintains a "Running total" for the percentage of the intervals produced so far that contain $\pi = 0.50$. Report this percentage here. That is, what percentage of the 100 random intervals you have generated succeeded in capturing the value of 0.50 inside the interval?Now press the Sample button a few times and watch how, if at all, the percentage reported under "Running total" changes. Now record what you think the percentage would be if you were to repeat this process forever.Based on your observations from the earlier parts, fill in the blanks: Thus, 95% confidence means that if we repeatedly sampled from a process and used the sample statistic to construct a 95% confidence interval, in the long run, roughly _____ % of all those intervals would manage to capture the actual value of the long-run proportion, and the remaining _____ % would not.

**Exercise 4.5.6** Now consider changing the confidence level to 90%.

Before you make this change, first predict what will happen to the resulting confidence intervals:How will the widths of the confidence intervals change (if at all)?How will the breakdown of successful/unsuccessful (green/red) intervals change? (What percentage of intervals do you suspect will be green in the long run?)Now change *Conf level* to 90% and (watch what happens as you)

[2] static-content.zybooks.com/applets/2021/confsim/ConfSim.html?hideExtras=1

press *Recalculate* and then *Sample*.How do the widths of the intervals change? Why does this make sense?How does the running total change? Why does this make sense?Now consider making the sample size four times as large, that is, 400 candies, while keeping the confidence level at 90%. Before you make this change, first predict what will happen to the resulting confidence intervals:How will the widths of the confidence intervals change (if at all)?How will the percentage of successful (green) intervals change?Now change $n$ to 400 and watch what happens as you press *Sample*.How do the widths of the intervals change? Why does this make sense?How does the running total change? Why does this make sense?To check your understanding about what 99% confidence means, suppose that you calculate a 99% confidence interval for the long-run proportion of orange to be (0.461, 0.589). For each of the following statements, indicate whether or not it is valid. If you think it is invalid, explain why.There is a 99% chance that the long-run proportion of Reese's Pieces candy that is orange is between 0.461 and 0.589.We are 99% confident that the long-run proportion of Reese's Pieces candy that is orange is between 0.514 and 0.566.If we were to repeat the process of taking random samples of 100 Reese's Pieces and then calculate a 99% confidence interval from the sample data for each of those samples, then in the long run, 99% of all those confidence intervals would contain the long-run proportion of Reese's Pieces candy that is orange.

**Key Idea 4.5.1** The **confidence level** indicates the long-run percentage of confidence intervals that would succeed in capturing the (unknown) value of the parameter if random samples were to be taken repeatedly from the population/process and a confidence interval produced from each sample. So, a 95% confidence level means that 95% of all samples would produce an interval that succeeded in capturing the unknown value of the parameter.

## 4.6 Tools, data, and formulas

### Tools

The standalone tools below are the previously introduced simulation zyTools for one proportion (2.12[1]) and one mean (3.10[2]) with expanded capabilities for computing confidence intervals. Additionally, a tool for simulating theory-based confidence intervals from population parameters is included as well.

**Table 4.6.1 One proportion simulator relevant features by section**

| Section | Feature |
|---|---|
| 2SD and theory-based confidence intervals for a single proportion | Calculate a 2SD 95% confidence i |
| Factors that affect the width of a confidence interval | Calculate a theory-based confider |

**Table 4.6.2 Table 4112: One mean simulator relevant features by section**

| Section | Feature |
|---|---|
| 2SD and theory-based confidence intervals for a single mean | Calculate a 2SD 95% confidence interv |
| Tools, data, and formulas | Calculate a theory-based confidence in |

**Table 4.6.3 Confidence intervals simulator relevant features by section**

| Section | Feature |
|---|---|
| Factors that affect the width of a confidence interval | Simulation of multiple confidence intervals for |
| Tools, data, and formulas | Simulation of multiple confidence intervals for |

### Data

**Table 4.6.4 Table 4114: References**

| Section | Data |
|---|---|
| Example: Statistical inference - confidence intervals | $\pi$=0.2 |
| Supplemental Exploration: Statistical inference - confidence intervals | $\pi$=0.5 |
| Example: 2SD and theory-based confidence intervals for a single proportion | $\pi$=0.5 |
| Supplemental Exploration: 2SD and theory-based confidence intervals for a single proportion | $\pi$=0.5 |
| Example: 2SD and theory-based confidence intervals for a single mean | Used( |
| Supplemental Exploration: 2SD and theory-based confidence intervals for a single mean | Sleep' |
| Example: Factors that affect the width of a confidence interval | n=41( |
| Supplemental Exploration A: Factors that affect the width of a confidence interval | n=15 |
| Supplemental Exploration B: Factors that affect the width of a confidence interval | $\pi$=0.5 |
| Investigation: Cell phones while driving | $\pi$=0.5 |

[1] learn.zybooks.com/zybook/GRACEMAT2000JohnsonFall2023/a9ead249-be3a-1cea-2e87-7bb963727c38

[2] learn.zybooks.com/zybook/GRACEMAT2000JohnsonFall2023/bc553747-b533-fd11-aeca-e2816608da2d

## Formulas for one proportion

- $\pi$ is the underlying process probability of success or population proportion of success.
- $n$ is the sample size.
- $\hat{p} = \frac{\text{number of successes in the sample}}{n}$ is the sample proportion.
- $SD(\hat{p})$ is the standard deviation of the statistic.
- $SD = \sqrt{\pi(1-\pi)/n}$ is the theory-based standard deviation of the null distribution.
- $SE = \sqrt{\hat{p}(1-\hat{p})/n}$ is the theory-based standard error, which approximates $SD$ and $SD(\hat{p})$.
- standardized statistic $= \frac{\text{statistic}-\text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
    - $z = \frac{\hat{p}-\pi}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one proportion.
    - $z = \frac{\hat{p}-\pi}{SD} = \frac{\hat{p}-\pi}{\sqrt{\pi(1-\pi)/n}}$ is the theory-based standardized statistic for one proportion.
- $z = \frac{\hat{p}-\pi}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one proportion.
- $z = \frac{\hat{p}-\pi}{SD} = \frac{\hat{p}-\pi}{\sqrt{\pi(1-\pi)/n}}$ is the theory-based standardized statistic for one proportion.
- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.
    - $\hat{p} \pm 2 \times SD(\hat{p})$ is a 2SD CI approximating a 95% confidence interval estimating $\pi$.
    - $\hat{p} \pm (\text{multiplier}) \times SE$ is a theory-based confidence interval estimating $\pi$, where the multiplier comes from the standard normal distribution.
- $\hat{p} \pm 2 \times SD(\hat{p})$ is a 2SD CI approximating a 95% confidence interval estimating $\pi$.
- $\hat{p} \pm$ (multiplier) $\times SE$ is a theory-based confidence interval estimating $\pi$, where the multiplier comes from the standard normal distribution.

[3] doi.org/10.1136/gut.2010.218305
[4] doi.org/10.1038/421711a
[5] www.pewresearch.org/internet/2018/09/27/a-majority-of-teens-have-experienced-some-form-of-cyberbullying/
[6] gss.norc.org/
[7] static-resources.zybooks.com/ISI/Ch04/UsedCarsSample.csv
[8] static-resources.zybooks.com/ISI/Ch03/SleepTimes.csv
[9] static-resources.zybooks.com/ISI/Ch04/SleepTimesPopulationBootstrap.csv
[10] www.avma.org/resources-tools/reports-statistics/us-pet-ownership-statistics
[11] news.gallup.com/poll/267914/americans-plan-spend-generously-christmas.aspx
[12] www.pewresearch.org/internet/wp-content/uploads/sites/9/2009/07/PI_2009.07.22_wireless-internet-use_FINAL.pdf

## Formulas for one mean

- $\mu$ is the population mean.
- $\sigma$ is the population standard deviation.
- $n$ is the sample size.
- $\overline{x}$ is the sample mean.
- $s$ is the sample standard deviation.
- $SD(\overline{x})$ is the standard deviation of the statistic.
- $SD = \sigma/\sqrt{n}$ is the theory-based standard deviation of the null distribution.
- $SE = s/\sqrt{n}$ is the theory-based standard error, which approximates $SD$ and $SD(\overline{x})$.
- standardized statistic $= \frac{\text{statistic}-\text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
    - $t = \frac{\overline{x}-\mu}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one mean.
    - $t = \frac{\overline{x}-\mu}{SE} = \frac{\overline{x}-\mu}{s/\sqrt{n}}$ is the theory-based standardized statistic for one mean.
- $t = \frac{\overline{x}-\mu}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for one mean.
- $t = \frac{\overline{x}-\mu}{SE} = \frac{\overline{x}-\mu}{s/\sqrt{n}}$ is the theory-based standardized statistic for one mean.
- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.
    - $\overline{x} \pm 2 \times SD(\overline{x})$ is a 2SD CI approximating a 95% confidence interval estimating $\mu$.
    - $\overline{x} \pm (\text{multiplier}) \times SE$ is a theory-based confidence interval estimating $\mu$, where the multiplier comes from the t-distribution with $n-1$ degrees of freedom.
- $\overline{x} \pm 2 \times SD(\overline{x})$ is a 2SD CI approximating a 95% confidence interval estimating $\mu$.
- $\overline{x} \pm (\text{multiplier}) \times SE$ is a theory-based confidence interval estimating $\mu$, where the multiplier comes from the t-distribution with $n-1$ degrees of freedom.

# Chapter 5

# Causation

## 5.1 Association and confounding

### Home court disadvantage?

Sports teams prefer to play in front of their own fans rather than at the opposing team's site. Having a sell-out crowd should provide even more excitement and lead to an even better performance, right? Well, consider the Oklahoma City Thunder, a National Basketball Association (NBA) team, in its second season (2008–2009) after moving from Seattle. This team had a win-loss record that was actually worse for home games with a sell-out crowd (3 wins and 15 losses) than for home games without a sell-out crowd (12 wins and 11 losses). (These data were noted in the April 20, 2009, issue of Sports Illustrated[1] in the Go Figure column.)

**Exercise 5.1.1** Identify the observational units and variables in this study. Also classify each variable as categorical (also binary?) or quantitative.

**Exercise 5.1.2** When did the Thunder have a higher winning percentage: in front of a sell-out crowd or a smaller crowd? Support your answer by calculating the proportion of sell-out games that they won and also the proportion of non-sell-out games that they won. (Write both these proportions as decimals.)

**Definition 5.1.1** Two variables are **associated** or related if the value of one variable gives you information about the value of the other variable. When comparing two groups this means that the proportions or means take different values in the two groups. ◊

**Exercise 5.1.3** Do the two variables appear to be associated?

Often, when a study involves two associated variables, it is natural to consider one the **explanatory variable** and the other the **response variable**.

**Definition 5.1.2** The **explanatory variable** is the variable we think is "explaining" the change in the response variable and the **response variable** is the variable we think is being impacted or changed by the explanatory variable. The explanatory variable is sometimes called the independent variable and the response variable is sometimes called the dependent variable. ◊

**Exercise 5.1.4** Which would you consider the explanatory variable in this study? Which is the response? (That is, what are the roles of these variables in this study?)

There are two possible explanations for this odd finding that the team had a better winning percentage with smaller crowds:

- The sell-out crowd *caused* the Thunder to play worse, perhaps because of pressure or nervousness.
- The sell-out crowd did *not* cause a worse performance, and some other issue (variable) explains why they had a worse winning percentage with sell-out crowds. In other words, a third variable is at play, which is related to both the crowd size and the game outcome.

(Of course, another explanation is random chance. Using methods you will learn later, we've determined that you can essentially rule out random chance in this case.)

**Exercise 5.1.5** Consider the second explanation. Suggest a plausible alternative variable that would explain why the team would be less likely to win in

[1] vault.si.com/vault/2009/04/20/1004202#%26gid=ci0258c0a3800826ef%26pid=1004202---021---image

front of a sell-out crowd than in front of a smaller crowd. (Make sure it's clear not just that your explanation would affect the team's likelihood of winning but also that the team would be less likely to win in front of a sell-out crowd compared to a smaller crowd.)

**Definition 5.1.3** A **confounding variable** is a variable that is related both to the explanatory and to the response variable in such a way that its effects on the response variable cannot be separated from the effects of the explanatory variable. ◊

**Exercise 5.1.6** Identify the confounding variable based on your suggested explanation in Exercise 5.1.5, p. 76. Explain how it is confounding—what is the link between this third variable and the response variable, and what is the link between this third variable and the explanatory variable? (Hint: Remember that this variable has to be recorded on the observational units: home games for the Thunder.)

Another variable recorded for these data was whether the opponent had a winning record the previous season. Of the Thunder's 41 home games, 22 were against teams that won more than half of their games. Let's refer to those 22 teams as strong opponents. Of these 22 games, 13 were sell-outs. Of the 19 games against opponents that won less than half of their games that season (weak opponents), only 5 of those games were sell-outs.

**Exercise 5.1.7** Was the Thunder more likely to have a sell-out crowd against a strong opponent or a weak opponent? Calculate the relevant (conditional) proportions to support your answer.

When the Thunder played a strong opponent, they won only 4 of 22 games. When they played a weak opponent, the Thunder won 11 of 19 games.

**Exercise 5.1.8** Was the Thunder less likely to win against a strong opponent than a weak one? Again calculate the relevant (conditional) proportions to support your answer.

**Exercise 5.1.9** Explain how your answers to Exercise 5.1.7, p. 77 and Exercise 5.1.8, p. 77 establish that strength of opponent is a confounding variable that prevents drawing a cause-and-effect conclusion between crowd size and game outcome.

**Exercise 5.1.10** Summarize your conclusion about whether these data provide evidence that a sell-out crowd caused the Thunder to play worse. Write as if to a friend who has never studied statistics. Be sure to address the fact that the Thunder had a much smaller winning percentage in front of a sell-out crowd.

Confounding explains why you cannot draw a cause-and-effect conclusion from association alone: The groups defined by the explanatory variable could differ in more ways than just the explanatory variable when confounding is present. The diagram in the figure below illustrates the confounding in the Thunder study. The top panel shows the study design: Observational units (home games) are sorted into groups according to the explanatory variable (whether the arena was sold out). Then the response (win/lose) was observed. The bottom panel shows the confounding: sell-out crowds tended to be against stronger opponents; weaker opponents tended not to sell out the arena.

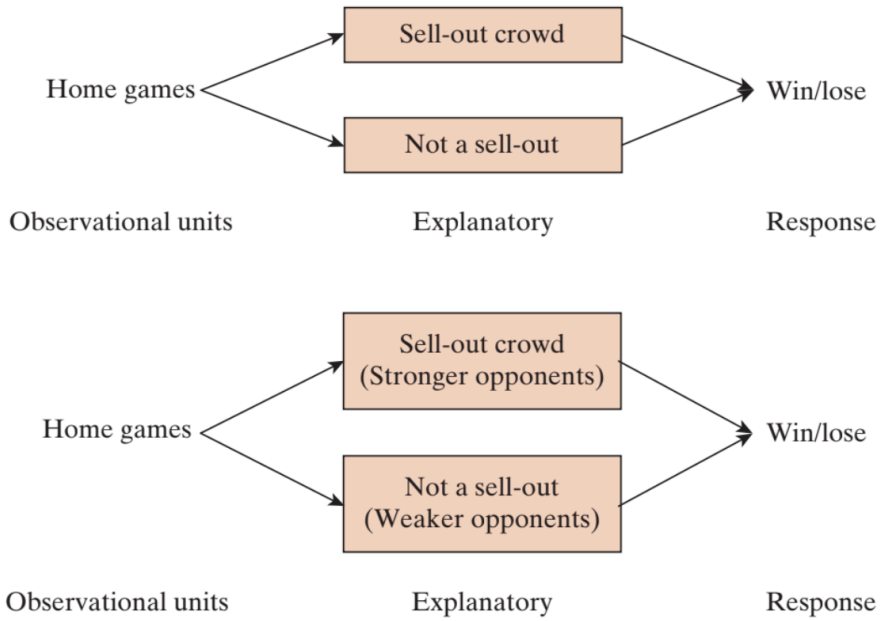



**Figure 5.1.4** Confounding in the study of the home court disadvantage

## 5.2 Observational studies versus experiments

### Have a nice trip

An area of research in biomechanics and gerontology concerns falls and fall-related injuries, especially for elderly people. Recent studies have focused on how individuals respond to large postural disturbances (e.g., tripping, induced slips). One question is whether subjects can be instructed to improve their recovery from such disturbances.

Suppose researchers want to compare two such recovery strategies, lowering (quickly stepping down with front leg and then raising back leg over the object) and elevating (lifting front leg over the object). Subjects will have first been trained on one of these two recovery strategies, and they will be asked to apply it after they feel themselves tripping. The researchers will then induce the subject to trip while walking (but harnessed for safety) using a concealed mechanical obstacle.

Suppose the following 24 subjects have agreed to participate in such a study:

| | |
|---|---|
| **Females** | Alisha, Alice, Betty, Martha, Audrey, Mary, Barbie, Anna |
| **Males** | Matt, Peter, Shawn, Brad, Michael, Kyle, Russ, Patrick, Bob, Kevin, Mitch, Marvin, Paul, Pedro, Roger, Sam |

**Exercise 5.2.1** One way to design this study would be to assign the 8 females to use the elevating strategy and the 16 males to use the lowering strategy. Would this be a reasonable strategy? Why not?

**Exercise 5.2.2** One way to deal with this issue is to assign 4 females and 8 males to each group. Show how the proportion of males in each group is the same.

**Exercise 5.2.3** Now, if you saw a difference in the proportion of trips in the two groups, could it be because of the sex of the subject? Why or why not? Could it be due to other variables, distinct from the recovery strategy? Why or why not?

**Exercise 5.2.4** Because there will always be more potential confounding variables which could be distributed unevenly between the groups being compared, identify a better method for deciding who uses which strategy.

**Definition 5.2.1** In an **observational study**, the groups you compare are "just there," that is, they are defined by what you see rather than by what you do. In an **experiment**, you actively create the groups by what you choose to do. More formally, you assign the conditions to be compared. These conditions may be one or more **treatments** or a **control** (a group you do nothing to). ◊

Note our change in terminology: We now will sometimes call the individuals in the study experimental units instead of observational units, because they are in an experiment instead of an observational study.

**Definition 5.2.2** In a **randomized experiment**, you use a chance device to make the assignments. The units in an experiment are often called experimental units. The role of the random assignment is to balance out potentially confounding variables among the explanatory variable groups, giving us the potential to draw cause-and-effect conclusions. ◊

**Exercise 5.2.5** Let's explore the process of random assignment to determine whether it does "work." First, let's focus on the sex (male vs. female) variable. Suppose we put each person's name on a slip, put those slips in a hat and mix them up thoroughly, and then randomly draw out 12 slips for names of people to assign to the elevating strategy. What proportion of this group do you expect will be male? What proportion of the lowering strategy do you expect will be male? Do you think we will always get an 8/8 split (8 males in each treatment group)?

**Exercise 5.2.6** To repeat this random assignment a large number of times to observe the long-run behavior, we will use the Randomizing Subjects applet[1]. Open the applet and press the *Randomize* button.

What proportion of subjects assigned to Group 1 are men? Of Group 2? What is the difference in these two proportions?

You will notice that the difference in proportions of males is shown in the dotplot in the bottom graph. In this graph, each dot represents one repetition of the random assignment process where we are recording the difference in proportions of men between the two groups.

**Exercise 5.2.7** Press the *Randomize* button again.

Was the difference in proportions of men the same this time?

**Exercise 5.2.8** Change the number of replications from 1 to 198 (for 200 total), uncheck the *Animate* option, and press the *Randomize* button.

The dotplot will display the difference between the two proportions of men for each of the 200 repetitions of the random assignment process. Where are these values centered?

**Exercise 5.2.9** Does random assignment always equally distribute/balance the men and women between the two groups? Is there a tendency for there to be a similar proportion of men in the two groups? Explain.

**Definition 5.2.3** A comparative study is **balanced with respect to a possible confounding variable** if the distribution of the variable is the same for each group in the study. ◊

**Exercise 5.2.10** Prior research has also shown that the likelihood of falling is related to variables such as walking speed, stride rate, and height, so we would like the random assignment to distribute these variables equally between the groups as well. In the applet, use the pull-down menu to switch from the sex-of-participant variable to the height variable. The dotplot now displays the differences in average height between Group 1 and Group 2 for these 200 repetitions. In the long run, does random assignment tend to equally distribute the height variable between the two groups? Explain how you are deciding.

**Exercise 5.2.11** Suppose there is a "balance gene" that is related to people's ability to recover from a trip. We didn't know about this gene ahead of time, but if you select the *Reveal gene?* button and then select gene from the pull-down menu, the applet shows you this gene information for each subject and also how the proportions with the *gene* differ in the two groups.

Does this variable tend to equalize between the two groups in the long run? Explain.

**Exercise 5.2.12** Suppose there were other "x-variables" that we could not measure such as stride rate or walking speed. Select the *Reveal both?* button and use the pull-down menu to display the results for the x-variable (X-var).

[1] `static-content.zybooks.com/applets/2021/randomassignment/Subjects.html?hideExtras=1`

Does random assignment generally succeed in equalizing this variable between the two groups or is there a tendency for one group to always have higher results for the x-variable? Explain.

**Key Idea 5.2.4** Randomly assigning experimental units to groups tends to balance out all other variables between the groups. Any variables that could have an effect on the response should be equalized between the two groups and therefore should not be confounding.

The diagram in the figure below illustrates the preceding key idea. The top panel shows the possible confounding effect if the study design had not been randomized. We want to know whether differences in the explanatory variable cause differences in the response (dotted arrow). However, a confounding variable would affect both the explanatory and response variables, making it impossible to separate the effects of explanatory and confounding variables on the response. The bottom panel shows how randomization removes the effect of the potential confounders.

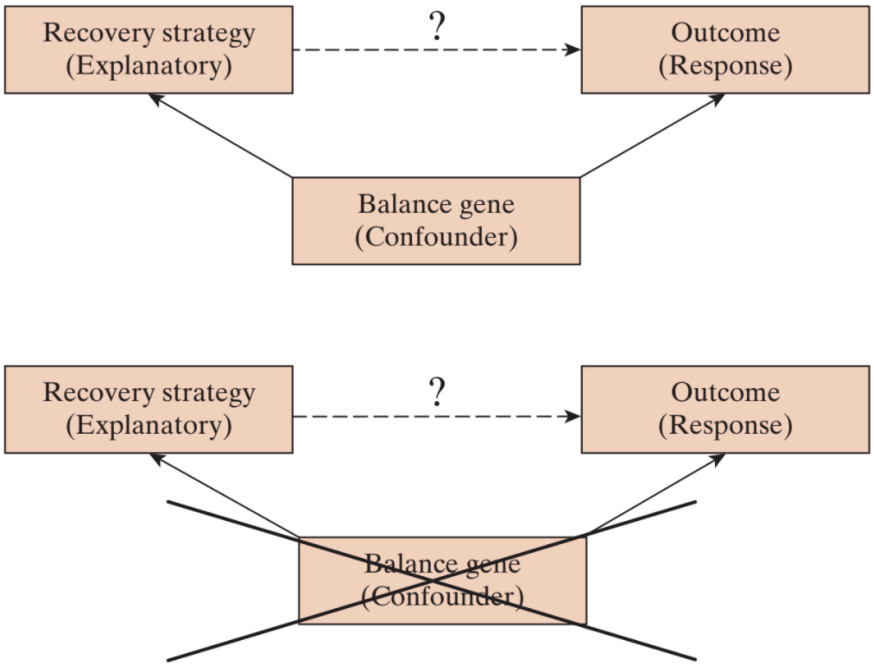



**Figure 5.2.5** Randomization creates approximate balance and tends to eliminate confounding effects

**Exercise 5.2.13** Suppose this study finds a statistically significant difference between the two groups. What conclusion would you draw? For what population? What additional information would you need to know?

**Exercise 5.2.14** As in Exercise 5.2.13, p. 81, if you obtain a statistically significant result, what does that suggest about the potential for a cause-and-effect relationship? Why?

## Another Study: Cursive Handwriting

An article about handwriting[2] appeared in the October 11, 2006, issue of the Washington Post. The article mentioned that among students who took the

[2] www.washingtonpost.com/wp-dyn/content/article/2006/10/10/AR2006101001475.html

essay portion of the SAT exam in 2005-2006, those who wrote in cursive style scored significantly higher on the essay, on average, than students who used printed block letters.

**Exercise 5.2.15** Identify the observational units in this study as well as the explanatory and response variables. Also classify each variable as categorical or quantitative.

**Exercise 5.2.16** Explain how you know that this was an observational study.

**Exercise 5.2.17** Is it reasonable to conclude that using a cursive writing style caused higher scores on the essay, or can you think of an alternative explanation for why students who wrote in cursive style scored higher on average than students who write with block letters? In other words, can you think of other ways in which the cursive and block letter groups might have systematically differed and identify a potential confounding variable?

The same *Washington Post* article also mentioned a different study in which the identical essay was shown to many graders, but some graders were randomly chosen to see a cursive version of the essay and the other graders were shown a version with printed block letters. The average score assigned to the essay with the cursive style was significantly higher than the average score assigned to the essay with the printed block letters.

**Exercise 5.2.18** How does this study differ from the original one? Explain.

**Exercise 5.2.19** Would you be willing to draw a cause-and-effect conclusion from this study, as opposed to the original study? Explain why.

The "placebo effect" has been found in numerous studies: When subjects are told something good is going to happen, they often have a positive response even if nothing is actually done to them. For this reason, subjects are often kept "blind" as to which treatment group they are placed in, for example by giving one group the actual treatment and the other group a "fake" treatment, like a placebo (an inert pill). Placebo treatments have even been used in studies about knee surgery!

**Exercise 5.2.20** Discuss why it was important in this second study for the scorers of the essays to not know that there was no difference in the essays given to different individuals other than the writing style.

**Exercise 5.2.21** Discuss how this study also reveals that those evaluating the response variable in a study, if it requires any subjective judgment, should be "blind" to which treatment group individuals are in.

**Definition 5.2.6** In **double-blind** studies, neither the subjects nor those evaluating the response variable know which treatment group the subject is in. ◊

## Exploring Further

Recall at the beginning of the exploration when we explicitly controlled the number of females to be the same in each group and the number of males to be the same in each group by randomly assigning 4 females to each treatment group and then randomly assigning 8 males to each treatment group. This is a technique called **blocking**.

**Definition 5.2.7** A **block study** design creates **blocks** of experimental units that are similar to each other, randomly assigns the treatments within each block, and then analyzes the data in a way which accounts for block to block variations. When there are only two treatment groups being compared, a block

study design is called a matched pairs design. The term block comes from the first block designs which were agricultural experiments in large fields where separate parts of the field were called blocks. ◊

**Exercise 5.2.22** In the Randomizing Subjects applet[3], select the *Randomized block* option on the left. Leave *Animate* checked and make sure that *sex* is the variable selected. Press the *Randomize* button.

What do you notice about the animated shuffling of cards to the two treatments?

**Exercise 5.2.23** Is the proportion of males the same in each group? Is the proportion of females the same in each group?

**Exercise 5.2.24** Uncheck *Animate* and do 999 more random shuffles.

Describe the shape, center, and variability of the graph displaying the differences in the proportions of males in each group for each of the 1,000 random shuffles.

We can see that blocking on sex has completely controlled sex as a confounding variable; that is to say there will always be the same proportions of males in each treatment and of females in each treatment, so sex could never account for any of the differences in the treatment groups. Might there be added benefits to controlling confounding variables when blocking is used?

**Exercise 5.2.25** Use the pull-down menu to select the variable *height*.

Report the SD of the distribution of differences in average heights between the two treatment groups.

**Exercise 5.2.26** Now select the *Completely Randomized* button to approximate the SD of the distribution of differences in average heights when the proportion of males in the treatment groups could differ by chance.

How does this SD compare to the SD of the distribution of differences in average heights when blocking on the variable sex?

**Key Idea 5.2.8** Why is the variability in the differences in average heights reduced in the block design compared to the completely randomized design?

When other variables (like height) are associated with the blocking variable, we see the added benefit that these variables are more evenly distributed across the treatments.

[3] `static-content.zybooks.com/applets/2021/randomassignment/Subjects.html?hideExtras=1`

# Chapter 6

# Comparing Two Proportions

## 6.1 Comparing two groups: Categorical response

### Murderous nurse?

For several years in the 1990s, Kristen Gilbert worked as a nurse in the intensive care unit (ICU) of the Veterans Administration Hospital in Northampton, Massachusetts. Over the course of her time there, other nurses came to suspect that she was killing patients by injecting them with the heart stimulant epinephrine. Gilbert was eventually arrested and charged with these murders. Part of the evidence presented against Gilbert at her murder trial was a statistical analysis of 1,641 eight-hour shifts during the time Gilbert worked in the ICU[1]. For each of these shifts, researchers recorded two variables: whether Gilbert worked on the shift and whether at least one patient died during the shift.

**Exercise 6.1.1** Identify the observational units in this study. (Hint: The observational units are not people. Think about what the sample size is.)

**Exercise 6.1.2** Classify each variable as categorical or quantitative: whether Gilbert worked on the shift and whether at least one patient died during the shift.

**Exercise 6.1.3** Which variable would you regard as explanatory and which as response?

**Exercise 6.1.4** Is this an observational study or an experiment? Justify your answer.

So what did the researchers find when they analyzed the 1,641 eight-hour shifts? They found that Gilbert worked on 257 of the shifts, and she did not work on 1,384 of them. They also found that a patient died during 74 of the shifts, and no deaths occurred on the other 1,567 shifts.

**Exercise 6.1.5** Explain why the information in the previous paragraph does not provide any clues about whether there is an association between Gilbert working on a shift and a patient dying during the shift. What additional information do you need to know?

Here is some additional information: Among the 74 shifts during which a patient died, 40 were shifts on which Gilbert worked and 34 were shifts on which Gilbert did not work.

**Exercise 6.1.6** Explain why simply comparing only the numbers 40 and 34 is not very informative.

When analyzing data on two categorical variables, a useful first step is to organize the data into a *two-way table of counts*. In this case both variables have only two categories, so we produce a $2 \times 2$ table.

**Exercise 6.1.7** Fill in the missing counts in the two-way table. (Hint: Start by entering 40 and 34 into the appropriate cells. Then make sure that both rows and both columns add up to the indicated totals.)

**Table 6.1.1**

| | Gilbert working on shift | Gilbert not working on shift | Total |
|---|---|---|---|
| Patient died during shift | | | 74 |
| No patient died during shift | | | 1,567 |
| Total | 257 | 1,384 | 1,641 |

[1] api.semanticscholar.org/CorpusID%3A159468911

**Exercise 6.1.8** Suggest a better way to compare the outcomes of the shifts on which Gilbert worked and the shifts on which she did not work than simply comparing 40 to 34. Explain why your answer is more informative.

A next step in analyzing categorical data in a two-way table is to calculate **conditional proportions**. A conditional proportion simply means that you consider the counts in only one category of the explanatory variable at a time.

**Exercise 6.1.9** Now let's calculate the conditional proportions.

**(a)** Calculate the conditional proportion of Gilbert shifts on which a patient died. (Hint: Put the total number of Gilbert shifts in the denominator of your fraction.)

**(b)** Calculate the conditional proportion of non-Gilbert shifts on which a patient died.

A **segmented bar graph** is an appropriate display for graphing data in a two-way table. Such a graph contains rectangles of total height 100% for each category of the explanatory variable. Then segments divide up each rectangle according to the conditional proportions of the response variable categories. A **mosaic plot** also makes the widths of the rectangles proportional to the sizes of the categories.

**Exercise 6.1.10** Fill in a segmented bar graph or mosaic plot to display the data from this study. First draw two (separated) rectangles, one for Gilbert shifts and one for non-Gilbert shifts, each with total height 100%. For the mosaic plot, first determine the proportion of shifts with Gilbert working. Then make the width of that rectangle smaller and the width of the Gilbert not working rectangle larger, to match those proportions. Then draw a line in each rectangle corresponding to the conditional proportion of shifts in which a patient died, as you calculated in #9.




**Exercise 6.1.11** Do the conditional proportions and graph appear to provide evidence that at least one patient was more likely to die on a Gilbert shift than on a non-Gilbert shift? In other words, does there appear to be a tendency for Gilbert shifts to have at least one death more often than non-Gilbert shifts? Explain.

**Exercise 6.1.12** Without doing any further analysis, do you consider the difference between the conditional proportions to be striking, worth reporting

to a jury?

A next step is to produce a single number (statistic) to summarize the data.

**Exercise 6.1.13** What arithmetic operation might you perform on the two conditional proportions to obtain a single statistic?

**Exercise 6.1.14** Calculate the difference in conditional proportions of death between Gilbert and non-Gilbert shifts. Does the value of this difference strike you as noteworthy?

**Exercise 6.1.15** Now calculate a different statistic: the ratio of conditional proportions of death between Gilbert and non-Gilbert shifts. This ratio is called a relative risk. Write a sentence interpreting this ratio value, as the prosecution might present to a jury. Does the value of this ratio strike you as noteworthy?

**Definition 6.1.2 Relative risk** is the ratio of two conditional proportions. It indicates how many times greater the risk of an outcome is for one group compared to another. ◊

**Exercise 6.1.16** Even if the difference between these conditional proportions of death is found to be highly statistically significant, can we absolutely conclude from this analysis that Kristen Gilbert was responsible for (i.e., caused) the increased chance of death on her shifts? Explain your answer. (Hint: Remember your answer to question 4 about what kind of study this is.)

**Exercise 6.1.17** Suggest an alternative explanation that the defense attorneys might use to explain the large difference in death proportions between the two groups.

## Exploring further

**1.** Suppose that the data had turned out as shown here:

**Table 6.1.3**

| | Gilbert working on shift | Gilbert not working on shift | Total |
|---|---|---|---|
| Patient died during shift | 100 | 357 | 457 |
| No patient died during shift | 157 | 1027 | 1,184 |
| Total | 257 | 1,384 | 1,641 |

**(a)** Calculate the conditional proportions of death for Gilbert shifts and for non-Gilbert shifts.

**(b)** Calculate the difference in these conditional proportions. How does this difference compare to the one in question 14? Are they very similar?

**(c)** Calculate the relative risk of these conditional proportions. How does this ratio compare to the one in question 15? Are they very similar?

**(d)** For which table of data, this new one or the original (real) data, would you consider the evidence against Gilbert to be more persuasive/incriminating? Explain.

**2.** Produce a hypothetical $2 \times 2$ table with the same marginal totals, but with the property that the data reveal virtually no association between the variables. In other words, produce a table so that the conditional proportions are very similar between the two groups.

**Table 6.1.4**

| | Gilbert working on shift | Gilbert not working on shift | Total |
|---|---|---|---|
| Patient died during shift | | | 457 |
| No patient died during shift | | | 1,184 |
| Total | 257 | 1,384 | 1,641 |

**3.** What would the segmented bar graph look like for a $2 \times 2$ table with virtually no association between the variables? Explain your answer.

## 6.2 Comparing two proportions: Simulation-based approach

### Is yawning contagious?

### Step 1: Ask a research question

Is yawning contagious? Conventional wisdom says yes: When we see someone else yawn, we're prone to let out a yawn ourselves. Has this happened to you or have you noticed it in others? Will data support this claim if we put it to a scientific test?

### Step 2: Design a study and collect data

The folks at *MythBusters*, a popular television program on the Discovery Channel, investigated this issue by using a hidden camera[1]. Fifty people attending a local flea market were recruited to participate. Subjects were ushered, one at a time, into one of three rooms by co-host Kari. She yawned (planting a yawn "seed") as she ushered subjects into two of the rooms, and for the other room she did not yawn. The researchers decided in advance, with a random mechanism, which subjects went to which room. As time passed, the researchers watched to see which subjects yawned.

**Exercise 6.2.1** Think about why the researchers made the decisions they did.

**(a)** Why did the researchers include a group that didn't see the yawn seed in this study? In other words, why didn't they just see how many yawned when presented with a yawn seed?

**(b)** Why did the researchers use random assignment to determine which subjects went to the "yawn seed" group and which to the control group?

**(c)** Is this an observational study or a randomized experiment? Explain how you are deciding.

**(d)** The researchers clearly used random assignment to put subjects into groups. Do you suspect that they also use random sampling to select subjects in the first place? What would random sampling entail if the population was all flea market patrons?

**Exercise 6.2.2** Identify the explanatory and response variables in this study. Also classify them as categorical or quantitative.

**Exercise 6.2.3** The two competing hypotheses to be tested are: "Yawning is not contagious" and "Yawning is contagious". Identify which is the null hypothesis and which is the alternative hypothesis remembering that the null hypothesis is typically a statement of no effect.

**Exercise 6.2.4** If yawning is not contagious, what does this say about whether there is an association between the explanatory and response variables?

**Exercise 6.2.5** If yawning is contagious, what does this say about whether there is an association between the explanatory and response variables?

**Exercise 6.2.6** State the null and alternative hypotheses in terms of association between the explanatory and response variables in this study.

[1] `mythresults.com/episode28`

An equivalent option to stating the null and alternative hypotheses for this study in terms of association is to talk about long-run probabilities.

**Exercise 6.2.7** There are two long-run probabilities (parameters) in this study. What are they?

**Exercise 6.2.8** Write the null and alternative hypotheses in terms of the two long-run probabilities in words and using appropriate symbols.

The researchers found that 11 of 34 subjects who had been given a yawn seed actually yawned themselves, compared with 3 of 16 subjects who had not been given a yawn seed.

**Exercise 6.2.9** Organize the information into a $2 \times 2$ table:

**Table 6.2.1**

| | **Yawn seed planted** | **Yawn seed not planted** | **Total** |
|---|---|---|---|
| **Subject yawned** | | | |
| **Subject did not yawn** | | | |
| **Total** | | | |

## Step 3: Explore the data

**Exercise 6.2.10** Calculate the conditional proportion of subjects who yawned in the "yawn seed" group, and then do the same for the control group. Then calculate the difference between these proportions, subtracting "yawn seed" - control.

Notice that the difference in conditional proportions is the statistic of interest now that we are comparing two groups.

**Key Idea 6.2.2** Produce a segmented bar graph or mosaic plot to compare the distributions of whether the subject yawned between the treatment (yawn seed) and control group.

**Exercise 6.2.11** Comment on what the graph and calculations reveal about the research question.

**(a)** Did a larger proportion of subjects yawn who were exposed to the yawn seed, as compared to the control group?

**(b)** Based on your analysis thus far, do you think these data provide much evidence that yawning is contagious? Explain.

## Step 4: Draw inferences beyond the data

We see that the proportion of subjects in the "yawn seed" group who yawned was greater than the proportion who yawned in the control group. But does this provide convincing evidence of a genuine difference in the long-run probabilities? To determine whether the *MythBusters* results provide convincing evidence that yawning is contagious, we will apply the same logic that we have used previously: We will use a simulation analysis to determine whether their results are typical or surprising for what we would find if yawning is not contagious.

**Exercise 6.2.12** As with earlier studies, there are two possible explanations for the MythBusters results. What are they? (Hint: These correspond to the null and alternative hypotheses stated earlier.)

The key to our simulation analysis is to assume that if yawning is not contagious (null hypothesis), then *the 14 yawners would have yawned regardless of whether they had seen the yawn seed*. Similarly, we'll assume that the 36 nonyawners would not have yawned, no matter which group they had been assigned. In other words, our simulation assumes the null hypothesis is true—that there is no association, no connection, between the yawn seed and actual yawning.

**Key Idea 6.2.3** To evaluate the statistical significance of the observed difference in our groups, we will investigate how large the difference in conditional proportions tends to be just from the random assignment of response outcomes to the explanatory variable groups.

We cannot use coins to conduct this simulation analysis, because we have two variables to consider: whether the yawn seed was planted and whether the subject yawned. Instead of coins, we will use index cards. Here's our strategy:

- Take a set of 50 cards, with 14 blue cards (to represent the yawners) and 36 green cards (to represent those who did not yawn).
- Shuffle the cards well and randomly deal out 34 to be the yawn seed group (the rest go to the control group).
- Count how many yawners (blue cards) you have in each group and how many nony- awners (green cards) you have in each group.
- Construct the two-way table to show the number of yawners and nonyawners in each group. (Shuffling and dealing the cards out into the two piles ensures that nothing different happened between your Yawn-seed group and your Non-yawn-seed group—any differences between the two groups that arise are due purely to the random assignment process.)

**Exercise 6.2.13** Do this shuffling and dealing once.

**(a)** Report the (simulated) $2 \times 2$ table that your shuffling and dealing produces:

**Table 6.2.4**

| | **Yawn seed planted** | **Yawn seed not planted** | **Total** |
|---|---|---|---|
| **Subject yawned** | | | **14** |
| **Subject did not yawn** | | | **36** |
| **Total** | **34** | **16** | **50** |

**(b)** Calculate the conditional proportions who yawned for your simulated data and the difference in those proportions.

**(c)** Is your simulated statistic (difference in conditional proportions) at least as large as the observed value of the statistic from the MythBusters study?

We need to perform a large number of repetitions (say, 1,000 or more) in order to assess whether the *MythBusters* result is typical or surprising when yawning is not contagious. To do this we will use an applet specifically designed for this purpose: the Two Proportions applet[2].

**Exercise 6.2.14** Open this applet and verify that the two-way table, segmented bar graph, and observed difference in simulated proportions are the same as those you found earlier.

[2] static-content.zybooks.com/applets/2021/chisqshuffle/ChiSqShuffle.htm?yawning=1&hideExtras=2

**(a)** Check the Show Shuffle Options box. Notice how the cards have been set up: 14 blue cards, with 11 in the yawners group. Press Shuffle to shuffle the cards and redistribute them to the two groups. How many yawners ended up in the seeded group? What is the corresponding difference in the conditional proportions for the shuffled data?

**(b)** Press the Shuffle button four more times. Record the difference in proportions each time. Was it always the same number?

Notice that each of these values has been added to the dotplot. (Technical detail: In this case, it's equivalent to look at the number of blue cards in the yawn group or the difference in conditional proportions as the statistic of interest.)

**Exercise 6.2.15** Use this applet to conduct 1,000 repetitions of this simulation: Change the Number of Shuffles from 1 to 995 (for a total of 1,000) and press Shuffle. The applet produces a dotplot showing the null distribution for the difference in proportions of yawners between the two groups. Map this simulation to the research study: What are the null hypothesis, one repetition, and the statistic?

**Exercise 6.2.16** Look more closely at the simulated null distribution for the difference in proportions of yawners between the two groups.

**(a)** Is this null distribution centered around 0? Explain why this makes sense. (Hint: Think about the choice of statistic and about the null hypothesis, which was the basis of the simulation analysis.)

**(b)** Is the observed value of the statistic from the MythBusters study (0.136) out in the tail of this null distribution or not so much? In other words, does the observed result appear to be typical or surprising when the null hypothesis (that yawning is not contagious) is true?

**(c)** As you know, you can assess the strength of evidence against the null hypothesis by calculating a p-value. To calculate a p-value from this null distribution, you will count the number of repetitions that produced a simulated difference in proportions equal to _____ or _____ ( more/less).

**Exercise 6.2.17** Enter the observed difference in the sample proportions in the Count Samples box and choose the direction from the pull-down menu and press Count. Check whether the shaded region of the dotplot corresponds with your answer to question 17c. Then report this approximate p-value.

**Exercise 6.2.18** Interpret this p-value by filling in the blanks: Under the assumption that _____, if we repeatedly _____ many, many times, the probability we would obtain a difference in conditional proportions as or more extreme than _____ is about _____.

**Exercise 6.2.19** Evaluate this p-value: Using guidelines for assessing the strength of evidence from p-values, would you conclude that the MythBusters result provides much evidence that yawning is contagious?

We have a small confession to make. We fudged the data from this study a bit. The actual data showed that 10 of the 34 subjects in the yawn seed group yawned themselves, and 4 of the 16 in the control group yawned.

**Exercise 6.2.20** Recalculate the conditional proportions who yawned for these data: Proportion who yawned in the "yawn seed" group, proportion who yawned in the control group, and difference in conditional proportions ("yawn

seed" - control).

**Exercise 6.2.21** How has your observed statistic changed from the originally presented data? Smaller, same, or larger?

**Exercise 6.2.22** Before you carry out the inference using this postconfession observed statistic, let's make some predictions: How do you expect the p-value and strength of evidence to change from your earlier analysis? Is hte p-value smaller, the same, or larger? Is the strength of evidence weaker, the same, or stronger? Explain why you answered what you did.

**Exercise 6.2.23** Conduct a simulation analysis to check your answer predictions from question 23. To do this, enter the new values into the four boxes of the table and press Use Table. Then ask for 1,000 repetitions in your simulation analysis. Enter the new observed value of the statistic and then press the Count button. Report the approximate p-value and summarize your conclusion in terms of strength of evidence that yawning is contagious.

## Estimation

Because the MythBusters data did not reveal a significant difference between the groups, it's less important than usual to estimate the parameter value. But we can still use the simulation results to estimate a 95% confidence interval for the relevant parameter. Keep in mind that the relevant parameter is now the difference in the long-run probabilities between the yawn seed treatment and the no-yawn seed treatment.

**Exercise 6.2.24** The parameter of interest in this yawning study can be expressed in symbols as $\pi_{\text{yawn seed}} - \pi_{\text{control}}$Describe what this parameter means in words.Report (again) the observed value of the statistic from the MythBusters actual data. (Hint: Recall that the statistic is the difference in conditional proportions between the two groups.)The observed value of this statistic is your best estimate of the unknown parameter value. But we should produce an interval estimate centered on this observed value. To do that we again need to consider the *chance variability* in the statistic.From the simulation analysis that you just conducted on the postconfession MythBusters data, what was the standard deviation of the null distribution?Determine endpoints of a 95% 2SD confidence interval for the difference by taking the observed value of the statistic $\pm$ 2 standard deviations.Does this confidence interval include only positive values, only negative values, or zero as well as positive and negative values?Explain the importance of your answer to part (e) in terms of whether the data provide evidence that subjects who experience a yawn seed are more likely to yawn than those who do not experience a yawn seed?

## Step 5: Formulate conclusions

**Exercise 6.2.25** The MythBusters hosts concluded from their study that there is "little doubt, yawning seems to be contagious." Based on your simulation analysis of their data, considering the issue of statistical significance, do you agree with this conclusion? Explain your answer, as if to the hosts, without using statistical jargon. Be sure to include in your answer an explanation for why you conducted the simulation analysis and what the analysis revealed about the research question.

**Exercise 6.2.26** If you had decided that the two groups differed significantly, would you have been justified in drawing a cause-and-effect conclusion between the yawn seed and an increased probability of yawning? Explain, based on how

the study was conducted.

**Exercise 6.2.27** Based on how the sample was selected, to what larger population would you feel comfortable generalizing the results of this study? Justify your answer.

**Exercise 6.2.28** Think about the difference between these two possible conclusions: (i) The data strongly suggest that yawning is not contagious and (ii.) The data do not provide convincing evidence to suggest that yawning is contagious. Explain why these conclusions are not the same. Also indicate which one is the more appropriate conclusion from your analysis and explain why.

**Exercise 6.2.29** As we discussed previously, in every statistical analysis it is possible to make a Type I or a Type II error. Identify which error you might be making here, describe it in context, and briefly identify the consequences.

## Step 6: Look back and ahead

**Exercise 6.2.30** Critique the design and conclusion of this study. Were there any limitations, such as how the subjects were selected or how the measurements were recorded? Was the sample size large enough? Did you observe a result of practical value? Discuss how you might address a few of these limitations. Suggest how you might design a follow-up study to investigate whether yawning is contagious. Address details such as how to ensure that the yawn seed is noticed, how long to wait for the subject's yawn to be connected to the seed, and how to select/recruit subjects.

## Exploring further

Now let's suppose that this yawning study had involved 500 people, 10 times as many as the actual study. Let's also suppose that the conditional proportions who yawned in each group were identical to the actual study. The $2 \times 2$ table would therefore be:

Now let's suppose that this yawning study had involved 500 people, 10 times as many as the actual study. Let's also suppose that the conditional proportions who yawned in each group were identical to the actual study. The $2 \times 2$ table would therefore be:

**Exercise 6.2.31** Before you conduct a simulation analysis, indicate how you expect the information in (a)-(d) to change:

**(a)** Difference in conditional proportions: smaller, the same, or larger?

**(b)** p-value: smaller, the same, or larger?

**(c)** Strength of evidence that yawning is contagious: weaker, the same, or stronger?

**(d)** Test decision at the 0.05 significance level: Reject null (unlike before) Fail to reject null (as before)

**Exercise 6.2.32** Use the Two Proportions applet, again entering the appropriate table of data first, to conduct a simulation analysis. Report the approximate p-value and summarize your conclusion in terms of strength of evidence that yawning is contagious. Also indicate whether you would find strong evidence against the null hypothesis at the 0.05 significance level.

**Exercise 6.2.33** Summarize how the 10-fold increase in the sample size has changed your conclusions. Do you have stronger evidence that yawning is

contagious? Is the evidence now very strong that yawning is contagious?

## Relative risk

A previous section introduced you to the relative risk as an alternative statistic to summarize the relationship between two binary variables.

**Exercise 6.2.34** What value of the relative risk indicates no association between the two variables in this study?

**Exercise 6.2.35** State the null and alternative hypotheses for the yawning study in terms of the relative risk.

We will now apply the 3S strategy using the relative risk. Use the sample size of 500 people you used in questions 32-34 for questions 37 and 38.

**Exercise 6.2.36** Statistic. Calculate, and then interpret, the observed value of the relative risk by dividing the conditional proportion of yawners in the yawn seed group compared to the conditional proportion of yawners in the no-yawn seed group.

In the Two Proportions applet[3], change the statistic to "Relative risk" in the Statistic pull-down menu. Confirm that the statistic matches your answer to question 37.

**Exercise 6.2.37** Simulate. If you still have your shuffles from question 33 use them; otherwise generate 1,000 new shuffles. Find the p-value using the relative risk. Compare this p-value to the p-value you obtained in question 33.

If you used the same simulated tables, you should have found that the p-value for the relative risk is exactly the same as for the difference in proportions! This is because there is a "one-to-one" correspondence between these statistics.

[3] static-content.zybooks.com/applets/2021/chisqshuffle/ChiSqShuffle.htm?yawning=1&hideExtras=2

# 6.3 Comparing two proportions: Theory-based approach

## Donating blood

Blood donations from volunteers serve a crucial role in the nation's health care system. Are Americans any more or less generous about donating blood in some years than others? Are women any more or less willing to give blood than men? To investigate these questions we can analyze data from the General Social Survey[1], which is a national survey conducted every two years on a nationwide random sample of adult Americans.

**Exercise 6.3.1** Did this study make use of random sampling, random assignment, both, or neither? Also explain what your answer means in terms of the scope of conclusions that can be drawn from these data.

Let's start by comparing the years 2002 and 2004.

**Exercise 6.3.2** State the appropriate null and alternative hypotheses, in words and in symbols, to address the research question of whether Americans were more generous with blood donations in one of these years than the other. (Hint: Remember that hypotheses are always about population parameters, and think about whether the alternative should be one- or two-sided before you see the data.)

Data from the 2002 and 2004 surveys are summarized in this two-way table:

**Table 6.3.1**

| | 2002 | 2004 |
|---|---|---|
| Donated blood in past year | 210 | 230 |
| Have not donated blood in past year | 1,152 | 1,106 |
| Total | 1,362 | 1,336 |

**Exercise 6.3.3** For each year, calculate the (conditional) proportion of adult Americans sampled who donated blood in the previous 12 months. Use appropriate symbols to represent them. Also calculate the difference between these sample proportions (taking the 2002 proportion minus the 2004 proportion).

**Exercise 6.3.4** Produce a segmented bar graph or mosaic plot to display the conditional proportions who gave blood in these two years. Comment on what the graph and your calculations from question 3 reveal about these two samples.

**Exercise 6.3.5** Use the Multiple Proportions applet[2] to conduct a simulation analysis to approximate a p-value for testing the hypotheses that you stated in question 2. You can enter the table of counts in the $2 \times 2$ table or type in the table in the **Sample data** box and press **Use Table**. Or you can copy the data from the text webpage and paste into the **Sample data** box and press **Use Data**. Verify the segmented bar graph that you created.

[1] gss.norc.org/

Sample data: (2×2: ☑)

| | Group A | Group B | Totals |
|---|---|---|---|
| Success | 210 | 230 | 440 |
| Failure | 1152 | 1106 | 2258 |

Sample data: (2×2: ☑)

(Explanatory, Response)

| | 2002 | 2004 |
|---|---|---|
| donated | 210 | 230 |
| didn't | 1152 | 1106 |

**Key Idea 6.3.2** Shuffling is an appropriate way to estimate a p-value for comparing groups regardless of the study design (observational study or experiment, random assignment or random sampling or neither). Of course, the study design is very important in determining the appropriate scope of conclusions. (Can you generalize to a larger population? Can you infer a cause-and-effect conclusion?)

**Exercise 6.3.6** Check the **Show Shuffle Options** box, ask for 1000 repetitions, and press **Shuffle**.

**(a)** Indicate how to find the p-value from the null distribution. (Hint: Remember whether the alternative hypothesis is one-sided or two-sided.)

**(b)** Report the approximate p-value.

**(c)** Based on this p-value, do the sample data provide much evidence that the population proportion who gave blood in 2002 differs from the population proportion who gave blood in 2004?

**Exercise 6.3.7** Does the null distribution of the simulated values appear to follow an approximately normal distribution? Centered around what value? Also report the standard deviation of these values.

**Exercise 6.3.8** Let's see what the standardized statistic tells us.

**(a)** Report the observed value in the sample for the statistic.

**(b)** How many standard deviations (based on your answer to question 7) above or below zero (the hypothesized value) is the observed value of the statistic?

**(c)** Is the strength of evidence against the null hypothesis based on this standardized statistic consistent with what the p-value revealed in question 6? Explain.

Just as you learned a theory-based approach for making inferences about a single proportion, so too will you now learn a theory-based approach for making inferences about comparing proportions between two groups. This theory-based approach is again based on an approximation from a normal distribution.

[2] static-content.zybooks.com/applets/2021/chisqshuffle/ChiSqShuffle.htm?hideExtras=2

**Exercise 6.3.9** Check the Overlay Normal distribution box. Is the behavior of the two distributions similar? How do the p-values compare?

**Exercise 6.3.10** Check the box to compute **95% CI(s) for difference in proportions**. (This box appears in the left-hand panel when the difference is selected as the statistic.)

**(a)** Report the endpoints of this interval.

**(b)** Interpret this interval. Be sure to mention clearly what parameter is being estimated by this interval. (Hint: In other words, you are 95% confident that this interval contains what?)

**(c)** Does this interval contain only negative values, only positive values, or does the interval include the value zero as well as both positive and negative values? Comment on the importance of what this means.

**Exercise 6.3.11** Another applet that can compute p-values and confidence intervals for tests of two proportions is the Theory-Based Inference applet[3]. Use the pull-down menu to select **Two proportions** and enter the relevant summary information, using the 2002 sample as Group 1. Check the **Test of Significance** box and press on the **inequality** sign button until the alternative hypothesis becomes two-sided. Press **Calculate**. Verify the p-value you obtained earlier (question 9) as well as the confidence interval (question 10).

## Validity Conditions for Theory-Based Approach (Two Proportions)

As with a single proportion, the theory-based inference approach for comparing two proportions is only valid when a normal distribution provides a good approximation to the distribution of the statistic $\hat{p}_1 - \hat{p}_2$. Conservatively, the theory-based significance test and confidence interval work well when there are at least 10 observations in each of the four cells of the two-way table. In other words, we require at least 10 "successes" and at least 10 "failures" in each sample in order to use this theory-based approach.

**Validity Conditions 6.3.3** The theory-based test and interval for the difference in two proportions (called a **two-sample z-test or interval**) work well when there are at least 10 observations in each of the four cells of the $2 \times 2$ table.

**Exercise 6.3.12** Is it valid to use the theory-based approach for two proportions in this study? Justify your answer.

**Exercise 6.3.13** Is it reasonable to generalize your conclusions from this analysis to the population of all adult Americans? Explain why or why not.

## Exploring further

**Exercise 6.3.14** Let's see what will happen if we switch which is Group 1 and which is Group 2.

**(a)** Before doing any calculations, what do you think would happen to the standardized statistic, p-value, and confidence interval if you entered the 2004 sample as Group 1 and the 2002 sample as Group 2?

**(b)** Make this change in the Theory-Based Inference applet (i.e., switch the roles of Groups 1 and 2), and then press Calculate. Report the new values

[3] `static-content.zybooks.com/applets/2021/tbia/TBIA.html?hideExtras=1`

of the standardized statistic, p-value, and confidence interval.

**(c)** Did these change as you predicted in part (a)?

Using the General Social Survey data to classify respondents by sex and combining the years 2002 and 2004 we can construct a two-way table:

**Table 6.3.4**

| | Male | Female |
|---|---|---|
| Donated blood in past year | 239 | 201 |
| Have not donated blood in past year | 1,032 | 1,226 |
| Total | 1,271 | 1,427 |

**Exercise 6.3.15** Analyze these data to address the question of whether American men and women differ with regard to donating blood.

**(a)** Are the validity conditions met?

**(b)** State the null and alternative hypothesis using full sentences. (Do males - females)

**(c)** Calculate the proportion of male blood donors and female blood donors.

**(d)** Create a chart that compares these two proportions.

**(e)** Calculate the standardized statistic and the p-value.

**(f)** Make a conclusion regarding your null and alternative hypothesis in light of the p-value you calculated.

**(g)** Calculate a 95% confidence interval for the difference between the proportion of each gender who donate blood.

**(h)** Interpret your confidence interval, giving all necessary context.

# 6.4 Formulas

## Formulas for two proportions

- $n_1$ and $n_2$ are the sample sizes for the two groups.
- $\hat{p}_2 = \frac{\text{number of successes in group 1}}{n_1}$ is the sample proportion for group 1.
- $\hat{p}_2 = \frac{\text{number of successes in group 2}}{n_2}$ is the sample proportion for group 2.
- $\hat{p}_1 - \hat{p}_2$ is the difference in sample proportions.
- $\hat{p} = \frac{\text{total number of successes}}{n_1+n_2}$ is the pooled sample proportion.
- $SD(\hat{p}_1 - \hat{p}_2)$ is the standard deviation of the statistic.
- SE using pooled $\hat{p} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}$ is a theory-based standard error that approximates SD of the null distribution.
- SE using $\hat{p}_1$ and $\hat{p}_2 = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ is a theory-based standard error that approximates $SD(\hat{p}_1 - \hat{p}_2)$.
- standardized statistic $= \frac{\text{statistic}-\text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
    - $z = \frac{\left(\hat{p}_1-\hat{p}_2\right)-0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two proportions.
    - $z = \frac{\left(\hat{p}_1-\hat{p}_2\right)-0}{\text{SE using pooled } \hat{p}} = \frac{\left(\hat{p}_1-\hat{p}_2\right)-0}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$ is the theory-based standardized statistic for two proportions.
- $z = \frac{\left(\hat{p}_1-\hat{p}_2\right)-0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two proportions.
- $z = \frac{\left(\hat{p}_1-\hat{p}_2\right)-0}{\text{SE using pooled } \hat{p}} = \frac{\left(\hat{p}_1-\hat{p}_2\right)-0}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$ is the theory-based standardized statistic for two proportions.
- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.
    - $(\hat{p}_1 - \hat{p}_2) \pm 2 \times SD(\hat{p}_1 - \hat{p}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\pi_1 - \pi_2$.
    - $(\hat{p}_1 - \hat{p}_2) \pm$ (multiplier) $\times$ (SE using $\hat{p}_1$ and $\hat{p}_2$) is a theory-based confidence interval estimating $\pi_1 - \pi_2$, where the multiplier comes from the standard normal distribution.
- $(\hat{p}_1 - \hat{p}_2) \pm 2 \times SD(\hat{p}_1 - \hat{p}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\pi_1 - \pi_2$.
- $(\hat{p}_1 - \hat{p}_2) \pm$ (multiplier) $\times$ (SE using $\hat{p}_1$ and $\hat{p}_2$) is a theory-based confidence interval estimating $\pi_1 - \pi_2$, where the multiplier comes from the standard normal distribution.

# Chapter 7

# Comparing Two Means

# 7.1 Comparing two groups: Quantitative response

A worksheet for Grace College from Zybooks

## Cancer Pamphlets

Researchers in Philadelphia investigated whether pamphlets containing information for cancer patients are written at a level that the cancer patients can compre- hend (Short, Moriarty, and Cooley, 1995[1]). First, they measured the readability of a sample of cancer pamphlets based on factors such as the length of sentences and the number of polysyllabic words, assigning each pamphlet a grade level. The results shown below are presented as a frequency table, reporting the number of pamphlets at each grade level.

**Table 7.1.1 Readability measures (grade level) for pamphlets aimed at cancer patients.**

| Pamphlets' readability levels | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Count (number of pamphlets) | 3 | 3 | 8 | 4 | 1 | 1 | 4 | 2 | 1 | 2 | 1 |

**Exercise 7.1.1** What are the observational units in the table above? How many observational units were measured?

**Exercise 7.1.2** Use the information in the table above to construct a histogram of the pamphlets' readability levels. How would you summarize the behavior of this histogram?

**Exercise 7.1.3** Using the information in the table above, calculate the mean pamphlet readability level. (Hint: Add up the values, 6 + 6 + 6 + 7 + … + 15 + 15 + 16 and divide by the number of observational units.)

**Exercise 7.1.4** Using the information in the table above, calculate the median pamphlet readability level. How many observations are on each side of this median value (including the repeat values)? (Hint: What is the position of the median?)

**Exercise 7.1.5** How do the mean and median compare? Is this what you would have predicted from the histogram? Explain briefly.

While the mean and median tell us about the center of the distribution, one way to summarize the behavior of the distribution of a quantitative variable is by dividing the distribution into four pieces of roughly equal size (number of observations). In other words, summarize the distribution by determining where the bottom 25% of the data are, the next 25%, the next 25%, and then the top 25%.

**Definition 7.1.2** The value for which 25% of the data lie below that value is called the **lower quartile** (or 25th percentile). Similarly, the value for which 25% of the data lie above that value is called the **upper quartile** (or 75th percentile). Quartiles can be calculated by determining the median of the values above/below the location of the overall median. The difference between the quartiles is called the **inter-quartile range (IQR)**, another measure of variability along with standard deviation. ◊

**Exercise 7.1.6** Take the 15 values below the position of the median, and find the median of those 15 values. This is the lower quartile.

[1] doi.org/10.1080/10691898.1995.11910492

**Exercise 7.1.7** Repeat for the 15 values above the position of the median. This is the upper quartile.

**Exercise 7.1.8** Calculate and interpret the inter-quartile range.

**Exercise 7.1.9** Explain why the inter-quartile range might be preferred to the standard deviation to summarize the variability in the pamphlet reading levels.

**Key Idea 7.1.3** The minimum, lower quartile, median, upper quartile, and maximum comprise the **five-number summary**. A visual representation of the five-number summary is a boxplot. The figure below shows a boxplot for these data.

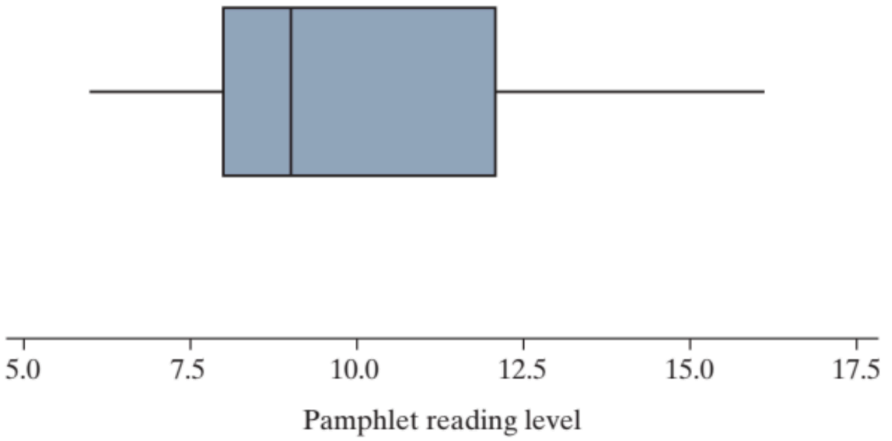



**Figure 7.1.4** Boxplot of cancer pamphlet reading levels

**Definition 7.1.5** A **boxplot** is a visual display of the five-number summary. The box displays the middle 50% of the distribution, and its width (the IQR) helps us visualize the spread of the distribution; the whiskers extend to the smallest and largest values in the dataset. ◊

**Exercise 7.1.10** How does the boxplot match up to the five-number summary? (Endpoint of lower whisker, lower edge of box, line within the box, upper edge of box, upper whisker).

The boxplot illustrates that there is some variability in the pamphlet reading levels. What the researchers wanted to know was how these different reading levels matched up to the reading ability of the cancer patients. The table below shows a frequency table for the reading level (again a grade level) for 63 cancer patients. Note that patient reading levels of under 3rd grade and above 12th grade are not determined exactly.

**Table 7.1.6 Comparing pamphlet readability scores and patient reading levels**

| **Patients' reading levels** | **<3** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **>12** | **Total** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Count (number of patients)** | **6** | **4** | **4** | **3** | **3** | **2** | **6** | **5** | **4** | **7** | **2** | **17** | **63** |

**Exercise 7.1.11** Explain why it is not possible to calculate the mean reading ability for these patients.

**Exercise 7.1.12** Explain why it is possible to calculate the median reading ability for these patients and do so.

**Exercise 7.1.13** How does the median reading level of the patients compare to the median reading level of the pamphlets? Does this indicate that the

pamphlets are a good match to the cancer patients? Explain.

**Exercise 7.1.14** Determine and interpret the lower quartile for the patient reading levels.

**Exercise 7.1.15** How does the lower quartile of patient reading levels compare to the lower quartile of the reading level of the cancer pamphlets? How does the lower quartile of patient reading levels compare to the minimum reading level of the cancer pamphlets?

**Exercise 7.1.16** Use your answer to question Exercise 7.1.15, p. 106 to decide whether the cancer pamphlets are a good match to patient reading levels. (Hint: Interpret the second comparison in Exercise 7.1.15, p. 106 in context.)

Later in this chapter, you will learn formal methods for assessing whether the centers of two distributions are statistically significantly different. But notice how such analysis may not be valid for a study like this:

- Means are not the most important feature to be comparing. For example, means will be influenced by outliers, and these data were not presented in a way to allow for calculation of means.
- We could focus on comparing the medians instead, but only comparing the centers of these distributions ignores the variability in the distributions, which is perhaps of more interest to these research questions.
- We don't know whether the pamphlets or patients are randomly sampled from larger populations.

This exploration reveals that measures of center do not always tell the whole story when you are analyzing data to address a particular research question. In this case, the research question of whether pamphlets' readability levels are well aligned with patients' reading levels requires looking at the entire distributions, not simply at measures of center. Examining displays of the distributions would be a better place to begin. In addition to dotplots and histograms, boxplots are another potential type of display that is especially useful for comparing distributions. But keep in mind, as shown in a previous exploration, that they can sometimes mask important features of distributions. Many software packages now allow you to overlay a boxplot on top of a histogram or dotplot to help see the entire distribution while highlighting key features like the quartiles.

## 7.2 Comparing two means: Simulation-based approach

A worksheet for Grace College from Zybooks

### Lingering effects of sleep deprivation

### STEP 1: Ask a research question.

Many students pull "all-nighters" when they have an important exam or a pressing assignment. Concerns that may arise include: Can you really function well the next day after a sleepless night? What about several days later: Can you recover from a sleepless night by getting a full night's sleep on the following nights?

**Exercise 7.2.1** What is your research conjecture about whether one can recover from a sleepless night by getting a full night's sleep on the following nights? What are some other related questions that you would be interested in investigating related to this issue?

### STEP 2: Design a study and collect data.

Researchers Stickgold, James, and Hobson investigated delayed effects of sleep deprivation on learning in a study published in Nature Neuroscience (2000).[1] Twenty-one volunteers, aged 18-25 years, were first trained on a visual discrimination task that involved watching stimuli appear on a computer screen and reporting what was seen.

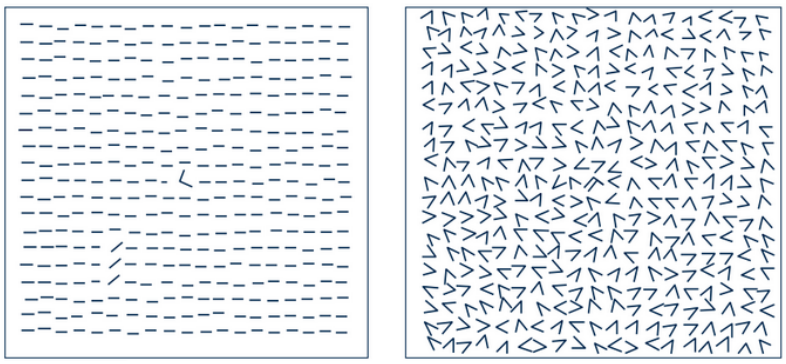


**Figure 7.2.1** Figure 741: Subjects were flashed the screen on the left and then it was masked by the screen on the right Then subjects were asked whether they had seen an L or a V and whether the slanted lines were placed vertically or horizontally

After the training period, subjects were tested. Performance was recorded as the minimum time (in milliseconds) between the appearance of stimuli and an accurate response. Following these baseline measurements, one group was randomly assigned to be deprived of sleep for 30 hours, followed by two full nights of unrestricted sleep, whereas the other group was allowed to get unrestricted sleep on all three nights. Following this, both groups were retested on the task to see how well they remembered the training from the first day. Researchers recorded the improvement in performance as the decrease in time required at retest compared to training. (Note: For example, if someone took

[1] doi.org/10.1038/81756

5 milliseconds (ms) to respond at the beginning of the study and then 2 ms to respond at the end, the improvement score is 3 ms. But if someone took 2 ms at the beginning and then 5 ms at the end, the improvement score is -3 ms.)

The goal of the study was to see whether the improvement scores tend to be higher for the unrestricted sleep treatment than for the sleep deprivation treatment.

**Exercise 7.2.2** Identify the explanatory and response variables in this study. Also classify them as either categorical or quantitative.

**Exercise 7.2.3** Was this an experiment or an observational study? Explain how you are deciding.

**Exercise 7.2.4** Let $\mu_u$ (u for unrestricted) be the long-run mean improvement on this task three days later when someone has had unrestricted sleep and let $\mu_d$(d for deprived) denote the long-run mean improvement when someone is sleep deprived on the first night. In words and symbols, state the null and the alternative hypotheses to investigate whether sleep deprivation has a negative effect on improvement in performance on visual discrimination tasks. (Hint: For the alternative hypothesis: Do you expect the people to do better or worse when sleep deprived? Based on your answer, what sign/direction should you choose for the alternative hypothesis?)

Here are the data, with positive values indicating better performance at retest than at training, and negative values indicating worse performance at retest than at training:

| Unrestricted-sleep group's improvement scores (milliseconds): |
|---|
| 25.20, 14.50, −7.00, 12.60, 34.50, 45.60, 11.60, 18.60, 12.10, 30.50 |

| Sleep-deprived group's improvement scores (milliseconds): |
|---|
| −10.70, 4.50, 2.20, 21.30, −14.70, −10.70, 9.60, 2.40, 21.80, 7.20, 10.00 |

## STEP 3: Explore the data.

**Exercise 7.2.5** To look at graphical and numerical summaries of the data from the study, go to the Multiple Means[2] applet. The sleep deprivation data have already been entered into the applet.Notice that the applet creates parallel dotplots, one for each study group. Based on these dotplots alone, which group (unrestricted or deprived) appears to have had the higher mean improvement? How are you deciding?Based on the dotplots alone, which group (unrestricted or deprived) appears to have had more variability in improvement? How are you deciding?Notice also that the applet also computes numerical summaries of the data, such as the mean and standard deviation (SD) for the improvements in each group. For the unrestricted group, record the sample size (n), mean, and SD. For the deprived group, record the sample size (n), mean, and SD.Recall from earlier in the course that the standard deviation is a measure of variability. Relatively speaking, smaller standard deviation values indicate less variability and a distribution whose data values tend to cluster more closely together, compared to a distribution with a larger standard deviation.Based on the numerical summaries reported in #5(c), which group (unrestricted or deprived) had the higher mean improvement?Based on the numerical summaries reported in #5(c), which group (unrestricted or deprived) had the higher variability in improvement?Notice that the applet also reports the observed difference in means for the improvements of the two groups. Record this value (and

its measurement units).Before you conduct an inferential analysis, does this difference in sample means (as reported in #5(f)) strike you as a meaningful difference? Explain your answer.

## STEP 4: Draw inferences beyond the data.

**Exercise 7.2.6** What are two possible explanations for why we observed the two groups to have different sample means for improvement in performance?

**Exercise 7.2.7** Describe how you might go about deciding whether the observed difference between the two sample means is statistically significant. (Hint: Think about how you assessed whether an observed difference between two sample proportions was statistically significant. Use the same strategy, with an appropriate modification for working with means instead of proportions.)

Once again the key question is how often random assignment alone would produce a difference in the groups at least as extreme as the difference observed in this study if there really were no effect of sleep condition on improvement score. You addressed similar questions before when you analyzed the dolphin therapy and yawing studies. The only change is that now the response variable is quantitative rather than categorical, so the relevant statistic is the difference in group means rather than the difference in group proportions. Also once again, we use simulation to investigate how often such an extreme difference would occur by chance (random assignment) alone (if the null hypothesis of no difference/no effect/no association were true).

In other words, we will again employ the 3S strategy.

## 1. Statistic:

**Exercise 7.2.8** A natural statistic for measuring how different the observed group means are from each other is the difference in the mean improvement scores between the two groups. Report the value of this statistic, as you did in #5(f).

## 2. Simulate:

**Exercise 7.2.9** You will start by using index cards to perform a tactile simulation of randomly assigning the 21 subjects between the two groups, assuming that sleep condition has no impact on improvement.

Because the null hypothesis asserts that improvement score is not associated with sleep condition, we will assume that the 21 subjects would have had exactly the same improvement scores as they did, regardless of which sleep condition group (unrestricted or deprived) the subject had been assigned.How many index cards do you need to conduct this simulation?What will you write on each index card?To conduct one repetition of this simulation:

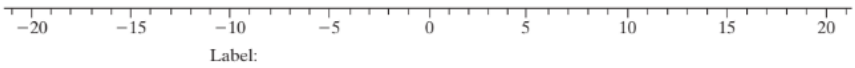
1. Shuffle the stack of 21 cards well and then randomly distribute cards into two stacks: one stack with 10 cards (the unrestricted group) and one with 11 (the sleep-deprived group).

2. Calculate and report the sample means for each rerandomized group:
   - Rerandomized unrestricted group's mean:
   - Rerandomized deprived group's mean:

[2] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

3. Calculate the difference in group means: unrestricted mean minus sleep-deprived mean. Report this value.

4. Combine this result with your classmates' to create a dotplot that shows the distribution of several possible values of difference in sample means that could have happened due to pure chance if sleep condition has no impact on improvement. Sketch a dotplot on an axis like the one shown being sure to label the horizontal axis.




At about what value is the dotplot centered? Explain why this makes sense. (Hint: What are we assuming to be true when we conduct the simulation?)Where is the observed difference in means from the original study (as reported in #8) on the dotplot? Did this value happen often, somewhat rarely, or very rarely? How are you deciding?

As before with simulation-based analyses, you would now like to conduct many, many more repetitions to determine what is typical and what is not for the difference in group means, assuming that sleep condition has no impact on improvement score. We think you would prefer to use a computer applet to do this rather than continue to shuffle cards for a very long time, calculating the difference of group means by hand.

**Exercise 7.2.10** Go back to the Multiple Means[3] applet, check the Show Shuffle Options box, select the Plot display, and press Shuffle Responses.Describe what the applet is doing and how this relates to your null hypothesis from #4.Record the simulated difference in sample means for the rerandomized groups, as given in the applet output. Is this difference more extreme than the observed difference from the study (as reported in #8)? How are you deciding?Click on Shuffle Responses again and record the simulated difference in sample means for the rerandomized groups. Did it change from #10(b)?Click on Shuffle Responses again and record the simulated difference in sample means for the rerandomized groups. Did it change from #10(b) and #10(c)?Now to see many more possible values of the difference in sample means, assuming sleep condition has no impact on improvement, use the Multiple Means applet:

- Change Number of Shuffles from 1 to 997.
- Press Shuffle Responses to produce a total of 1,000 shuffles and rerandomized statistics.Consider the histogram of the 1,000 could-have-been values of difference in sample means, assuming that sleep condition has no effect on improvement.What does one observation on the histogram represent? (Hint: Think about what you would have to do to put another observation on the graph.)Describe the overall shape of the null distribution displayed in this histogram.Where does the observed difference in sample means (as reported in #8) fall in this histogram: near the middle or out in a tail? Are there a lot of observations that are even more extreme than the observed difference, assuming sleep condition has no impact on improvement? How are you deciding?To estimate a p-value, continue with the Multiple Means applet.
- Type in the observed difference in group means (as reported in #8) in the Count Samples box (for the one-sided alternative hypothesis) and press Count.
- Record the approximate p-value.Fill in the blanks of the sentence to complete the interpretation of the p-value. The p-value of _____ is the probability of observing _____ _____ assuming _____.

## 3. Strength of evidence:

**Exercise 7.2.11** Based on the p-value, evaluate the strength of evidence provided by the experimental data against the null hypothesis that sleep condition has no effect on improvement score: not much evidence, moderate evidence, strong evidence, or very strong evidence?

**Exercise 7.2.12** Significance: Summarize your conclusion with regard to strength of evidence in the context of this study.

**Exercise 7.2.13**

**(a)** Use the 2SD method to approximate a 95% confidence interval for the difference in long-run mean improvement score for subjects who get unrestricted sleep minus the long-run mean improvement score for subjects who are sleep deprived. (Hints: Remember the observed value of the difference in group means and obtain the SD of the difference in group means from the applet's simulation results. The interval should be observed difference in means ±2SD, where SD represents the standard deviation of the null distribution of the difference in group means.)

**(b)** Interpret what this confidence interval reveals, paying particular atten-

[3] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

tion to whether the interval is entirely positive, entirely negative, or contains zero. (Hint: Be sure to convey "direction" in your interpretation by saying how much larger improvement scores are on average for the treatment you find to have the larger long-run mean: I'm 95% confident that the long-run mean improvement score is _____ to _____ higher with the _____ treatment as opposed to the _____ treatment.)

## STEP 5: Formulate conclusions.

**Exercise 7.2.14** Generalization: Were the participants in this study randomly selected from a larger population? Describe the population to which you would feel comfortable generalizing the results of this study.

**Exercise 7.2.15** Causation: Were the participants in the study randomly assigned to a sleep condition? How does this affect the scope of conclusion that you can draw?

## Exploring Further: Another Statistic

Could we have chosen a statistic other than the difference in group means to summarize how different the two groups' improvement scores were? Yes, for example we could have used the difference in group medians. Why might we do this? For one reason, the median is less affected by outliers than the mean.

To analyze the difference in group medians, we carry out the 3S strategy as before, except:

1. We calculate the observed value of the difference in medians as the statistic.
2. After we conduct the rerandomizing, we calculate the difference in medians for the rerandomized data. Then we repeat this process a large number of times.
3. We determine the p-value by counting how many of the simulated statistics are at least as large as the observed value of the difference in medians.

Return to the Multiple Means[4] applet and use the Statistic pull-down menu (on the left) to select Difference in Medians.

**Exercise 7.2.16** From the Summary Statistics for the original data, record the median improvement score for each group. Also record the difference between the medians (unrestricted median minus deprived median).

**Exercise 7.2.17** Enter 1000 for Number of Shuffles and press Shuffle Responses. Describe the resulting null distribution of difference in group medians. Does this null distribution appear to be centered near zero? Does it seem to have a bell-shaped distribution?

**Exercise 7.2.18** To calculate a p-value based on the difference in medians, enter the observed value in the Count samples box. Then press Count. Report both the value that you enter into the applet and the resulting p-value.

**Exercise 7.2.19** Does this p-value indicate strong evidence that sleep deprivation has a harmful effect on improvement score? Explain how you are deciding.

[4] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

**Exercise 7.2.20** With which statistic (difference in means or difference in medians) do the data provide stronger evidence that sleep deprivation has a harmful effect on improvement score? Explain how you are deciding.

## STEP 6: Look back and ahead.

**Exercise 7.2.21** Looking back: Did anything about the design and conclusions of this study concern you? Issues you may want to critique include: any mismatch between the research question and the study design, how the experimental units were selected, how the treatments were assigned to the experimental units, how the measurements were recorded, the number of experimental units in the study, and whether what we observed is of practical value. Looking ahead: What should the researchers' next steps be to fix the limitations or build on this knowledge?

## 7.3 Comparing two means: Theory-based approach

A worksheet for Grace College from Zybooks

### Close Friends

### Step 1: Ask a research question

How many close friends do you have? You know, the kind of people you like to talk to about important personal matters. Do men and women tend to differ on the number of close friends? And if so, by how much do men and women differ with regard to how many close friends they have?

### Step 2: Design a study and collect data

One of the questions asked of a random sample of adult Americans on the 2004 General Social Survey (GSS)[1] was: "From time to time, most people discuss important matters with other people. Looking back over the last six months—who are the people with whom you discussed matters important to you? Just tell me their first names or initials." The interviewer then recorded how many names each person gave, along with keeping track of the person's sex. The GSS is a survey of a representative sample of U.S. adults who are not institutionalized.

**Exercise 7.3.1** Identify the variables recorded. Also classify each as either categorical or quantitative, and identify each variable's role: explanatory or response.

**Exercise 7.3.2** Did this study make use of random assignment, random sampling, both, or neither?

**Exercise 7.3.3** Was this an experiment or an observational study? Explain how you are deciding.

**Exercise 7.3.4** In words, state the null and the alternative hypotheses to test whether American men and women differ with regard to how many friends they have.

**Exercise 7.3.5** Define the parameters of interest and assign symbols to them.

**Exercise 7.3.6** State the null and the alternative hypotheses in symbols.

### Step 3: Explore the data

The survey responses are summarized in the table below (and the data can be found in expanded form in the file CloseFriends[2]).

**Table 7.3.1 Number of close friends reported by sex**

| Number of close friends reported | 0 | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|---|
| Men | 196 | 135 | 108 | 100 | 42 | 40 | 33 | 654 |
| Women | 201 | 146 | 155 | 132 | 86 | 56 | 37 | 813 |

**Exercise 7.3.7** Now you will obtain numerical summaries (statistics, such as mean and SD) of the survey data using the Multiple Means applet[3].

- Open the data file CloseFriends[4] to access the raw data. Copy the data (e.g., CTRL-A and CTRL-C).

[1] gss.norc.org/

[2] static-content2.zybooks.com/isi/data/chap6/CloseFriends.txt

- Open the Multiple Means applet[5] and press *Clear*. Click inside the *Sample data* box and paste (e.g., CTRL-V). Then press *Use Data*.

**(a)** Report the sample size, sample mean, and sample SD of the number of close friends reported for each sex. Based on the sample statistics, who tends to have more close friends, on average: men or women? How are you deciding?

**(b)** Based on the sample statistics, who tends to have more variability with respect to how many close friends they have: men or women? How are you deciding?

**(c)** Based on the data as presented in the table, are the distributions of number of close friends symmetric or skewed? If skewed, in which direction?

**(d)** Calculate the observed difference in the mean number of close friends between these two groups (men - women). (Verify your calculation by checking the Observed diff output.)

**(e)** Notice that the value recorded in (d) is a negative number. Well, number of friends can only be 0 or more. Why is the number recorded in (d) less than 0?

## Step 4: Draw inferences beyond the data

The question we want to answer is, Is the difference in mean number of close friends between men and women, as seen in the GSS sample data, something that could plausibly have happened by chance, by random sampling alone, if, on average, men and women have the same number of close friends?

**Exercise 7.3.8** Using the same approach you did previously with the Multiple Means applet, you can use this applet to generate possible values of the difference in sample means under the null hypothesis by shuffling which response values go with which explanatory variable values. Check the *Show Shuffle Options* box and enter 1,000 for the *Number of Shuffles*. Press the *Shuffle Responses* button (there will be a pause; this is a large dataset!). The histogram on the right is the simulated null distribution for the difference in sample means. Enter the observed difference in sample means in the *Count Samples* box and press *Count*.

**(a)** Record your estimated p-value.

**(b)** Fill in the blanks of the sentence, to complete the interpretation of the p-value: The p-value of _____ is the probability of observing _____ assuming_____.

**Exercise 7.3.9** If the difference in means was larger (more different than 0), how would this impact the size of the p-value?

**Exercise 7.3.10** How would increasing the sample size (all else remaining the same) change the p-value? Why?

**Exercise 7.3.11** How would increasing the standard deviation of the number of close friends for both males and females (all else remaining the same) change

[3] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

[4] static-content2.zybooks.com/isi/data/chap6/CloseFriends.txt

[5] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

the p-value? Why?

As we saw earlier, another measure of the strength of evidence against the null hypothesis would be to standardize the statistic by subtracting the hypothesized value and dividing by the standard error of the statistic.

**Exercise 7.3.12** On the left side of the applet (below the Sample data window) use the Statistic pull- down menu to select the t-statistic (again, there will be a pause). Record the value of the t-statistic that is computed for your data. Write a one-sentence interpretation of this value.

Notice that the standardized statistic uses the letter $t$ instead of the $z$ we saw when testing proportions. This is because the theoretical distribution used is now a t-distribution instead of a normal distribution. These t-distributions are very similar to normal distributions especially when sample sizes are large. The t-statistic, like the z-statistic, tells us how many standard deviations our sample difference is above or below the mean and it can be judged in the same manner.

**Exercise 7.3.13** In light of the value of the standardized statistic, should you expect the p-value to be large or small? How are you deciding?

**Exercise 7.3.14** The histogram on the far right now displays the null distribution of simulated t-statistics. Describe the behavior of this distribution.

**Exercise 7.3.15** Below the null distribution check the box Overlay t distribution. Does the t-distribution appear to adequately predict the behavior of the shuffled t-statistics? What do you think this suggests about whether the validity conditions will be met for these data?

**Exercise 7.3.16** Enter the observed value of the t-statistic in the Count Samples box and press Count. (Remember to use the pull-down menu to specify what direction(s) you want to consider more extreme based on the alternative hypothesis.) How does the p-value from the t-distribution (in orange) compare to the simulation-based p-value (from Part 7.3.8.a, p. 115)?

## Validity Conditions

The validity conditions required for this theory-based approach (a "two-sample t-test") to be valid are shown in the box.

**Validity Conditions 7.3.2** The quantitative variable should have a symmetric distribution in both groups or you should have at least 20 observations in each group and the sample distributions should not be strongly skewed.

**Exercise 7.3.17** Do the validity conditions appear to be satisfied for these data? Justify your answer.

## Step 5: Formulate conclusions

**Exercise 7.3.18** Based on the analysis you have just carried out, state your conclusion in the context of the study. Be sure to comment on the following:

**(a)** Statistical significance: Do the data provide evidence that the mean number of close friends that men have is different from the mean number of friends women have? How are you deciding?

**(b)** Causation: Do the data provide evidence that how many friends one has is caused by one's sex? How are you deciding?

**(c)** Generalization: To whom can you apply the results of this study? All people? All adults? How are you deciding?

**Exercise 7.3.19** When you have selected the t-statistic and the theory-based overlay, you can also use this applet for estimating the parameter of interest. (You can also go directly to the Theory-Based Inference applet[6] which will also allow you to change the confidence level.) Check the box next to *95% CI(s) for difference in means*.

**(a)** Identify, in words related to the context of this study, the relevant parameter to be estimated here.

**(b)** Report the 95% confidence interval for this parameter.

**(c)** Does the 95% confidence interval calculated from the GSS sample data contain the value 0? What does that imply? (Hint: Recall that a confidence interval is an interval of plausible values for the parameter of interest, and the interval is calculated using the sample statistics.)

**(d)** Does the 95% confidence interval agree with your conclusion in Exercise 7.3.16, p. 116? How are you deciding?

**(e)** Fill in the blanks of the sentence to complete the interpretation of the 95% confidence interval:

We are _____% confident that the mean number of close friends men have is _____ than the mean number of close friends that women have, by between _____ and _____ friends.

**(f)** How would the width of the interval change if you increased the confidence level to 99%? Why?

**(g)** How would the width of the interval change if you increased the sample size? Why?

**(h)** How would the width of the interval change if the variability of the number of close friends increased for both males and females? Why?

## Step 6: Look back and ahead

**Exercise 7.3.20** Looking back: Did anything about the design and conclusions of this study concern you? Issues you may want to critique include: the match between the research question and the study design, how the observational units were selected, how the measurements were recorded, the number of observational units in the study, whether what we observed is of practical value.

**Exercise 7.3.21** Looking ahead: What should the researchers' next steps be to fix the limitations in this study and/or build on this knowledge?

[6] static-content.zybooks.com/applets/2021/tbia/TBIA.html?hideExtras=1

## 7.4 Data and Formulas

### Learning Goals

- *Compare two sample means numerically and graphically.*
- *Carry out a simulation-based analysis to investigate the difference between two population means.*
- *Carry out a theory-based analysis to investigate the difference between two population means.*

### Data

**Table 7.4.1 References**

| Section | Data | Reference |
|---|---|---|
| Example: Comparing two groups: Quantitative response | OldFaithful.csv[1] | GOSA (1978)[2]GOSA (2003)[3] |
| Supplemental Exploration: Comparing two groups: Quantitative response | | Short et al. (JSE 2017)[4] |
| Example: Comparing two means: Simulation-based approach | DungBeetles.csv[5] | Dacke et al. (Curr. Biol. 2013)[6] |
| Supplemental Exploration: Comparing two means: Simulation-based approach | Sleep.csv[7] | Stickgold et al.(Nat. Neurosci. 2000)[8] |
| Example: Comparing two means: Theory-based approach | VideogameAgression.csv[9] | Hollingdale and Greitemeyer (PLOS ONE 2014)[10] |
| Supplemental Exploration: Comparing two means: Theory-based approach | CloseFriends.csv[11] | General Social Survey (2004)[12] |
| Investigation: Memorizing letters | memorizing_letters.csv[13] | |

[1] static-resources.zybooks.com/ISI/Ch07/OldFaithful.csv
[2] www.geyserstudy.org/
[3] www.geyserstudy.org/
[4] doi.org/10.1080/10691898.1995.11910492
[5] static-resources.zybooks.com/ISI/Ch07/DungBeetles.csv
[6] www.doi.org/10.1016/j.cub.2012.12.034
[7] static-resources.zybooks.com/ISI/Ch07/Sleep.csv
[8] doi.org/10.1038/81756
[9] static-resources.zybooks.com/ISI/Ch07/VideogameAgression.csv
[10] dx.doi.org/10.1371%2Fjournal.pone.0111790
[11] static-resources.zybooks.com/ISI/Ch07/CloseFriends.csv
[12] gss.norc.org/
[13] static-resources.zybooks.com/ISI/Ch07/memorizing_letters.csv

## Formulas for two means

- $\mu_1$ and $\mu_2$ are the population means for the two groups.
- $n_1$ and $n_2$ are the sample sizes for the two groups.
- $\overline{x}_1$ and $\overline{x}_2$ are the sample means for the two groups.
- $s_1$ and $s_2$ are the sample standard deviations for the two groups.
- $SD(\overline{x}_1 - \overline{x}_2)$ is the standard deviation of the statistic.
- SE $= \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ is the theory-based standard error that approximates SD of the null distribution and $SD(\overline{x}_1 - \overline{x}_2)$.
- standardized statistic $= \frac{\text{statistic} - \text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
  - $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two independent means.
  - $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{SE} = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ is the theory-based standardized statistic for two independent means.
- $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two independent means.
- $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{SE} = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ is the theory-based standardized statistic for two independent means.
- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.
  - $(\overline{x}_1 - \overline{x}_2) \pm 2 \times SD(\overline{x}_1 - \overline{x}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_1 - \mu_2$.
  - $(\overline{x}_1 - \overline{x}_2) \pm (\text{multiplier}) \times (SE)$ is a theory-based confidence interval estimating $\mu_1 - \mu_2$, where the multiplier comes from a t distribution with $\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$ degrees of freedom.
- $(\overline{x}_1 - \overline{x}_2) \pm 2 \times SD(\overline{x}_1 - \overline{x}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_1 - \mu_2$.
- $(\overline{x}_1 - \overline{x}_2) \pm (\text{multiplier}) \times (SE)$ is a theory-based confidence interval estimating $\mu_1 - \mu_2$, where the multiplier comes from a t distribution with $\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$ degrees of freedom.

# Chapter 8

# Paired Data

# 8.1 Paired designs

## Rounding first base

Imagine you are at the plate in baseball and have hit a hard line drive. You want to try to stretch your hit from a single to a double. Does the path that you take to "round" first base make much of a difference? For example (see the figure below), is it better to take a "narrow angle" or a "wide angle" around first base? (This exploration is based on an actual study reported in a master's thesis by W. F. Woodward in 1970.[1])

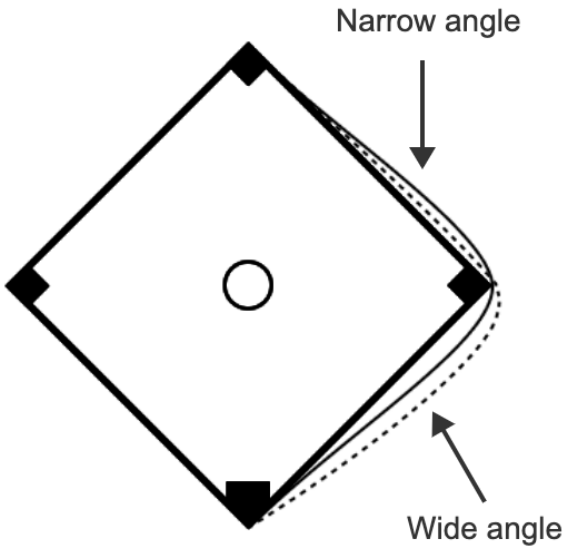



**Figure 8.1.1** Narrow angle versus wide angle

Think about designing a study to investigate this question.

**Exercise 8.1.1** Identify the explanatory and response variables in this study.

**Exercise 8.1.2** How would you design an observational study to investigate this question? Explain why an observational study would not allow you to decide which base-running angle is better than the other.

**Exercise 8.1.3** Suppose 20 baseball players volunteered to participate in an experiment. Suppose that you also plan to assign a single angle, either wide or narrow, to each player. How would you decide which player ran with which base-running angle?

A reasonable experimental design would be to randomly assign 10 of the 20 players to run with the wide angle and the other 10 to run with the narrow angle.

**Exercise 8.1.4** Some runners are faster than others. Explain how random assignment controls for this, so that speed is not likely to be a confounding variable in this study.

Even though random assignment tends to balance out other variables (such as speed) between the two groups, there's still a chance that most of the fast runners could be in one group and most of the slow runners in the other group. More importantly, there's likely to be a good bit of variability in the runners'

[1] fsu.catalog.fcla.edu/permalink.jsp?23FS027575641

speeds, and that variability would make it harder to spot a difference between the base-running angles even if one angle really is superior to the other.

**Exercise 8.1.5** Suggest a different way of conducting the experiment to make sure that speed is completely balanced between the two groups.

**Definition 8.1.2** For a **paired design**, response values come in pairs, with one response value in the pair for Group 1 and the other for Group 2. Sometimes the pairs come from matching individuals to create groups of two (we call this **paired design using matching**); sometimes the pairs come from measuring the same individual twice, once under each condition (we call this **paired design using repeated measures**). For an **independent groups design**, each individual in a group is unrelated to all the other individuals in the study. Each individual provides only one response value. ◊

In this study each runner can use both base-running angles. That way we can be sure that neither treatment has more of the fast or slow runners, and we can also expect that differences in times for each runner will show considerably less variability than individual running times.

**Exercise 8.1.6** What aspect of this experiment should be determined randomly? (Hint: The treatment is not determined randomly, because each runner experiences both treatments. But what other factor could still have an effect on the response unless it was randomized?)

**Exercise 8.1.7** What do you suggest using as the variable to be analyzed with this paired-design experiment? (Hint: Think of a better option than simply analyzing the set of times with the wide angle and the set of times with the narrow angle separately the way you would for an independent groups design of the sort described in #3 and #4.)

With a paired design, we analyze the differences in the response between the two treatments. In this case we would calculate the difference in running times between the wide and narrow angles for each player and then analyze the sample of differences. You will learn how to do this in future sections.

The order in which the players run with the two angles should be determined randomly; otherwise, the order could be a confounding variable. Perhaps players would be slower with their second angle because of fatigue or perhaps they would be faster with their second angle if they were slow to get warmed up. Randomizing the order takes away any worries about an order effect.

So far you have seen three designs for this study. The first (Exercise 8.1.2, p. 122) was observational. The second (Exercise 8.1.3, p. 122 and Exercise 8.1.4, p. 122) was a randomized experiment with independent groups. The third (Exercise 8.1.5, p. 123 and Exercise 8.1.6, p. 123) used a paired design with pairs created by repeated measures.

## Exploring Further

**Exercise 8.1.8** Consider a fourth design: Suppose you have 20 players, as before, and you have the time for each player in a 100-yard dash. Explain how you could use this information to create pairs of runners that you expect to be similar and how you would assign one runner in each pair to the narrow angle and the other to a wide angle.

**Exercise 8.1.9** Of the four designs (observational with independent groups, experimental with independent groups, paired design using repeated measures, and paired design using matching), which do you think is best for this context? Explain why. (Hint: Pairing works best when the units in a pair are as similar

to each other as possible.)

**Exercise 8.1.10** As noted in Exercise 8.1.9, p. 123, pairing works best when the two units in a pair are as similar as possible. When the units in a pair are not similar, pairing will not be effective. Invent and describe an example where you think pairing will be effective. Give the response and explanatory variable, how you would create the pairs, and why you think pairing will be effective.

**Exercise 8.1.11** Invent and describe an example where you think pairing will not be effective. Again, give the response and explanatory variable, how you would create the pairs, and this time why you think pairing will not be effective.

**Exercise 8.1.12** For some contexts, the strategy of using repeated measures to create pairs is either impossible or a bad idea. Invent and describe such a context.

**Exercise 8.1.13** Pairing can also be used in the design of an observational study. Revisit the observational study in #2, and suppose also that you have times in the hundred yard dash for each of the 20 players. Explain how you could use those times to create pairs. (Suppose that, of the 20 players, 12 used the wide angle and only 8 used the narrow angle. This means you can create only 8 pairs.)

## 8.2 Simulation-based approach for analyzing paired data

### Exercise and heart rate

Raising your heart rate is one of the major goals of exercise. Do some exercises raise your heart rate more than others? Which common exercise, jumping jacks or bicycle kicks, raises your heart rate more? In this exploration, you are going to compare heart rates of you and your classmates doing jumping jacks and bicycle kicks. If you don't want to collect your own data, you may use data already collected on a class where jumping jacks were compared to bicycle kicks. The dataset is JJvsBicycle[1]. Below are links to YouTube videos explaining how to do each of these exercises.

- Jumping jack: http://www.youtube.com/watch?v=dmYwZH_BNd0[2]
- Bicycle kicks: https://www.youtube.com/watch?v=9FGilxCbdz8[3]

**Exercise 8.2.1** You will first randomize the order (jumping jacks first or bicycle kicks first) in which you measure your heart rate. If heads:

- Do jumping jacks for 30 seconds. Then, measure your jumping jack heart rate.
- Sit down and take a break for two minutes.
- Now do bicycle kicks for 30 seconds. Then, measure your bicycle kick heart rate.

If tails:

- Now do bicycle kicks for 30 seconds. Then, measure your bicycle kick heart rate.
- Sit down and take a break for two minutes.
- Do jumping jacks for 30 seconds. Then, measure your jumping jack heart rate.

**(a)** Report your heart rate for bicycle kicks and jumping jacks in beats per minute.

**(b)** Find the difference in your two heart rates (jumping jack heart rate-bicycle kick heart rate) in beats per minute.

Your instructor will give you and your classmates instructions on where to record your jumping jack heart rate, bicycle kick heart rate, and difference (jumping jacks-bicycle kick).

**Exercise 8.2.2** Explain why it is reasonable to say that the two heart rates you collected should not be treated as independent data.

**Exercise 8.2.3** Why do you think we randomized the order of jumping jacks and bicycle kicks before measuring heart rates?

As seen in previous chapters, we can summarize the quantitative data on heart rates using averages or means. But, because the data are paired, instead of

[1] static-content2.zybooks.com/isi/data/chap7/JJvsBicycle.txt
[2] www.youtube.com/watch?v=dmYwZH_BNd0
[3] www.youtube.com/watch?v=9FGilxCbdz8

comparing mean jumping jack heart rate to mean bicycle kick heart rate, we will instead look at the mean difference in heart rate between doing jumping jacks and bicycle kicks. Thus, we can define our parameter of interest to be

**Exercise 8.2.4** State the null and alternative hypotheses to test whether the mean difference in heart rate between the two exercises is not 0.

**Key Idea 8.2.1** When the parameter of interest is the long-run mean difference or population mean difference, the corresponding statistic is the sample mean difference.

**Exercise 8.2.5** Find the average of the differences between the two heart rates for the entire class. This is the statistic we will use to summarize the data.

Your null hypothesis should essentially state that there is no difference in the heart rates between the two exercises, on average. If that is the case, it doesn't really matter if we swap someone's jumping jack heart rate with his or her bicycle kick heart rate. This is how we will model the null to develop a null distribution. To randomly swap some of the values we can just use a coin flip. If the coin lands heads you will swap the two heart rates. If the coin lands tails you won't swap the heart rates.

**Exercise 8.2.6** Flip a coin for each pair of heart rates and switch the appropriate ones. Recalculate the difference in heart rates and find the new simulated mean difference. Plot this value on the board in the classroom along with those from the rest of the class. Where does the actual statistic you found in question 5 fit in this null distribution? Is it out in the tail?

**Exercise 8.2.7** As you know, it would be better to have many more simulations than what your class just did. We will do this by using an applet.

- Go to the Matched Pairs applet[4].
- Press *Clear* to erase the default data and then copy and paste the JJvsBicycle[5] data (both columns— jacks and kicks) into the *Data* window. Then press *Use Data*.
- Notice that the applet graphs the individual heart rates in each group, along with the means and standard deviations for the two groups.
- Below that, the applet provides a dotplot of the differences in the heart rates in the sample. Please note that these difference values can be negative numbers because you are looking at *change or difference* in heart rates. The "Differences" graph also shows the mean of the differences and the standard deviation of the differences.
- Write down these values in the following table:

**Table 8.2.2**

| **Condition** | **Sample mean $\overline{x}$** | **Sample standard deviation $s$** |
|---|---|---|
| **Jumping jacks** | $\overline{x}_{jj} =$ | $s_{jj} =$ |
| **Bicycle kicks** | $\overline{x}_{bicycle} =$ | $s_{bicycle} =$ |
| **Diff = JJ - BK** | $\overline{x}_d =$ | $s_d =$ |

**Exercise 8.2.8** The Matched Pairs applet will perform the simulation similar to what you did with flipping a coin.

- Check the *Randomize* box.

[4] static-content.zybooks.com/applets/2021/matchedpairs/MatchedPairs.htm?hideExtras=1

[5] static-content2.zybooks.com/isi/data/chap7/JJvsBicycle.txt

- Set the number of times to *Randomize* to 1 and press *Randomize*.
- Once the coin tosses have determined which heart rate will be in which column, the applet displays the rerandomized data (the colors show you the original column for each observation, so you should see a mix in each group now).
- This could-have-been value for the mean difference is added to the Average Difference graph.
- Write down these could-have-been values for the re-randomized data:

**Table 8.2.3**

| **Condition** | **Rerandomized sample mean $\overline{x}$** | **Rerandomized sample standard deviation $s$** |
|---|---|---|
| **Jumping jacks** | $\overline{x}_{jj} =$ | $s_{jj} =$ |
| **Bicycle kicks** | $\overline{x}_{bicycle} =$ | $s_{bicycle} =$ |
| **Diff = JJ - BK** | $\overline{x}_d =$ | $s_d =$ |

**(a)** How does the actual mean difference compare to your simulated mean difference: More extreme/Less extreme/Similar.

**(b)** How are you deciding?

**Exercise 8.2.9** Update the number of times to *Randomize* to 99 (for a total of 100 repetitions), and press *Randomize*. Consider the graph "Average Difference" that the applet has created

**(a)** How many dots are in this graph?

**(b)** What does each dot represent?

The table below summarizes the key aspects of the simulation:

**Exercise 8.2.10** To see many more possible values of mean difference in sample means that could have been, If jumping jacks and bicycle kick rates were swappable, update the number of times to *Randomize* to 900 and press *Randomize* (for a total of 1,000 repetitions). Describe the updated "Average Difference" graph with the 1,000 samples or repetitions, with regard to the following features.

**(a)** Shape:

**(b)** About what number is this graph centered? Explain why you were expecting this.

**(c)** This graph also reports a value for standard deviation, SD. Report this value and give a simple interpretation of this value, as in, "What is this value measuring?"

**Exercise 8.2.11** You now should have generated 1,000 possible values of the mean difference in jumping jacks and bicycle kick heart rates that were simulated assuming the null hypothesis was true and these rates were the same, on average. How does the observed mean difference from your data (as reported in question 5 and question 7) compare to these simulated values? Is an average difference in heart rates like that observed in the actual study unlikely to happen by chance alone if jumping jacks and bicycle kick heart rates are the same, on average? How are you deciding?

**Exercise 8.2.12** To quantify the strength of evidence against the null hypothesis, you can find the p-value. Go back to the Matched Pairs applet. In the

Count Samples box, make an appropriate selection from the drop-down menu. (Hint: In what direction does your alternative hypothesis look?) and enter the appropriate number in the box (Hint: At least as extreme as what number?). Report the approximate p-value.

**Exercise 8.2.13** Use the p-value to state a conclusion in the context of the problem. Be sure to comment on statistical significance. Can you conclude that there is strong evidence that jumping jack heart rate and bicycle kick heart rate differ? Why or why not? Can you draw a cause-and-effect conclusion? To what population are you willing to generalize the results?

**Exercise 8.2.14** Alternatively, you can summarize the strength of evidence using a standardized statistic. Find the standardized statistic and confirm that the strength of evidence you receive from the p-value is approximately the same as with the standardized statistic.

**Exercise 8.2.15** We can again use the 2SD method to approximate a 95% confidence interval for the mean difference in heart rates between those that do the two exercises. The overall structure of the formula is the same: statistic $\pm$ 2 (SD), where the statistic is the sample mean difference in heart rates for your class and SD is the standard deviation of your null distribution when you did 1,000 repetitions in the applet (not the standard deviation from the data). Use these numbers to find an approximate 95% confidence interval for the mean difference in heart rates for those that do the two exercises your class did.

**Exercise 8.2.16** Provide an interpretation of this confidence interval, being sure to explain the parameter in this context.

As you may have already noticed, the strategy we used to find the p-value for this study is the same 3S strategy that is found in Step 4 of our statistical investigation method and has been used in analyses involving one or two groups:

- **Statistic**: Compute the statistic in the sample. In this case, the statistic you looked at was the observed mean difference in heart rates.
- **Simulate**: Identify a chance model that reflects the null hypothesis. To simulate what could have been if the null hypothesis is true, you can toss a coin for each student, and if it lands heads, swap the two heart rates recorded for that student. If the coin lands tails, do not swap the heart rates. Repeat this process 1,000 times, recording the mean difference in heart rates each time and thus obtaining a distribution of these mean differences that were simulated assuming the null hypothesis were true.
- **Strength of evidence**: If your actual observed statistic falls in the tail of the null distribution, then you have strong evidence that there is a genuine difference in the average heart rates between the two exercises.

Let's check out how things would have worked had we ignored the pairing and analyzed the data as if the jumping jacks heart rates and bicycle kicks heart rates had come from two totally different samples that were independent of each other.

**Exercise 8.2.17** Go to the Multiple Means applet[6] and analyze the data as though we have two independent samples, as you did in a previous chapter. (The heart rate data are unstacked, meaning the heart rates for the two exercises are given in two columns. Make sure you have the unstacked box checked before you paste in your data.)

**(a)** With regard to the graph of the distribution of "Shuffled Differences in

Means": What are the mean and SD?

**(b)** Find and report the approximate p-value.

**Exercise 8.2.18** Compare the SD of the null distribution obtained using the two-independent-samples method to that obtained using the paired samples method. Which SD is larger?

**Exercise 8.2.19** Compare the p-value obtained using the two-independent-samples method to that obtained using the paired samples method. Which p-value is smaller and hence provides stronger evidence against the null hypothesis of no difference?

Note: Using a paired samples method will often give a smaller p-value and hence stronger evidence against the null hypothesis than the two-independent-samples method. Perhaps this is what you found in Question 19. This would happen if students with high heart rates after doing jumping jacks tended to have high heart rates after doing bicycles, and students with low heart rates after doing jumping jacks tended to have low heart rates after doing bicycles. This would make the variability of the differences in heart rates small. However this may not have happened for students in your class. If the variability within your jumping jack data or within your bicycle data is small but there is a lot of variability in the differences, you could get a smaller p-value using the two-independent-samples method.

[6] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

## 8.3 Theory-based approach for analyzing paired data

### Comparing auction formats

An economist at Vanderbilt University devised a study[1] to compare different types of online auctions. In one experiment he compared a Dutch auction to a first-price sealed bid auction. In the Dutch auction the item for sale starts at a very high price and is lowered gradually until someone finds the price low enough to buy. In the first-price sealed bid auction each bidder submits a single sealed bid before a particular deadline. After the deadline, the person with the highest bid wins. The researcher auctioned off collectible trading cards from the game Magic: The Gathering. He placed pairs of identical cards up for auction; one would go into the Dutch auction and the other to the first-price sealed bid auction. He then looked at the difference in the prices he received on the pair. He repeated this for a total of 88 pairs.

**Exercise 8.3.1** Before we look at the data that were collected and start the analysis, let's make sure you understand the study design.

**(a)** Explain why the price data should be analyzed using paired samples as opposed to two independent samples.

**(b)** What makes a pair?

**(c)** What is the explanatory variable? Is it categorical or quantitative?

**(d)** What is the response variable? Is it categorical or quantitative?

**Exercise 8.3.2** State the relevant hypotheses in words. (Use a two-sided alternative.)

**Exercise 8.3.3** Define the parameter of interest and give the symbol that should be assigned to it.

**Exercise 8.3.4** State the relevant hypotheses in symbols.

**Exercise 8.3.5** The data for the auction can be found in the file Auction[2]. The selling prices for the Dutch auction are labeled Dutch and the prices for first-price sealed bid auction are labeled FP. Paste these data into the Matched Pairs applet[3].

**(a)** What is the sample mean price for the Dutch auction?

**(b)** What is the sample mean price for the first-price sealed bid auction?

**(c)** What are the mean and standard deviation for the difference in the two prices?

**(d)** Determine a p-value using the applet. Explain how you did so.

Notice that the simulated null distribution is bell-shaped. This is no coincidence. A theory-based method exists that predicts this to occur when certain validity conditions are met.

---

[1] doi.org/10.1257/aer.89.5.1063

[2] static-content2.zybooks.com/isi/data/chap7/Auction.txt

[3] static-content.zybooks.com/applets/2021/matchedpairs/MatchedPairs.htm?hideExtras=1

**Validity Conditions 8.3.1** Validity conditions for theory-based analysis of paired data: Theory-based methods of inference will work well for paired data if the distribution of differences has a symmetric distribution, or you have at least 20 pairs (i.e., at least 20 differences) and the distribution of the sample differences is not strongly skewed. This test is known as **a paired t-test**.

**Exercise 8.3.6** Are the validity conditions met for these data? Explain.

**Exercise 8.3.7** Because the sample is large enough without strong skewness in the distribution of differences, we can use a theory-based approach. We will do this by using the Theory-Based Inference applet[4]. To do this:

- Open the Theory-Based Inference applet[5].
- Choose *One mean* from the pull-down menu.
- Enter the sample size, sample mean, and sample standard deviation for the differences as you found in question 5.
- Check the box for *Test of Significance*.
- Enter the appropriate information for the hypotheses.
- Make sure the appropriate sign for the alternative hypotheses is chosen.
- Press *Calculate*.

**(a)** What is the value of the standardized statistic? What does that number tell you?

**(b)** In the light of the value of the standardized statistic, do you expect the p-value to be small or not small? How are you deciding?

**(c)** What is the value of the p-value? Is it similar to the p-value found using the simulation-based method?

**Exercise 8.3.8** Recall that the Theory-Based Inference applet[6] can also produce confidence intervals for the parameter of interest.

- Go to the applet, check the *Confidence interval* box and let the confidence level be 95%.
- Press *Calculate CI*.

**(a)** Report the 95% confidence interval.

**(b)** Interpret the 95% confidence interval in context.

**(c)** Does the 95% confidence interval agree with your conclusion when using the p-value? How are you deciding?

## Exploring Further

**Exercise 8.3.9** Based on the analysis in Exercise 8.3.7, p. 131 and Exercise 8.3.8, p. 131, state your conclusion in the context of the study. Be sure to comment on:

- Statistical significance: Do the data provide evidence that selling cards in

[4] static-content.zybooks.com/applets/2021/tbia/TBIA.html?hideExtras=1
[5] static-content.zybooks.com/applets/2021/tbia/TBIA.html?hideExtras=1
[6] static-content.zybooks.com/applets/2021/tbia/TBIA.html?hideExtras=1

a Dutch auction differ than when sold in a first-price sealed bid auction on average? How are you deciding?

- Estimation: Find and interpret a 95% confidence interval.
- Causation: Can you conclude causation? If yes, what causes what? If not, how are you deciding?
- Generalization: Can you extend the results of this study? Other kinds of cards? Other types of items? Anything sold in an auction format on the Internet? How are you deciding?

# 8.4 Data and Formulas

## Learning Goals

- *Distinguish between the following study designs for comparing two groups on a quantitative response: observational, randomly assigned independent groups, matched pairs, repeated measures.*
- *Perform and interpret a simulation-based analysis of paired data for a quantitative response.*
- *Perform and interpret a theory-based analysis of paired data for a quantitative response.*

## Data

**Table 8.4.1 References**

| Section | Data |
|---|---|
| Example: Paired designs | |
| Supplemental Exploration: Paired designs | FirstBase.csv[2] |
| Example: Simulation-based approach for analyzing paired data | FirstBase.csv[4] |
| Supplemental Exploration: Simulation-based approach for analyzing paired data | JJvsBicycle.csv[6] |
| Example: Theory-based approach for analyzing paired data | DadJokes.csv[7] |
| Supplemental Exploration: Theory-based approach for analyzing paired data | Auction.csv[9] |
| Investigation: Filtering water in Cameroon | Ecoli_time.csv[11] Ecoli_sand.c |

## Formulas for two means

Independent samples:

- $\mu_1$ and $\mu_2$ are the population means for the two groups.
- $n_1$ and $n_2$ are the sample sizes for the two groups.
- $\overline{x}_1$ and $\overline{x}_2$ are the sample means for the two groups.
- $s_1$ and $s_2$ are the sample standard deviations for the two groups.
- $SD(\overline{x}_1 - \overline{x}_2)$ is the standard deviation of the statistic.
- SE $= \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ is the theory-based standard error that approximates SD of the null distribution and $SD(\overline{x}_1 - \overline{x}_2)$.

[1] www.washingtonpost.com/people/valerie-strauss/
[2] static-resources.zybooks.com/ISI/Ch08/FirstBase.csv
[3] fsu.catalog.fcla.edu/permalink.jsp?23FS027575641
[4] static-resources.zybooks.com/ISI/Ch08/FirstBase.csv
[5] fsu.catalog.fcla.edu/permalink.jsp?23FS027575641
[6] static-resources.zybooks.com/ISI/Ch08/JJvsBicycle.csv
[7] static-resources.zybooks.com/ISI/Ch08/DadJokes.csv
[8] doi.org/10.1016/j.cub.2019.05.073
[9] static-resources.zybooks.com/ISI/Ch08/Auction.csv
[10] doi.org/10.1257/aer.89.5.1063
[11] static-resources.zybooks.com/ISI/Ch08/Ecoli_time.csv
[12] static-resources.zybooks.com/ISI/Ch08/Ecoli_sand.csv
[13] static-resources.zybooks.com/ISI/Ch08/Ecoli_flow.csv
[14] hope.edu/news/2007/04/17/hope-project-in-cameroon-designed-to-save-lives.html

- standardized statistic $= \frac{\text{statistic}-\text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
    - $t = \frac{\left(\overline{x}_1-\overline{x}_2\right)-0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two independent means.
    - $t = \frac{\left(\overline{x}_1-\overline{x}_2\right)-0}{SE} = \frac{\left(\overline{x}_1-\overline{x}_2\right)-0}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$ is the theory-based standardized statistic for two independent means.

- $t = \frac{\left(\overline{x}_1-\overline{x}_2\right)-0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two independent means.

- $t = \frac{\left(\overline{x}_1-\overline{x}_2\right)-0}{SE} = \frac{\left(\overline{x}_1-\overline{x}_2\right)-0}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$ is the theory-based standardized statistic for two independent means.

- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.
    - $(\overline{x}_1 - \overline{x}_2) \pm 2 \times SD(\overline{x}_1 - \overline{x}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_1 - \mu_2$.
    - $(\overline{x}_1 - \overline{x}_2) \pm (\text{multiplier}) \times (SE)$ is a theory-based confidence interval estimating $\mu_1 - \mu_2$, where the multiplier comes from a t distribution with $\frac{\left(\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{s_1^2}{n_1}\right)^2+\frac{1}{n_2-1}\left(\frac{s_2^2}{n_2}\right)^2}$ degrees of freedom.

- $(\overline{x}_1 - \overline{x}_2) \pm 2 \times SD(\overline{x}_1 - \overline{x}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_1 - \mu_2$.

- $(\overline{x}_1 - \overline{x}_2) \pm (\text{multiplier}) \times (SE)$ is a theory-based confidence interval estimating $\mu_1 - \mu_2$, where the multiplier comes from a t distribution with $\frac{\left(\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{s_1^2}{n_1}\right)^2+\frac{1}{n_2-1}\left(\frac{s_2^2}{n_2}\right)^2}$ degrees of freedom.

Matched pairs:

- $d = (group\ 1) - (group\ 2)$ is the variable of differences for each matched pair.

- $\mu_d$ is the population mean difference.

- $n$ is the sample size (number of matched pairs).

- $\overline{x}_d$ is the sample mean of the differences.

- $s_d$ is the sample standard deviation of the differences.

- $SD(\overline{x}_d)$ is the standard deviation of the statistic.

- SE $= s_d/\sqrt{n}$ is the theory-based standard error that approximates SD of the null distribution and $SD(\overline{x}_d)$.

- standardized statistic $= \frac{\text{statistic}-\text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.

  - $t = \frac{\overline{x}_d - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for matched pairs.
  - $t = \frac{\overline{x}_d - 0}{SE} = \frac{\overline{x}_d - 0}{s_d/\sqrt{n}}$ is the theory-based standardized statistic for matched pairs.

- $t = \frac{\overline{x}_d - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for matched pairs.
- $t = \frac{\overline{x}_d - 0}{SE} = \frac{\overline{x}_d - 0}{s_d/\sqrt{n}}$ is the theory-based standardized statistic for matched pairs.
- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.
  - $\overline{x}_d \pm 2 \times SD(\overline{x}_d)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_d$.
  - $\overline{x}_d \pm$ (multiplier) $\times$ $(SE)$ is a theory-based confidence interval estimating $\mu_d$, where the multiplier comes from a t distribution with $n - 1$ degrees of freedom.
- $\overline{x}_d \pm 2 \times SD(\overline{x}_d)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_d$.
- $\overline{x}_d \pm$ (multiplier) $\times$ $(SE)$ is a theory-based confidence interval estimating $\mu_d$, where the multiplier comes from a t distribution with $n - 1$ degrees of freedom.

# Chapter 9

# Comparing More Than Two Proportions

# 9.1 Comparing multiple proportions: Simulation-based approach

### Recruiting organ donors

### Step 1: Ask a research question

Previously we considered a study[1] that investigated how to encourage people to be more likely to become organ donors. At the time, we did not formally analyze the data from the study. Now we are studying methods for comparing three groups, so we can further analyze the data and draw inferential conclusions from this study.

### Step 2: Design a study and collect data

Recall that researchers asked volunteer subjects to imagine they moved to a new state, applied for a driver's license, and needed to decide whether to become an organ donor. Researchers created three different versions of the survey, each with a different donor recruiting strategy. Subjects were randomly assigned to one of the three types of recruiting strategies:

- The default option is to be an organ donor, and individuals have to opt out of organ donation (opt-out group)
- The default option is to not be an organ donor, and individuals have to opt in to organ donation (opt-in group)
- There is no default option, and individuals have to choose whether they will be an organ donor (neutral group)

**Exercise 9.1.1** What are the observational units? What are the variables? Are the variables categorical or quantitative? If categorical, how many categories do they have? Also, identify the roles (explanatory or response) of the variables.

**Exercise 9.1.2** Did this study make use of random sampling, random assignment, both, or neither? Also describe the implication of your answer in terms of scope of conclusions that can potentially be drawn from this study.

**Exercise 9.1.3** Did this study pair subjects in one group with subjects in another group or are the subjects in the different recruiting groups independent of each other?

**Exercise 9.1.4** Let's write out the hypotheses.

**(a)** Write the appropriate null and alternative hypotheses (in words) using the language of association between the explanatory and response variables.

**(b)** Express the null hypothesis in symbols. (Hint: Remember that hypotheses are always about parameters, so define and use appropriate symbols for parameters.)

[1] papers.ssrn.com/sol3/papers.cfm?abstract_id=1324774

## Step 3: Explore the data

**Exercise 9.1.5** Researchers found that 23 of 55 subjects in the opt-in group chose to be organ donors, 41 of 50 in the opt-out group chose to be organ donors, and 44 of 56 in the neutral group chose to be organ donors.

**(a)** Based on these counts, produce a two-way table of counts, putting the explanatory variable in columns. (Hint: You have seen this table earlier!)

**(b)** For each of the three groups, calculate the conditional proportion who agreed to become an organ donor.

**(c)** Produce a segmented bar graph (by hand or with the Multiple Proportions applet[2]) to compare these conditional proportions across the three groups.

**(d)** Comment on what your calculations and graph reveal about whether the default option used in the question appears to affect how likely the person is to become an organ donor.

## Step 4: Draw inferences beyond the data

We see some differences in the sample proportions of individuals who chose to be organ donors across the three treatments, but are these differences large enough to be statistically significant? In other words, do these data provide strong evidence of an association between donor recruiting strategy (opt-in, opt-out, neutral) and choosing to become an organ donor in the population?

**Exercise 9.1.6** Suggest two possible explanations for how these observed proportions could have turned out to be as different from each other as they are. (Hint: Think about what our null and alternative hypotheses are.)

## Applying the 3S Strategy

To investigate, we will see how we can apply the 3S strategy to these data. Because researchers are attempting to minimize (or at least control) the probability of Type I errors, in cases with multiple groups they first do an overall test. However, this requires computing a single statistic and considering the null distribution of that statistic. Let's look at this approach now.

**Exercise 9.1.7** Statistic: What statistic would you propose?

**(a)** Propose a formula for a statistic that could be used to measure how different the three sample proportions are. (Hint: The key here is that the statistic needs to provide a single number that somehow focuses on differences between/among the observed sample proportions.)

**(b)** For the statistic you propose, would a large value or small value give evidence for the alternative hypothesis that at least one recruiting strategy has a different probability of someone agreeing to be a donor?

When we compared two proportions, we used the difference in two proportions as the statistic. Here there are multiple proportions. There are actually several reasonable ways of summarizing the differences among the groups in one number. Once you have settled on a statistic, you apply the 3S strategy

[2] static-content.zybooks.com/applets/2021/chisqshuffle/ChiSqShuffle.htm?hideExtras=1

as before—simulate the distribution of the statistic under the null hypothesis and then see where the observed value of the statistic falls in that distribution.

**Exercise 9.1.8** One possible statistic is $\hat{p}$ (max) - $\hat{p}$ (min). Review the conditional proportions you determined in question 5b and calculate the observed value of this statistic for these data. Why might this be a less than satisfactory choice?

However, this statistic ignores some of the groups. Another reasonable statistic to calculate is the mean of the absolute values of group differences for each pair of conditional proportions. (We could call this statistic **Mean Group Diff** for the mean of group differences.)

**Exercise 9.1.9** Let's construct the Mean Group Diff statistic for these data by going through the following steps.

**(a)** Review the conditional proportions of organ donors you determined in question 5b and calculate the group differences in these proportions for each pair: Opt-out minus opt-in, opt-out minus neutral, and opt-in minus neutral.

**(b)** Calculate the mean of the absolute values of these group differences.

So which statistic should you use? And how can you use the statistic to estimate a p-value? We'll consider those questions later.

Simulation: We first saw how we can simulate a null hypothesis of no association by shuffling the response variable outcomes across the explanatory variable groups. This models the random assignment process used in the data collection process, and, assuming the null hypothesis is true, "breaks any potential association" between the response and explanatory variables. We modeled this with playing cards previously.

**Exercise 9.1.10** Describe how you would model a simulation of the null hypothesis with red and black playing cards now that there are three explanatory variable groups instead of two. Make sure to tell how many black cards you would need and how many red ones. Also tell what each color of cards would represent. How many cards would you deal out to each group? What would you calculate after dealing them out?

**Exercise 9.1.11** Now let's use the Multiple Proportions applet[3] to simulate a null distribution of these Mean Group Diff statistics. This null distribution is again used to measure the strength of evidence provided by the observed statistic against the null hypothesis. Open the Organ Donor data file[4] and paste the two columns (with column names) into the applet and press *Use Data*. (Or enter/paste in the two-way table of counts in the *Sample data* window and press *Use Table*.) Check the *Show table* box and verify that the sample proportions of organ donors match your answers from #5(b) and the observed Mean Group Diff statistic shown matches your answer from #9(b). If you cannot see the whole table, click on the lower right corner and drag it to make the table larger.

**(a)** Now check Show Shuffle Options and leave Number of Shuffles at 1. Press Shuffle Response to perform one shuffle of the response variable values. You should see the shuffled response variable, the new two-way table for the simulated data, and the value of the simulated statistic (Most Recent Shuffled Mean Group Diff) which is also placed in blue on the graph on the right. Select the Plot radio button to see the shuffled segmented bar graph. How does the distribution across the groups for the shuffled data compare to the original data? Is the simulated Mean Group Diff

statistic value closer to zero than the observed value of the Mean Group Diff statistic?

**(b)** Now enter 999 for the Number of Shuffles and press Shuffle Responses. This repeats the shuffling of the response variable 999 more times for a total of 1000 repetitions. You should see a graph of a null distribution of the Mean Group Diff statistics. What is the shape of this distribution? Why is it not a bell-shaped curve centered at zero?

**Exercise 9.1.12** Strength of evidence: To estimate a p-value, determine how often the observed value of the statistic, or something even larger, occurred in the null distribution. (Hint: Enter the observed value for the Mean Group Diff statistic from the research study in the Count Samples box and press Count.)

**Exercise 9.1.13** Based on the p-value, summarize the strength of evidence that the sample data provide against the null hypothesis.

## Step 5: Formulate conclusions

**Exercise 9.1.14** Generalization: Comment on how the sample was obtained and to which population your conclusion can be generalized.

**Exercise 9.1.15** Causation: Comment on how the study was designed and whether a cause-and-effect conclusion is warranted from this study.

## Step 6: Look back and ahead

**Exercise 9.1.16** Summarize your conclusion for the researchers. Do you have any concerns about the study design? Any comments on sample size? Are there other limitations that you feel need to be addressed? Are there specific improvements you would suggest to make this study better? Based on what you concluded from this study, can you suggest one or two follow-up studies that could be carried out?

[3] `static-content.zybooks.com/applets/2021/chisqshuffle/ChiSqShuffle.htm?hideExtras=1`

[4] `static-content2.zybooks.com/isi/data/prelim/OrganDonor.txt`

## 9.2 Comparing multiple proportions: Theory-based approach (Part A)

### Conserving hotel towels

### Step 1: Ask a research question

If you have stayed at a hotel recently, you may have noticed how guests are increasingly encouraged to practice conservation habits by not having their towels washed every day—instead hanging them back on the rack to be reused. Are some "encouragement strategies" more effective than others?

### Step 2: Design a study and collect data

A recent study (Goldstein et al., 2008[1]) conducted a randomized experiment to investigate how different phrasings on signs placed on bathroom towel racks impacted guests' towel reuse behavior. In particular, the researchers were interested in evaluating how messages that communicated different types of social norms impacted towel reuse. One week prior to a guest staying in the room, rooms at a particular hotel were randomly assigned to receive one of the following five messages on a sign hung on the towel bar in the room:

**Message 1:** HELP SAVE THE ENVIRONMENT. You can show your respect for nature and help save the environment by reusing your towels during your stay. (No social norm)

**Message 2:** JOIN YOUR FELLOW GUESTS IN HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 75% of the guests participated in our new resource savings program by using their towels more than once. You can join your fellow guests in this program to help save the environment by reusing your towels during your stay. (Guest identity norm)

**Message 3:** JOIN YOUR FELLOW GUESTS IN HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 75% of the guests who stayed in this room (#xxx) participated in our new resource savings program by using their towels more than once. You can join your fellow guests in this program to help save the environment by reusing your towels during your stay. (Same room norm)

**Message 4:** JOIN YOUR FELLOW CITIZENS IN HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 75% of the guests participated in our new resource savings program by using their towels more than once. You can join your fellow citizens in this program to help save the environment by reusing your towels during your stay. (Citizens norm)

[1] doi.org/10.1086/586910

**Message 5:** JOIN THE MEN AND WOMEN WHO ARE HELPING TO SAVE THE ENVIRONMENT. In a study conducted in fall 2003, 76% of the women and 74% of the men participated in our new resource savings program by using their towels more than once. You can join the other men and women in this program to help save the environment by reusing your towels during your stay." (Gender identity norm)

Data were collected on 1,595 instances of potential towel reuse. For each of the 1,595 instances, room attendants recorded whether the hotel guest reused their towels.

**Exercise 9.2.1** In your own words, state the research question of interest in this study.

**Exercise 9.2.2** Identify the explanatory and response variables in this study, along with their types (categorical or quantitative). For the explanatory variable, indicate how many categories it involves.

**Exercise 9.2.3** Did the study design make use of any pairing of subjects or are the subjects who receive the different signs independent of each other?

**Exercise 9.2.4** State the appropriate null and alternative hypotheses for testing the researchers' conjecture using both words and symbols.

## Step 3: Explore the data

**Exercise 9.2.5** Paste the two-way table from the Towels[2] file into the Multiple Proportions applet[3], then in the *Sample Data* box and press *Use Table*.

**(a)** Check the box next to Show Table in order to present the two-way table of counts and calculate conditional proportions of those who reused towels. (You can grab the bottom right corner of the table in the applet to extend it.) Report these five proportions.

**(b)** Looking at the segmented bar graph as well as the conditional proportions, conjecture on whether you think there will be a significant difference among the five groups with regard to towel reuse.

## Step 4: Draw Inferences beyond the data

**Exercise 9.2.6** Let's apply the 3S strategy.

**(a)** Statistic: Report the observed value of the Mean Group Diff (the differences in group means for all possible pairs of groups) statistic for this study, as given by the applet.

**(b)** Simulate: Use the applet to conduct 1000 shuffles of the response variable. (First check Show Shuffle Options and then enter 1000 for the Number of Shuffles. Press Shuffle Responses.) Then enter the observed value of the Mean Group Diff statistic in the Count Samples box and press Count. Report the p-value.

**(c)** Strength of evidence: Based on the p-value, evaluate the strength of evidence provided by the data against the null hypothesis. Summarize

[2] `static-content2.zybooks.com/isi/data/chap8/Towels.txt`

[3] `static-content.zybooks.com/applets/2021/chisqshuffle/ChiSqShuffle.htm?hideExtras=1`

your conclusion in the context of this study.

## Another Choice of Statistic: Chi-square Statistic

The Mean Group Diff statistic is fairly easy to understand but is not widely used in part because it cannot be easily predicted theoretically. A more commonly used statistic is called a chi-square ($\chi^2$) statistic. You can think of this statistic as measuring how far the group proportions of success are from the overall proportion of success. These (standardized) deviations from the overall proportion are then squared and summed together across all of the explanatory variable groups. We'll do this calculation now.

**Exercise 9.2.7** Determine the overall proportion, p̂, of rooms that reused towels.

**Exercise 9.2.8** We will calculate the standard error of a group proportion by computing $\sqrt{\frac{\hat{p}(1-\hat{p})}{n_i}}$ where $\hat{p}$ is the overall proportion found in #7 and n is the sample size for the ith group. Find the standard error of the group proportion for each of the five groups.

**Exercise 9.2.9** Create a standardized z-statistic for each conditional proportion compared to this overall proportion. To do this take each of the five group proportions, subtract the overall proportion (question 7), and then divide the result by the standard deviation you found for that group in question 8.

**Exercise 9.2.10** Which of the five standardized statistics are larger than 2 or smaller than -2, suggesting strong evidence that the group proportion is significantly different than the overall proportion?

**Exercise 9.2.11** Square each of these values and sum them to calculate the chi-square statistic.

**Exercise 9.2.12** What kinds of values (large, small, positive, negative) will the chi-square statistic have if the null hypothesis is true? What kinds of values will provide evidence against the null hypothesis?

From the *Statistic* pull-down menu, select $\chi^2$. What does the applet report for the observed value of the chi-square statistic? Confirm that this matches your answer to question 11. (*Note*: You may be a bit different due to rounding.)

**Exercise 9.2.13** The applet will also now display the null distribution for the $^2$ statistic rather than the Mean Group Diff statistic. Determine the p-value based on the simulated $^2$ statistics. (Hint: Change the value in the Count Samples box to the observed value of the $^2$ statistic.)

**Exercise 9.2.14** How does the p-value based on the $^2$ statistic compare to the one based on the Mean Group Diff statistic? Are they similar? Is your conclusion about strength of evidence provided by the data against the null hypothesis similar?

## Theory-based approach

The primary advantage of using the $\chi^2$ statistic is that its null distribution can be predicted well by a theoretical distribution. So, once again a theory-based approach could be used without conducting a simulation in the first place.

**Exercise 9.2.15** Below the graph of the simulated chi-square statistics, check the box to Overlay Chi-square distribution. Does the theoretical distribution match the distribution of simulated statistics reasonably well? How does the theory-based p-value compare to what you found in question 13? Does it lead

you to a similar conclusion about strength of evidence?

## Validity conditions

Like all theory-based tests, this one also comes with the condition of having a large sample size. We will (very conservatively) consider a sample size large if the sample data include at least 10 observations in each cell of the two-way table.

**Exercise 9.2.16** Go back to the applet and make sure the *Show Table* box is checked.

**(a)** How many cells are in this table? In other words, how many counts need to be checked that they are at least 10?

**(b)** Is the validity condition for a theory-based chi-square test satisfied for these data? Justify your answer.

## Follow-up analysis

When a chi-square test produces a significant result, we conclude that at least one probability or population proportion differs from at least one of the others. A sensible next step is to try to identify which group proportion differs significantly from which others. We can do this by producing confidence intervals for pairwise differences in population proportions.

**Exercise 9.2.17** Below the p-value output, check the box to *Compute 95% CI(s) for difference in proportions.*

**(a)** How many intervals are produced?

**(b)** Which conditions are significantly different from each other? How are you deciding?

**(c)** For one of the intervals you identified in part (b), write a one-sentence interpretation of the interval, being very clear what is supposed to be captured inside the interval and which condition has a more positive effect on towel use and by how much.

## Step 5: Summarize conclusions

**Exercise 9.2.18** Is this an observational study or a randomized experiment?

**Exercise 9.2.19** Explain how the conclusions of this study are limited by conducting the study at a single hotel.

## Step 6: Look back and ahead

**Exercise 9.2.20** Summarize your conclusion for your analysis of the data from this study.

- Summarize the research question, the methods, and the data.
- Refer to the p-values from the $\chi^2$-test and the confidence intervals to make a recommendation for the hotel manager about whether it matters what message is used, which message should be used, and how large of an impact the best message would be expected to have.
- Think about the study and its design. Were there any limitations you

feel need to be addressed? Did the design match the research question? Can you suggest a better sampling method of hotel guests—a better way to collect data on towel use? Share some ideas for a logical next study.

# 9.3 Comparing multiple proportions: Theory-based approach (Part B)

## Nearsightedness and nightlights revisited

Recall that a previous example described a study[1] investigating whether there is a relationship between use of night lights in a child's room before age 2 and the child's eyesight condition a few years later. Before, we presented a two-way table of counts from the study examining whether the child was nearsighted as the response variable:

**Table 9.3.1**

| **Observed** | **Darkness** | **Near light** | **Room light** |
|---|---|---|---|
| Nearsighted | 18 | 78 | 41 |
| Not nearsighted | 154 | 154 | 34 |
| Total | 172 | 232 | 75 |

The conditional proportions of nearsightedness in the three lighting groups are 0.105, 0.336, and 0.547, respectively, which suggest that the more light used in the child's room, the more likely the child is to become nearsighted. Now we know how to calculate a standardized statistic and estimate a p-value to measure the strength of evidence provided by these sample data for the conjecture that eyesight is associated with the type of lighting used in the child's room.

**Exercise 9.3.1** State null and alternative hypotheses in terms of population proportions.

**Exercise 9.3.2** State null and alternative hypotheses in terms of association between variables.

**Exercise 9.3.3** Are the validity conditions necessary for a chi-square test met for these data? Explain.

## Alternative Formula for Chi-square Statistic

Another popular and more general way to formulate the chi-square statistic is in terms of observed and expected cell counts. The **observed counts** are what you actually observed in the study. The *expected cell counts* are what you would expect for the count in a cell if the null hypothesis were true.

**Exercise 9.3.4** To find out how many children we expect to be in the (darkness, nearsighted) cell of the table, answer (a)-(d):

**(a)** Find the overall proportion of children who were nearsighted in the sample.

**(b)** How many children in the sample slept in darkness?

**(c)** Multiply these two numbers together to find the "expected" number of children who would be nearsighted in the "slept in darkness" group if the null hypothesis were true (the probability of nearsightedness is the same in each condition).

**(d)** How different is this expected cell count from what was observed for that cell? What does this suggest about the strength of evidence against the null hypothesis?

[1] doi.org/10.1038/20094

A similar calculation is performed to find the expected cell count for each of the six cells of the table. The chi-square statistic is obtained by squaring the differences in the observed and expected counts, dividing by the expected count (a form of standardizing), and then summing the six values. This can be thought of as $(O-E)^2/E$, where where $O$ = observed count and $E$ = expected count.

**Exercise 9.3.5** Enter the two-way table of counts into the Multiple Proportions applet[2] (remember to use one-word variable names and to not include the totals). Press *Use Table* and check the box to *Show* $\chi^2$ *output*.

Report the chi-square statistic and theory-based p-value. What conclusion do you draw about the statistical significance of the sample results?

**Exercise 9.3.6** Examine the Chisq Cell Contributions (remember you can expand that table by pulling out the lower right corner). (Verify that the first term in the table is the square of [18 -the number you calculated in #4(c)]/ number from #4(c).) These six values are added together to yield the chi-square statistic; in other words, they are the six $(O-E)^2/E$ values. Which cell contributions are largest and so are contributing the most to the chi-square statistic value? According to the segmented bar graph, how are the observed counts in these cells deviating from what the null hypothesis would predict? (In other words, which is larger, the observed count or the expected count? How does that relate to our research question?)

The original research study actually classified the response variable into three categories of eyesight, as shown in the following table:

**Table 9.3.2**

| Observed | Darkness | Near light | Room light |
|---|---|---|---|
| Nearsighted | 18 | 78 | 41 |
| Normal refraction | 114 | 115 | 22 |
| Farsighted | 40 | 39 | 12 |

One advantage to the more general chi-square statistic formula using observed and expected counts is that it works for any size table (any number of categories for the explanatory variable and the response variable).

**Exercise 9.3.7** Create a segmented bar graph to summarize this table. Discuss what the graph reveals, including what this graph tells you that we did not learn from the initial table showing only nearsighted vs. not nearsighted.

**Exercise 9.3.8** With this new table, do you need to change your null and alternative hypotheses from question 2?

**Exercise 9.3.9** Are the validity conditions for the chi-square test met for this table?

**Exercise 9.3.10** Enter this table into the Multiple Proportions applet and carry out the chi-square test. Report the chi-square statistic and p-value. Is the strength of evidence of a genuine association between the variables in the population stronger, weaker, or essentially the same as before? Explain your answer.

**Exercise 9.3.11** Carry out a randomization test (shuffling) for this table using the chi-square statistic. Overlay the chi-square distribution on the simulated null distribution of chi-square statistic values. Does it appear to be a reasonable model for the null distribution?

[2] `static-content.zybooks.com/applets/2021/chisqshuffle/ChiSqShuffle.htm?hideExtras=1`

**Exercise 9.3.12** Examine the cell contributions. Is the pattern of deviations from the null similar to before?

**Exercise 9.3.13** Does this study suggest that use of night lights and room lights causes an increase to the chance that a child is nearsighted? Why or why not?

## 9.4 Chi-square goodness-of-fit test

### Are birthdays equally distributed throughout the week?

Are the days of the week that women give birth distributed equally throughout the week? Is someone just as likely to be born on a Tuesday as a Saturday? In this Exploration, you will investigate whether there is convincing evidence that birthdays are not distributed equally throughout the week. You will be using a chi-square goodness-of-fit test to help make your conclusion.

### Step 1: Ask a research question

**Exercise 9.4.1** State your research question.

### Step 2: Design a study and collect data

**Exercise 9.4.2** What's the categorical variable of interest, and what are the possible categories?

**Exercise 9.4.3** State the null and alternative hypotheses.

**Exercise 9.4.4** Write down your own birthday (month/day/year) and those of up to nine of your family members or friends. (Try not to write down a birthday that someone else in the class would write down.)

**Exercise 9.4.5** Do an Internet search for "day of birth calculator." Use the calculator to find out on what day of the week each of your family members/ friends was born. Summarize your results in a table.

**Table 9.4.1**

| | **Sun** | **Mon** | **Tue** | **Wed** | **Thurs** | **Fri** | **Sat** | **Total** |
|---|---|---|---|---|---|---|---|---|
| **Observed count** | | | | | | | | |

**Exercise 9.4.6** Combine your results with those for the rest of the class and enter those results in the table.

**Table 9.4.2**

| | **Sun** | **Mon** | **Tue** | **Wed** | **Thurs** | **Fri** | **Sat** | **Total** |
|---|---|---|---|---|---|---|---|---|
| **Observed count** | | | | | | | | |

### Step 3: Explore the data

**Exercise 9.4.7**

**(a)** If the null hypothesis is true and birthdays are distributed equally over the seven days in a week, then the expected count for each day will just be the total number of birthdays divided by 7. Find the expected count for each day.

**Table 9.4.3**

| | **Sun** | **Mon** | **Tue** | **Wed** | **Thurs** | **Fri** | **Sat** | **Total** |
|---|---|---|---|---|---|---|---|---|
| **Observed count** | | | | | | | | |

**(b)** Is the validity condition of all observed cell counts being at least 10 met? If not, what can you do so it is met?

**Definition 9.4.4** A **goodness-of-fit test** is used to compare observed counts for a categorical variable to hypothesized probabilities. ◊

**Exercise 9.4.8** Comparing the observed and expected counts, does it seem that the data gathered would be likely, assuming the birthdays are distributed equally?

## Step 4: Draw inferences

## Simulation-based Goodness-of-fit Test: Using the MAD Statistic

**Exercise 9.4.9** You will now conduct a simulation analysis using the mean absolute difference (MAD) statistic to investigate whether the data provide evidence that birthdays are not equally likely on all seven days of the week.

**(a)** *Statistic*: Here, we will use a MAD-type statistic to compare the observed and expected counts. To find the MAD statistic value for your data, complete the table:

**Table 9.4.5**

| | **Sun** | **Mon** | **Tue** | **Wed** | **Thurs** | **Fri** | **Sat** | **Total** |
|---|---|---|---|---|---|---|---|---|
| $\lvert O - E \rvert$ | | | | | | | | |

Where $O$ is the observed count and $E$ is the expected count, and $|\cdot|$ is the absolute value function.

**(b)** The MAD statistic is obtained by finding the average of the absolute values of the differences in observed and expected counts. Calculate the MAD statistic.

**(c)** *Simulate*: You could spin a spinner with 7 equal sections to find out what values of the MAD statistic could happen by chance alone, or you can use an applet. Open the Goodness of Fit applet[1].

- Press the *Clear* button, enter two column names (e.g., category count), and then specify each day of the week followed by a space and the count from your sample (e.g., Mon 16). Hit return and repeat for each day of the week. Then press *Use Table*. This should update the bar graph under "Sample Data."
- In the right panel, check the box next to *Show Sampling Options* and enter the hypothesized probabilities to three decimal places, separated by commas.
- Set the *Number of Samples* to be 1 and push *Sample*.
- What is the resulting "Most Recent Sample MAD"? Is this value more or less extreme than the one from the observed data? How are you deciding?
- Now change the *Number of Samples* to be 999, for a total of 1000 samples, and push *Sample*. This will produce a simulated null distribution of 1,000 MAD statistics.

**(d)** **Strength of evidence**:

- Where does your observed MAD statistic lie on the simulated null distribution of MAD statistics? Does the observed MAD statistic appear surprising and unlikely to have happened by chance alone,

that is, if the null hypothesis of all days being equally likely were to be true? How are you deciding?

- To find the p-value to quantify the strength of evidence against the null hypothesis, enter the observed value of the MAD statistic in the **Count Samples** box and press **Count**.

**Key Idea 9.4.6 You may be thinking.** Earlier we subtracted proportions, and now we are using counts. Can we not use proportions anymore?

Sure, you can still work with proportions. If you used proportions, your MAD statistic would subtract the observed proportions from the hypothesized proportions (in this case, 1/7 or 0.143 for each day) for each day of the week and then average the absolute values of those differences.

**Exercise 9.4.10** Based on the p-value you obtained in Exercise 9.4.9, p. 151 do you have strong evidence against the null hypothesis? How are you deciding?

## Simulation-based Goodness-of-fit Test: Using the Chi-square Statistic

**Exercise 9.4.11** In this situation you can also conduct a simulation analysis using the chi-square statistic to investigate whether the data provide evidence that birthdays are not equally likely on all seven days of the week.

**(a)** *Statistic*: To find the chi-square statistic value for your data, complete the table:

**Table 9.4.7**

| | **Sun** | **Mon** | **Tue** | **Wed** | **Thurs** | **Fri** | **Sat** | **Total** |
|---|---|---|---|---|---|---|---|---|
| $O - E$ | | | | | | | | |
| $(O - E)^2$ | | | | | | | | |
| $\frac{(O-E)^2}{E}$ | | | | | | | | |

Where $O$ is the observed count and $E$ is the expected count. Recall that the chi-square statistic is given by the formula

$$\chi^2 = \sum \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$

Thus, the chi-square statistic is the Total from the last row of the table.

**(b)** *Simulate*: Just as you ran the simulation using the MAD statistic, you can follow the same steps this time using the Statistic pull-down menu to select the chi-square statistic ($\chi^2$). Create a null distribution for the chi-square statistic.

**(c)** *Strength of evidence*

- Where does the observed chi-square statistic lie on the simulated null distribution of the chi-square statistic? Does it appear surprising and unlikely to have happened by chance alone, that is, if the null hypothesis of all days being equally likely were to be true? How are you deciding?
- Find the corresponding p-value. How does it compare to what you found with the MAD statistic?

[1] static-content.zybooks.com/applets/2021/gof/GOF.html?hideExtras=1

**Exercise 9.4.12** Based on the p-value reported in question 11c from the simulated null distribution for a chi-square statistic, do you have strong evidence against the null hypothesis? How are you deciding?

### Step 5: Formulate conclusions

**Exercise 9.4.13** State your complete conclusions. In particular, to what population are you willing to generalize your conclusions?

**Exercise 9.4.14** If you found convincing evidence that birthdays are not equally distributed throughout the week, which days seemed to be different than the others and why do you think that is the case?

### Step 6: Look back and ahead.

**Exercise 9.4.15** If you did not find convincing evidence that birthdays are not distributed equally throughout the week, what could you do differently in a follow-up study to try to support your case?

### Revisiting step 4: Draw inferences

### Theory-based Chi-square Goodness-of-fit Test

Earlier, for Step 4, you used simulation to find a p-value using the chi-square statistic. Recall that one advantage of using the chi-square statistic (instead of the MAD) is that the distribution of the chi-square statistic can be predicted by a theory-based distribution called the chi-square distribution.

**Exercise 9.4.16** Go back to the Goodness of Fit applet, and picking up where you left off, check the "Overlay chi-square distribution" box under the null distribution for the chi-square statistics. Report the "theory-based p-value."

**Exercise 9.4.17**

**(a)** How does the theory-based p-value compare to the one from the simulation-based p-value using the chi-square test statistic? Do you still arrive at the same conclusion about whether there is convincing evidence that not all seven days of the week are equally likely to be birthdays?

**(b)** Remember that there are always certain checks to be made before you can decide that it is valid to use a theory-based method to test hypotheses. In particular, are all observed cell counts at least 10?

**Exercise 9.4.18** Does the theory-based chi-square goodness-of-fit test appear to be valid in this context? Explain how you are deciding.

### Exploring Further

One of the authors collected data on which day of the week each of her 49 students was born. The data appear in the table.

**Table 9.4.8**

| | **Sun** | **Mon** | **Tue** | **Wed** | **Thurs** | **Fri** | **Sat** | **Total** |
|---|---|---|---|---|---|---|---|---|
| **Observed count** | **4** | **7** | **7** | **7** | **8** | **6** | **10** | **49** |

**Exercise 9.4.19** Do you think these data will provide more or less evidence against the null hypothesis than the data you collected? Explain your reasoning.

We will use the data from this class to determine whether there is convincing evidence that not all seven days of the week are equally likely to be birthdays.

**Exercise 9.4.20** Using a significance level of $\alpha = 0.05$, do the data provide convincing evidence that the days of the week that women give birth are not equally distributed throughout the week?

**(a)** State the hypotheses using either words or symbols

**(b)** Name the theory-based test you will use and verify validity conditions for that test.

**(c)** Calculate the test statistic (standardized statistic) and corresponding p-value.

**(d)** State your conclusion (with appropriate justification based on your p-value and stated significance level) in the context of the research question.

**(e)** If there is not sufficient evidence to support the research hypothesis that all seven days of the week are not equally likely to be birthdays, is this proof that birthdays are evenly distributed throughout the days of the week? Explain.

**(f)** Were these data stronger or weaker evidence against the null hypothesis than the data you observed? Is this what you predicted?

**(g)** Based on your conclusion in (d), are you potentially making a conclusion error? Explain the potential consequences of this type of error in this context.

# 9.5 Data and Formulas

## Data

**Table 9.5.1 References**

| Section | Data |
|---|---|
| Example: Comparing multiple proportions: Simulation-based approach | Stop.csv[1] |
| Supplemental Exploration: Comparing multiple proportions: Simulation-based approach | OrganDonation.csv[3] |
| Example: Comparing multiple proportions: Theory-based approach | Acupunture.csv[5] |
| Supplemental Exploration A: Comparing multiple proportions: Theory-based approach | Towels.csv[7] |
| Supplemental Exploration B: Comparing multiple proportions: Theory-based approach | NightLight1.csv[9] Night |
| Example: Chi-square goodness of fit | DieRolls.csv[12] |
| Supplemental Exploration: Chi-square goodness of fit | Birthdays.csv[13] |
| Investigation: Who yields to pedestrians? | Pedestrians.csv[14] |

## Formulas for multiple proportions and goodness-of-fit

Difference of proportions (two samples):

- $n_1$ and $n_2$ are the sample sizes for the two groups.
- $\hat{p}_2 = \frac{\text{number of successes in group 1}}{n_1}$ is the sample proportion for group 1.
- $\hat{p}_2 = \frac{\text{number of successes in group 2}}{n_2}$ is the sample proportion for group 2.
- $\hat{p}_1 - \hat{p}_2$ is the difference in sample proportions.
- $\hat{p} = \frac{\text{total number of successes}}{n_1+n_2}$ is the pooled sample proportion.
- $SD(\hat{p}_1 - \hat{p}_2)$ is the standard deviation of the statistic.
- SE using pooled $\hat{p} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}$ is a theory-based standard error that approximates SD of the null distribution.
- SE using $\hat{p}_1$ and $\hat{p}_2 = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ is a theory-based standard error that approximates $SD(\hat{p}_1 - \hat{p}_2)$.
- standardized statistic $= \frac{\text{statistic} - \text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.

[1] static-resources.zybooks.com/ISI/Ch09/Stop.csv
[2] www.scribd.com/document/25779288/Observational-Study-for-Determining-the-Percentage-of-Vehicles-Completely-S
[3] static-resources.zybooks.com/ISI/Ch09/OrganDonation.csv
[4] papers.ssrn.com/sol3/papers.cfm?abstract_id=1324774
[5] static-resources.zybooks.com/ISI/Ch09/Acupunture.csv
[6] doi.org/10.1001/archinte.167.17.1892
[7] static-resources.zybooks.com/ISI/Ch09/Towels.csv
[8] doi.org/10.1086/586910
[9] static-resources.zybooks.com/ISI/Ch09/NightLight1.csv
[10] static-resources.zybooks.com/ISI/Ch09/NightLight2.csv
[11] doi.org/10.1038/20094
[12] static-resources.zybooks.com/ISI/Ch09/DieRolls.csv
[13] static-resources.zybooks.com/ISI/Ch09/Birthdays.csv
[14] static-resources.zybooks.com/ISI/Ch09/Pedestrians.csv
[15] doi.org/10.1073/pnas.1118373109

  - $z = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two proportions.
  - $z = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - 0}{\text{SE using pooled } \hat{p}} = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - 0}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$ is the theory-based standardized statistic for two proportions.

- $z = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two proportions.
- $z = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - 0}{\text{SE using pooled } \hat{p}} = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - 0}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$ is the theory-based standardized statistic for two proportions.
- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.
  - $(\hat{p}_1 - \hat{p}_2) \pm 2 \times SD(\hat{p}_1 - \hat{p}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\pi_1 - \pi_2$.
  - $(\hat{p}_1 - \hat{p}_2) \pm$ (multiplier) $\times$ (SE using $\hat{p}_1$ and $\hat{p}_2$) is a theory-based confidence interval estimating $\pi_1 - \pi_2$, where the multiplier comes from the standard normal distribution.
- $(\hat{p}_1 - \hat{p}_2) \pm 2 \times SD(\hat{p}_1 - \hat{p}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\pi_1 - \pi_2$.
- $(\hat{p}_1 - \hat{p}_2) \pm$ (multiplier) $\times$ (SE using $\hat{p}_1$ and $\hat{p}_2$) is a theory-based confidence interval estimating $\pi_1 - \pi_2$, where the multiplier comes from the standard normal distribution.

Mean Group Diff:

- $p_i$ are the population proportions for the groups.
- $n_i$ are the sample sizes for the groups.
- $\hat{p}_i$ are the sample proportions for the groups.
- $l$ is the number of groups.
- The Mean Group Diff is the mean of the absolute value of the differences of sample proportions, and can be computed directly as $\frac{\sum_{\text{groups } i,j} |\hat{p}_i - \hat{p}_j|}{(l-1)(l/2)}$.

Chi-square for proportions:

- $p_i$ are the population proportions for the groups.
- $n_i$ are the sample sizes for the groups with $n$ as the overall sample size.
- $\hat{p}_i$ are the sample proportions for the groups with $\hat{p}$ as the overall sample proportion.
- $l$ is the number of groups.
- Chi-square $\chi^2$ is the sum of squared standardized proportions and can be calculated directly as $\chi^2 = \sum_{\text{groups } i} \left( \frac{\hat{p}_i - \hat{p}}{\sqrt{\hat{p}(1-\hat{p})/n_i}} \right)^2$.

- The mean of a chi-square distribution is equal to the degrees of freedom $df = l - 1$.
- The SD of a chi-square distribution is equal to $\sqrt{2\ df}$.

Mean Absolute Difference (MAD):

- $O$ are the observed counts for the cells.
- $E$ are the expected counts for the cells.
  - one-way table: $E =$ (Expected probability $p_i$) $\times$ (Total).
  - two-way table: $E =$ (Row total) $\times$ (Column total)/(Grand total).
- The Mean Absolute Difference ($MAD$) is the mean of the absolute value of the differences of cell counts, and can be computed directly as $\frac{\sum_{\text{cells}} |O - E|}{\#\text{Cells}}$.
- one-way table: $E =$ (Expected probability $p_i$) $\times$ (Total).
- two-way table: $E =$ (Row total) $\times$ (Column total)/(Grand total).

Chi-square for tables:

- $O$ are the observed counts for the cells.
- $E$ are the expected counts for the cells.
  - one-way table: $E =$ (Expected probability $p_i$) $\times$ (Total).
  - two-way table: $E =$ (Row total) $\times$ (Column total)/(Grand total).
- The cell contributions are the values $\frac{(O-E)^2}{E}$.
- Chi-square $\chi^2$ is the sum of cell contributions and can be calculated directly as $\chi^2 = \sum_{\text{cells}} \left( \frac{(O-E)^2}{E} \right)$.
- The mean of a chi-square distribution is equal to the degrees of freedom.
  - one-way table: $df = \#\text{Groups} - 1$.
  - two-way table: $df = (\#\text{Groups} - 1) \times (\#\text{Outcomes} - 1)$.
- The SD of a chi-square distribution is equal to $\sqrt{2\ df}$.
- one-way table: $E =$ (Expected probability $p_i$) $\times$ (Total).
- two-way table: $E =$ (Row total) $\times$ (Column total)/(Grand total).
- one-way table: $df = \#\text{Groups} - 1$.
- two-way table: $df = (\#\text{Groups} - 1) \times (\#\text{Outcomes} - 1)$.

# Chapter 10

# Comparing More Than Two Means

# 10.1 Comparing multiple means: Simulation-based approach

## Exercise and brain volume

Brain size typically shrinks as people age past adulthood, and such shrinkage may be linked to dementia. Therefore, any intervention that can protect against brain shrinkage could help to protect the elderly against dementia and Alzheimer's disease. Researchers in China recently investigated whether different kinds of exercise/activity might help to prevent brain shrinkage or perhaps even lead to an increase in brain size (Mortimer et al., 2012[1]).

The researchers randomly assigned elderly adult volunteers into four activity groups: tai chi, walking, social interaction, and no intervention. Except for the group with no intervention, each group met for about an hour three times a week for 40 weeks to participate in their assigned activity. The tai chi group was led by a tai chi master and an assistant, the walking group walked around a track, the social interaction group met at a community center and discussed topics that interested them, and the no-intervention group just received four phone calls during the study period. A total of 120 participants started the study, and 13 dropped out along the way, so 107 completed the study.

Each participant had an MRI to determine brain size before the study began and again at its end. The researchers measured the percentage increase or decrease in brain size during that time. They thought that physical activity would help increase brain size; hence they anticipated that the tai chi and walking groups would tend to show larger increases in brain size during the study than the other activity groups.

## Step 1: Ask a research question

**Exercise 10.1.1** What is the main research question that the researchers hoped to answer?

## Step 2: Design a study and collect data

**Exercise 10.1.2** Identify the observational units in this study.

**Exercise 10.1.3** Identify the explanatory and response variables in this study. Indicate whether they are quantitative or categorical. If they are categorical, indicate how many categories.

**Exercise 10.1.4** What type of study was this: an experiment or an observational study? Are the responses of the participants in one treatment group paired with or independent of the responses of the participants in other treatment groups?

We need a procedure that considers information from across all four groups to allow for comparing all four groups simultaneously. When we compared two groups, the null hypothesis can be written in terms of no association between the explanatory and response variables, whereas the alternative asserts that there is an association between the explanatory and response variables. More specifically, the null hypothesis asserts that all four long-run means are equal, and the alternative states that at least one long-run mean differs from the others.

[1] doi.org/10.3233/JAD-2012-120079

**Exercise 10.1.5** State the null and the alternative hypotheses, in symbols and in words, in the context of this study.

The researchers calculated the percentage change in size of each participant's brain. If a person's brain size increased, then this percentage change was positive; if brain size decreased, then this percentage change was negative. These data can be found in the file Brain[2].

## Step 3: Explore the data

**Exercise 10.1.6** Paste the data into the Multiple Means applet[3] and press *Use Data*. The applet will produce a dotplot and descriptive statistics. Then check the *Show Groups* box. Check the *Boxplots* box to overlay those as well.

**(a)** Record the means and standard deviations in the table and provide estimates of the five-number summaries of each group from the boxplots.

**Table 10.1.1**

| Group | Sample size | Mean | SD | Minimum | Lower quartile | Median | Upper qua |
|---|---|---|---|---|---|---|---|
| **No interaction** | | | | | | | |
| **Social** | | | | | | | |
| **Walking** | | | | | | | |
| **Tai chi** | | | | | | | |

**(b)** Which activity group tended to have the largest increase in brain size percentage change? Which tended to have the smallest increase (i.e., largest decrease)?

**Exercise 10.1.7** Describe what the graphs and statistics reveal about whether the four activities appear to differ with regard to percentage change in brain size for the participants in this study.

**Key Idea 10.1.2** Suppose for the moment that there really is no effect of the different activities on percentage change in brain size.

**Exercise 10.1.8** Would it be possible to obtain sample means as far apart as these four are in this sample?

**Exercise 10.1.9** Describe a general process for assessing how unlikely it would be, when the null hypothesis is true, to obtain sample means as far apart as these four are.

## Step 4: Draw inferences (3S Strategy)

**Statistic**: To minimize the chance of making a Type I error, we won't run six different tests comparing each group to each other. So, instead we need a single statistic to help test the overall null hypothesis of no association.

**Exercise 10.1.10** The first step in applying the 3S strategy is to decide on a statistic that measures how far apart the sample means are. We could focus on the difference between the max $\bar{x}$ and the min $\bar{x}$ values that you identified in question 6b. Why might this not be the best statistic to use?

**Exercise 10.1.11** Propose an alternative statistic to use for this purpose. In other words, come up with a formula based on the four sample means that will result in a single number that summarizes how far apart the four sample

[2] static-content2.zybooks.com/isi/data/chap9/Brain.txt

[3] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

means are. (Hint: You should be able to calculate the value of your statistic from the observed sample data.)

When we compared means between two groups in Chapter 6 using simulation-based approaches, we used the difference between the two means for the statistic. But in this study involving four groups, there are six different pairs of groups for which we could take the differences in means.

**Exercise 10.1.12** Suggest how you might combine those six differences into one statistic.

One suggestion for this statistic is to calculate the mean of the absolute values of the differences in means (called Mean Group Diff, for mean group differences):

**Exercise 10.1.13** Explain why it's important to take the absolute values before averaging the six differences.

**Exercise 10.1.14** What is the value of the Mean Group Diff statistic for the exercise data, as shown in the applet?

**Exercise 10.1.15** What types of values will the Mean Group Diff statistic have if the null hypothesis is true (large, small, positive, negative)? If there is strong evidence against the null hypothesis?

**Simulate**

**Exercise 10.1.16** Suppose that you were to conduct a tactile simulation to generate many possible values of the Mean Group Diff statistic that could have happened if the null hypothesis were true, that is, by random chance alone. As you didin the simpler case with comparing just two means, you can use index cards here. Answer (a)-(f) about what else needs to be kept in mind:

**(a)** How many cards will you need?

**(b)** Do the cards just need to be certain colors, or do they need something written on them, and if so, what?

**(c)** When you shuffle and deal, how many piles will you make?

**(d)** How many cards will you place in each pile?

**(e)** What should you be recording after the completion of each shuffle and deal?

**(f)** What else do you need to do to find the p-value?

Instead of doing the tactile simulation (which would be fun, of course, but time consuming) let us use the Multiple Means applet[4] to run the simulation with the Brain[5] data. Check the *Show Shuffle Options* box in the applet. Enter 1 for *Number of Shuffles* and press *Shuffle Responses*. Notice in the *Most Recent Shuffle* data window, the *Treatment* column has not changed, but the *BrainChange* column has mixed up all its values. If you select the *Plot* radio button, you will see the new dotplots.

**Exercise 10.1.17** Notice that the new dotplots and statistics are produced for the shuffled data. The individual with the smallest percentage change, originally in the walking group, is now in which group? Report the value of the Mean Group Diff statistic for the shuffled data. Is this value more extreme than the observed value of the Mean Group Diff statistic from the observed

[4] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

[5] static-content2.zybooks.com/isi/data/chap9/Brain.txt

data?

**Exercise 10.1.18** Press Shuffle Responses again, with the Plot option selected. Notice the applet pooling all the responses together and then randomly reassigning them to their new groups. What did you get for the Mean Group Diff value for these shuffled responses?

**Exercise 10.1.19** Press Shuffle Responses three more times. Record the three new values of the Mean Group Diff statistic for the shuffled data. Are these all the same? Are any of them more extreme than the observed value of the Mean Group Diff statistic from the experimental data?

**Exercise 10.1.20** Now enter 995 for the *Number of Shuffles*, producing a total of 1,000 shuffles.

**(a)** Describe the shape of the null distribution of the Mean Group Diff statistic. Is it symmetric or skewed? If it is skewed, in which direction is it skewed?

**(b)** Is the null distribution centered at zero? Explain why your answer makes sense.

**Strength of evidence**

**Exercise 10.1.21** Now you will calculate a p-value, as you have done from simulated null distributions many times before, in order to assess the strength of evidence that the experimental data provide against the null hypothesis that activity has no effect on change in brain size.

**(a)** To calculate the p-value, you will count how many of the simulated Mean Group Diff statistics are equal to _____ or (larger/smaller). (Review your answer to question 14 to consider what values are "more extreme" than those observed.)

**(b)** Check the Count Samples box in the applet and then enter the observed value of the Mean Group Diff statistic. Report the p-value.

**(c)** You should find that the p-value is fairly close to a common cutoff value for assessing statistical significance. So, go ahead and produce 9,000 more shuffles to produce a more accurate estimate of the p-value (there will be a pause). Report this p-value.

**(d)** Interpret this p-value: It is the probability of obtaining a Mean Group Diff statistic of _____ or _____, assuming that _____.

**(e)** Based on this p-value, evaluate the strength of evidence: The experimental data provide _____ evidence against the null hypothesis that the activity group has no effect on percentage change in brain size.

**Exercise 10.1.22** Significance:

**(a)** Summarize your conclusion based on the p-value.

**(b)** Does this analysis allow you to determine which activities differ significantly from which others with regard to percentage change in brain size? Explain why or why not.

**Exercise 10.1.23** Estimation: Based on what you learned in earlier chapters, suggest how you might go about estimating the magnitude of the difference in the long-run average percentage change in brain size between the tai chi and no-intervention groups. (You will see a method for doing this later)

### Step 5: Formulate conclusions

**Exercise 10.1.24** Generalization: To what population would you feel comfortable generalizing these results?

**Exercise 10.1.25** Causation: Would you feel comfortable with drawing a cause-and-effect conclusion from this study? Between what two variables? Explain why or why not.

### Step 6: Look back and ahead

**Exercise 10.1.26** Looking back: Did anything about the design and conclusions of this study concern you? Issues you may want to critique include: The match between the research question and the study design, how the experimental units were selected, how the treatments were assigned to the experimental units, how the measurements were recorded, the number of experimental units in the study, whether what we observed is of practical value.

**Exercise 10.1.27** Looking ahead: What should the researchers' next steps be to fix the limitations or build on this knowledge?

## 10.2 Comparing multiple means: Theory-based approach

### Comparing popular diets

Because about two-thirds of Americans are considered overweight, weight loss is big business. There are many different types of diets, but do some work better than others? Is low fat better than low carb or is some combination best? Researchers (Garnder et al. 2007[1]) conducted a study involving four popular diets: Atkins (very low carb), Zone (40:30:30 ratio of carbs, protein, fat), LEARN (high carbohydrate, low fat), and Ornish (low fat). They randomly assigned women aged 25-50 with a body mass index (BMI) of 27-40 (overweight and obese) to one of the four diets. The 311 women who volunteered for the program were educated on their assigned diet and were observed periodically as they stayed on the diet for a year. At the end of the year, the researchers calculated the change in BMI (e.g., negative means reduction in BMI) for each woman and compared the results across the four diets.

### Step 1: Ask a research question

**Exercise 10.2.1** What is the overarching research question the researchers hoped to answer?

### Step 2: Design a study and collect data

**Exercise 10.2.2**

**(a)** What are the observational units in this study?

**(b)** Identify the explanatory and response variables. Classify them as categorical or quantitative. For categorical variables, indicate how many categories are used.

**Exercise 10.2.3**

**(a)** Does this study make use of random sampling, random assignment, both, or neither? What are the implications of your answer with regard to scope of inference?

**(b)** Did the researchers collect the data as paired data or as independent samples? In other words, according to the study design, are the responses from one treatment group paired with or independent of the responses from other treatment groups?

**Exercise 10.2.4** State the null and alternative hypotheses, both in words and symbols, for testing the research conjecture. (Recall that the response variable is change in BMI; positive values indicate an increase in BMI and negative values indicate a decrease in BMI from the beginning to the end of the study.)

### Step 3: Explore the data

**Exercise 10.2.5** The data for this investigation appear in a file called Diets[2]. Copy and paste the data into the Multiple Means applet[3]. (Recall that the response variable is change in BMI.) Press *Use Data*, and the applet will produce

[1] doi.org/10.1001/jama.297.9.969

dotplots and summary statistics.

**(a)** Report the mean and standard deviation of the BMI values in this sample.

**(b)** Check the Show Groups box. You can also check the Boxplots box to overlay those as well. Report the means, standard deviations, and sample sizes for the change in BMI amounts for each diet.

**(c)** Describe what the graphs and statistics reveal about whether the diets appear to differ with regard to change in BMI amounts and, if so, which diet appears to be best and which worst.

## Step 4: Draw inferences beyond the data

**Exercise 10.2.6** Conduct a simulation-based randomization test, as was done in the last section, to compare the four group means using the mean of the absolute values of the group differences (Mean Group Diff) as the statistic. Report the observed value of this statistic. Then determine and report a p-value based on at least 1,000 shuffles.

**Exercise 10.2.7** Based on this p-value, evaluate the strength of evidence against the null hypothesis provided by the experimental data. In other words, do the data provide strong evidence that at least one of the four diets differs with regard to average change in BMI?

Although the Mean Group Diff statistic is fairly easy to understand and calculate, it is not commonly used, in part because there is no theory-based model for the null distribution. Another downside is that the Mean Group Diff statistic is not standardized so it is not comparable across studies (e.g., a Mean Group Diff statistic of 1 in one study might be strong evidence but in another study might not show convincing evidence against the null hypothesis).

**Key Idea 10.2.1** Instead of comparing only the differences in means, a statistic could standardize those differences by comparing to the amount of within-group or natural variability in the response variable while also taking sample sizes into account.

A much more commonly used statistic for comparing multiple groups on a quantitative response, which does have a theory-based distribution, is called an **F-statistic**. As with the Mean Group Diff statistic, the F-statistic equals zero only when the group means are all identical. Otherwise the F-statistic is positive, with larger values indicating larger differences across the group means.

The F-statistic (see Formula section at the end of Example 9.2 for details) is a ratio of "between-group" and "within-group" variability. Thus,

$$F = \frac{\textbf{Between-group variability}}{\textbf{Within-group variability}}$$

**Exercise 10.2.8** Select the *Statistic* pull-down menu above the null distribution to select *F-statistic*, and the null distribution changes from the Mean Group Diff statistic to the F-statistic.

**(a)** Report the observed value of the F-statistic, which appears on the left side of the applet.

**(b)** Use the simulation results and the observed value of the F-statistic to estimate the p-value.

[2] static-content2.zybooks.com/isi/data/chap9/Diets.txt

[3] static-content.zybooks.com/applets/2021/anovashuffle/AnovaShuffle.htm?hideExtras=2

**(c)** Is this p-value similar to the one based on the Mean Group Diff statistic? Is your conclusion about strength of evidence the same with both statistics?

The F-statistic has a known theoretical distribution when the null hypothesis is true, called (surprisingly enough) an **F-distribution**.

**Exercise 10.2.9** Check the box to Overlay F distribution.

**(a)** Would you say there is good agreement between this theoretical prediction and your simulated null distribution?

**(b)** How does the theory-based p-value compare to what you found in Part 10.2.8.b, p. 166

But what does F = 3.80 really mean? Check the box by *Show ANOVA Table* to see more output related to the calculation of the value of the F-statistic.

**Exercise 10.2.10** Confirm that the F-statistic is the ratio of the mean square for treatment to the mean square for error. These are the numbers in the `MS` column of the ANOVA table. The mean square for treatment is in the `Model` row, and the mean square for error is in the `Error` row.

The next series of exercises will take a closer look at how the F-statistic is calculated.

**Exercise 10.2.11** Use the Descriptive Statistics applet[4] to find the standard deviation of the group averages:

- Type in the four group means for the original data. (Clear the existing data first and uncheck the Includes header box.)
- Have the applet calculate the standard deviation of these four group means.

**(a)** Report the number you found.

**(b)** Write a one-sentence interpretation of this number you calculated.

**Exercise 10.2.12** To estimate the natural variation in these BMI values, we want to basically average the "within-group" variability across the four groups.

**(a)** Take the four group standard deviations and average them. (You can again use the Descriptive Statistics applet.)

**(b)** Write a one-sentence interpretation of this number.

**Exercise 10.2.13** Now let $s_{\bar{x}_i}$ be your answer to Exercise 10.2.11, p. 167 and let $\bar{s_i}$ be your answer to Exercise 10.2.12, p. 167.

**(a)** Calculate

$$77.75 \times \frac{{s_{\bar{x}_i}}^2}{{\bar{s_i}}^2}.$$

The 77.75 is the average of the group sizes. Recall that variance is the squared of standard deviation.

**(b)** Is your answer to part a above the approximately the same as your answer to Exercise 10.2.10, p. 167?

Another way to summarize the usefulness of the explanatory variable, is the "coefficient of determination," or $R^2$ value.

[4] `static-content.zybooks.com/applets/2021/descstats/Dotplot.htm?hideExtras=1`

**Definition 10.2.2** The proportion of the variation in the response variable which is explained (accounted for) by the treatment (groups) is called

$$R^2 = 1 - \frac{\text{within-group variability}}{\text{total variability}}.$$

$\Diamond$

**Exercise 10.2.14** Again let $\bar{s_i}$ be your answer to Exercise 10.2.12, p. 167, and let $s$ be the standard deviation of all the BMI's.

**(a)** Compute

$$R^2 \approx 1 - \frac{\bar{s_i}^2}{s^2}.$$

This approximation works when the sizes of the groups are almost the same. Again recall that variance is the square of standard deviation.

**(b)** $R^2$ is the proportion of the variability in BMI values that can be explained by diet type. Is the percentage of variation in BMI explained high (close to 100%) or low (close to 0%) in this case?

**(c)** How much variability in BMI is "left over" (not explained by the type of diet)?

**(d)** Even though we found our F-statistic to be statistically significant, what implications does our $R^2$ have interpreting our conclusions about the impact of the diets?

**Exercise 10.2.15** You have now calculated a p-value in three different ways, two using a simulation-based approach (but with different statistics: Mean Group Diff and F) and one using a theory-based approach (using the F-statistic). Are they all similar? What is your conclusion about whether the diets differ with regard to average BMI changes?

## Follow-up analysis

The ANOVA F-test is an overall test. A significant test result tells us that we have strong evidence that at least one population mean is different from the others but does not tell us which one(s) differ from which other one(s). To determine this we need to apply a follow-up procedure to ANOVA. There are many different follow-up tests that can be done. We will use pairwise confidence intervals for the difference in two means, as we did before.

**Key Idea 10.2.3** If an ANOVA test finds evidence that there is at least one long-run or population mean that is different, there are many different follow-up tests designed to help pinpoint where difference(s) occur. One option is to look at all the pairwise theory-based confidence intervals for the difference in long-run or population means.

**Exercise 10.2.16** When you have the F-statistic selected and the F-distribution overlaid (or are showing the ANOVA table) you can check the box to Compute 95% CI(s) for difference in means. Report the intervals for (a)-(f).

**(a)** Atkins diet vs. Zone diet

**(b)** Atkins diet vs. Ornish diet

**(c)** Atkins diet vs. LEARN diet

**(d)** Zone diet vs. Ornish diet

**(e)** Zone diet vs. LEARN diet

**(f)** Ornish diet vs. LEARN diet

**Exercise 10.2.17** Which of the confidence intervals calculated in Exercise 10.2.16, p. 168 indicate a significant difference between groups? How are you deciding?

**Exercise 10.2.18** Based on your analysis, would you conclude that there is one diet that works significantly better than the other three? If so, which one?

## Validity conditions

As with other theory-based approaches of conducting tests of significance, certain validity conditions must be met in order to conduct an ANOVA F-test.

**Validity Conditions 10.2.4 for the ANOVA F-test.** The F-distribution is a good approximation to the null distribution of the F-statistic as long as:

- Either the sample size is at least 20 for all the groups, without strong skewness or outliers; or if the sample size is less than 20, then the distribution of the response variable is approximately symmetric in all the samples (examine the dotplots for skewness or outliers).
- The standard deviations of the samples are approximately equal to each other (largest standard deviation is not more than twice the value of the smallest standard deviation).

**Exercise 10.2.19** Do the sample data for the diet study appear to satisfy the validity conditions for conducting an ANOVA F-test? How are you deciding?

## Step 5: Formulate conclusions

**Exercise 10.2.20** Are you comfortable with concluding from this study that the diet used causes a difference in BMI change? Justify your answer.

**Exercise 10.2.21** To what population are you comfortable generalizing the results of this study? Justify your answer.

## Step 6: Look back and ahead

**Exercise 10.2.22** Looking back: Did anything about the design and conclusions of this study concern you? Issues you may want to critique include: The match between the research question and the study design, how the experimental units were selected, how the treatment was assigned to the experimental units, how the measurements were recorded, the number of experimental units in the study, whether what we observed is of practical value.

**Exercise 10.2.23** Looking ahead: What should the researchers' next steps be to fix the limitations or build on this knowledge?

## 10.3 Data and Formulas

### Data

**Table 10.3.1 References**

| Section | Data |
|---|---|
| Example: Comparing multiple means: Simulation-based approach | ProseComprehe |
| Supplemental Exploration: Comparing multiple means: Simulation-based approach | Brain.csv[4] |
| Example: Comparing multiple means: Theory-based approach | ProseRecall.csv |
| Supplemental Exploration: Comparing multiple means: Theory-based approach | Diets.csv[9] |
| Investigation: Aggression | Agression.csv[11] |

### Formulas for multiple means

Difference of means (two independent samples):

- $\mu_1$ and $\mu_2$ are the population means for the two groups.
- $n_1$ and $n_2$ are the sample sizes for the two groups.
- $\overline{x}_1$ and $\overline{x}_2$ are the sample means for the two groups.
- $s_1$ and $s_2$ are the sample standard deviations for the two groups.
- $SD(\overline{x}_1 - \overline{x}_2)$ is the standard deviation of the statistic.
- SE $= \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ is the theory-based standard error that approximates SD of the null distribution and $SD(\overline{x}_1 - \overline{x}_2)$.
- standardized statistic $= \frac{\text{statistic} - \text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.
  - $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two independent means.
  - $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{SE} = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ is the theory-based standardized statistic for two independent means.
- $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for two independent means.

[1] static-resources.zybooks.com/ISI/Ch10/ProseComprehension.csv
[2] doi.org/10.1016/S0022-5371%2872%2980006-9
[3] doi.org/10.1016/S0022-5371%2872%2980006-9
[4] static-resources.zybooks.com/ISI/Ch10/Brain.csv
[5] doi.org/10.3233/JAD-2012-120079
[6] static-resources.zybooks.com/ISI/Ch10/ProseRecall.csv
[7] doi.org/10.1016/S0022-5371%2872%2980006-9
[8] doi.org/10.1016/S0022-5371%2872%2980006-9
[9] static-resources.zybooks.com/ISI/Ch10/Diets.csv
[10] doi.org/10.1001/jama.297.9.969
[11] static-resources.zybooks.com/ISI/Ch10/Agression.csv
[12] doi.org/10.1177%2F1948550612436984

- $t = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{SE} = \frac{\left(\overline{x}_1 - \overline{x}_2\right) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ is the theory-based standardized statistic for two independent means.

- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.

  - $(\overline{x}_1 - \overline{x}_2) \pm 2 \times SD(\overline{x}_1 - \overline{x}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_1 - \mu_2$.
  - $(\overline{x}_1 - \overline{x}_2) \pm$ (multiplier) $\times (SE)$ is a theory-based confidence interval estimating $\mu_1 - \mu_2$, where the multiplier comes from a t distribution with $\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$ degrees of freedom.

- $(\overline{x}_1 - \overline{x}_2) \pm 2 \times SD(\overline{x}_1 - \overline{x}_2)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_1 - \mu_2$.

- $(\overline{x}_1 - \overline{x}_2) \pm$ (multiplier) $\times (SE)$ is a theory-based confidence interval estimating $\mu_1 - \mu_2$, where the multiplier comes from a t distribution with $\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$ degrees of freedom.

Mean difference (matched pairs):

- $d = (group\ 1) - (group\ 2)$ is the variable of differences for each matched pair.

- $\mu_d$ is the population mean difference.

- $n$ is the sample size (number of matched pairs).

- $\overline{x}_d$ is the sample mean of the differences.

- $s_d$ is the sample standard deviation of the differences.

- $SD(\overline{x}_d)$ is the standard deviation of the statistic.

- SE $= s_d/\sqrt{n}$ is the theory-based standard error that approximates SD of the null distribution and $SD(\overline{x}_d)$.

- standardized statistic $= \frac{\text{statistic} - \text{mean of the null distribution}}{\text{SD of the null distribution}}$ is the number of standard deviations away from the mean of the sampling distribution a statistic is.

  - $t = \frac{\overline{x}_d - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for matched pairs.
  - $t = \frac{\overline{x}_d - 0}{SE} = \frac{\overline{x}_d - 0}{s_d/\sqrt{n}}$ is the theory-based standardized statistic for matched pairs.

- $t = \frac{\overline{x}_d - 0}{\text{Simulated SD of null distribution}}$ is the simulation-based standardized statistic for matched pairs.

- $t = \frac{\overline{x}_d - 0}{SE} = \frac{\overline{x}_d - 0}{s_d/\sqrt{n}}$ is the theory-based standardized statistic for matched pairs.

- statistic $\pm$ margin of error = statistic $\pm$ (multiplier) $\times$ (SD of statistic) is a confidence interval.

    - $\overline{x}_d \pm 2 \times SD(\overline{x}_d)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_d$.
    - $\overline{x}_d \pm$ (multiplier) $\times$ $(SE)$ is a theory-based confidence interval estimating $\mu_d$, where the multiplier comes from a t distribution with $n-1$ degrees of freedom.

- $\overline{x}_d \pm 2 \times SD(\overline{x}_d)$ is a 2SD CI approximating a 95% confidence interval estimating $\mu_d$.
- $\overline{x}_d \pm$ (multiplier) $\times$ $(SE)$ is a theory-based confidence interval estimating $\mu_d$, where the multiplier comes from a t distribution with $n-1$ degrees of freedom.

Mean Group Diff:

- $\mu_i$ are the population means for the groups.
- $n_i$ are the sample sizes for the groups.
- $\overline{x}_i$ are the sample means for the groups.
- $s_i$ are the sample standard deviations for the groups.
- The Mean Group Diff is the mean of the absolute value of the differences of sample means, and can be computed directly as $\frac{\sum_{\text{groups } i,j} |\overline{x}_i - \overline{x}_j|}{(l-1)(l/2)}$, where $l$ is the number of groups.

F-statistic (ANOVA test):

- $\mu_i$ are the population means for the groups.
- $n_i$ are the sample sizes for the groups with $n$ as the overall sample size.
- $\overline{x}_i$ are the sample means for the groups.
- $s_i$ are the sample standard deviations for the groups.
- $l$ is the number of treatments (groups).
- $df$ are the degrees of freedom for the F-distribution.
    - $df_{Treatment} = l - 1$
    - $df_{Error} = n - l$
    - $df_{Total} = df_{Treatment} + df_{Error} = n - 1$
- $SS$ is the sum of squares.
    - $SS_{Treatment} = \sum n_i(\bar{x}_i - \bar{x})^2$
    - $SS_{Error} = \sum (n_i - 1)(s_i^2)$
    - $SS_{Total} = SS_{Treatment} + SS_{Error}$
- $MS$ is the mean squares.
    - $MS_{Treatment} = \frac{SS_{Treatment}}{df_{Treatment}} = \frac{\sum n_i(\bar{x}_i - \bar{x})^2}{l-1}$
    - $MS_{Error} = \frac{SS_{Error}}{df_{Error}} = \frac{\sum (n_i-1)(s_i^2)}{n-l}$.

- $F$ is calculated as the ratio $F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{MS_{Treatment}}{MS_{Error}}$.
- $R^2$ is the "coefficient of determination" the proportion of the variation in the response variable which is explained by the treatment, calculated as $R^2 = \frac{SS_{Treatment}}{SS_{Error}}$.
- The mean of an F-distribution is $\frac{df_{Error}}{df_{Error}-2}$.
- The SD of an F-distribution is $\sqrt{\frac{2(df_{Error})^2(df_{Treatment}+df_{Error}-2)}{df_{Treatment}(df_{Error}-2)^2(df_{Error}-4)}}$.
- $df_{Treatment} = l - 1$
- $df_{Error} = n - l$
- $df_{Total} = df_{Treatment} + df_{Error} = n - 1$
- $SS_{Treatment} = \sum n_i(\bar{x}_i - \bar{x})^2$
- $SS_{Error} = \sum (n_i - 1)(s_i^2)$
- $SS_{Total} = SS_{Treatment} + SS_{Error}$
- $MS_{Treatment} = \dfrac{SS_{Treatment}}{df_{Treatment}} = \dfrac{\sum n_i(\bar{x}_i - \bar{x})^2}{l - 1}$
- $MS_{Error} = \frac{SS_{Error}}{df_{Error}} = \frac{\sum (n_i-1)(s_i^2)}{n-l}$.

# Chapter 11

# Two Quantitative Variables

# 11.1 Two quantitative variables: Scatterplot and correlation

## Height and winning at tennis

### Step 1: Ask a research question

Based on trends observed in other sports, researchers were interested in exploring how heights of winning tennis players might be changing over time, given that height may be an advantage to winning at tennis.

### Step 2: Design a study and collect data

To investigate this question, researchers gathered data[1] on heights of the winning tennis players at Wimbledon, a prestigious tennis tournament held each year from the 1880s through 2019

**Exercise 11.1.1** Identify (be specific) the observational units in this study.

**Exercise 11.1.2** What is the explanatory variable? Is it categorical or quantitative?

**Exercise 11.1.3** What is the response variable? Is it categorical or quantitative?

**Exercise 11.1.4** In your own words, explain what it would mean for there to be an association between the explanatory and response variables in this study.

**Exercise 11.1.5** Is this study an experiment or an observational study? Why?

### Step 3: Explore the data

The table below represents a subset of the data values reported in the article.

**Table 11.1.1**

| | | | | | |
|---|---|---|---|---|---|
| Year | 1881 | 1882 | 1883 | 1884 | 1885 |
| Height (cm) | 177 | 177 | 177 | 177 | 177 |

**Table 11.1.2**

| | | | | | |
|---|---|---|---|---|---|
| Year | 2017 | 2018 | 2018 | 2019 | 2019 |
| Height (cm) | 185 | 188 | 173 | 188 | 168 |

**Exercise 11.1.6** Name one possible reason some of the heights are the same for different years.

**Exercise 11.1.7** Name one reason some years may have multiple heights.

## Graphical summary of two quantitative variables: Scatterplots

This study is different than studies we've looked at before because both of the variables of interest are quantitative. For this reason, the graphical and numerical techniques we will use to summarize the data are different from those we've seen before.

**Key Idea 11.1.3** A **scatterplot** is a graph showing a dot for each observational unit, where the location of the dot indicates the values of the observa-

[1] www.topendsports.com/sport/tennis/anthropometry-wimbledon.htm

tional units for both the explanatory and response variables. Typically, the explanatory variable is placed on the $x$-axis, and the response variable is placed on the $y$-axis.

**Exercise 11.1.8** Paste the Wimbledon[2] data into the Corr/Regression applet[3].

Which variable is on the x-(horizontal) axis? Which variable is on the y-(vertical) axis? If you want to reverse those roles, press the (Explanatory, Response) button above the data window.

**Exercise 11.1.9** Create a rough sketch of the scatterplot from the applet. Be sure the axes are well labeled. Rather than drawing each dot, you could just draw an ellipse (oval) shape where most of the points lie. Show that you understand how a scatterplot is created by making a dot on the scatterplot corresponding to a Wimbledon winner in 2019 that is 188 cm tall.

When describing a scatterplot, we look for three aspects of association: direction, form, and strength. The **direction** of association between two quantitative variables is either positive or negative, depending on whether the response variable (size) increases as the explanatory variable (year) tends to increase (positive association) or decrease (negative association).

**Exercise 11.1.10** Direction: Is the association between year and winning height positive or negative? Interpret what this means in context.

The **form** of association between two quantitative variables is described by indicating whether a line would do a reasonable job summarizing the overall pattern in the data or if a curve would be better. It is important to note that—especially when the sample size is small— you don't want to let one or two points on the scatterplot change your interpretation of whether the form of association is linear. In general, assume that the form is linear unless there is compelling (strong) evidence in the scatterplot that the form is not linear.

**Exercise 11.1.11** Form: Does the association between year and height appear to be linear, or is there strong evidence (over many observational units) suggesting the relationship is nonlinear?

In describing the **strength** of association revealed in a scatterplot, we see how closely the points follow the form; that is, how closely do the points follow a straight line or curve? If all of the points fall pretty close to this straight line or curve, we say the association is strong. Weak associations will show little pattern in the scatterplot, and moderate associations will be somewhere in the middle.

**Exercise 11.1.12** Strength: In your opinion, would you say that the association between height and year appears to be strong, moderate, or weak?

It is also important to consider any **unusual observations** that do not follow the overall pattern.

**Exercise 11.1.13** Are there any observational units (dots on the scatterplot, representing individual Wimbledon champions) that seem to fall outside of the overall pattern? If yes, identify it/them on the graph you just made. If not, provide a hypothetical example of a year and height combination that would represent an unusual observation.

*Note:* There are two kinds of unusual observations seen in scatterplots: influential observations and outliers. An observation is *influential* if removing it from the dataset dramatically changes our perception of the association. Typically influential observations are extreme in the explanatory variable. *Outliers*

[2] `static-content2.zybooks.com/isi/data/chap10/Wimbledon.txt`

[3] `static-content.zybooks.com/applets/2021/regshuffle/regshuffle.htm?hideExtras=2`

are observations that don't follow the overall pattern of the relationship. They may or may not be influential and they may or may not be extreme in either variable individually but are unusual in terms of the combination of values.

**Key Idea 11.1.4** Association between quantitative variables can be described with direction, form, and strength.

- If above-average values of one variable tend to correspond to above-average values of the other variable, the **direction** is positive. If, however, above-average values of one variable are associated with below-average values of the other, the **direction** is negative.
- The **form** of the association is whether the data follow a linear pattern or some more complicated pattern.
- The **strength** of an association refers to how closely the data follow a particular pattern. When an association is strong, the value of one variable can be accurately predicted from the other.
- Also look for, and investigate further, any **unusual observations** that do not fit the pattern of the rest.

## Numerical summaries

Describing the direction, form, and strength of association based on a scatterplot, along with investigating unusual observations, is an important first step in summarizing the relationship between two quantitative variables. Another approach is to use a statistic. One of the statistics most commonly used for this purpose is the correlation coefficient.

The **correlation coefficient**, often denoted by the symbol r, is a single number that takes a value between -1 and 1, inclusive. Negative values of r indicate a negative association, whereas positive values of r indicate a positive association. The stronger the linear association is between the two variables, the closer the value of the correlation coefficient will be to either -1 or 1, whereas weaker linear associations will have correlation coefficient values closer to 0. Moderate linear associations will typically have correlation coefficients in the range of 0.3 to 0.7 or -0.3 to -0.7.

**Exercise 11.1.14** Will the value of the correlation coefficient for the year and height data be negative or positive? Why?

**Key Idea 11.1.5** The correlation coefficient uses a rather complex formula that is rarely computed by hand; instead, people almost always use a calculator or computer to calculate the value of the correlation coefficient. But you should be able to apply the above properties to interpret the correlation coefficient that is found.

**Exercise 11.1.15** Without using the applet, give an estimated value of the correlation coefficient between year and height based on the scatterplot.

**Exercise 11.1.16** Now, check the Correlation coefficient box in the applet to reveal the actual value of the correlation coefficient. Report the value.

**Key Idea 11.1.6** The correlation coefficient is only applicable for data which has a linear form; nonlinear data are not summarized well by the correlation coefficient. In fact, we could say that the correlation coefficient is a numerical summary of the strength and direction of a linear association between two quantitative variables.

Another point about correlation is that it is sensitive to influential observations. Influential observations are unusual observations which substantially impact the value of the correlation coefficient based on whether or not they are included in the dataset.

**Exercise 11.1.17** In the Add/Remove Observations section of the applet, enter a year of 2020 and height of 140 cm and press Add. Note how this is an unusual observation on the scatterplot. How did the correlation coefficient change as a result?

**Exercise 11.1.18** This correlation coefficient is not particularly strong because there is a lot of scatter/variation about the regression line. Suggest another variable we haven't considered yet that might explain some of this variability in heights, even in the same year.

Earlier, you brainstormed reasons there might be more than one winner in a given year. How can there be two winners in the same year? Although initially Wimbledon was a males-only tournament, since 1887, both a female and male winner have been crowned. (Note: A few years don't have data due to the championship not being played that year or height data not being available on the winner.)

The figure below color codes the scatterplot depending on whether the height corresponds to the Male or Female champion that year.

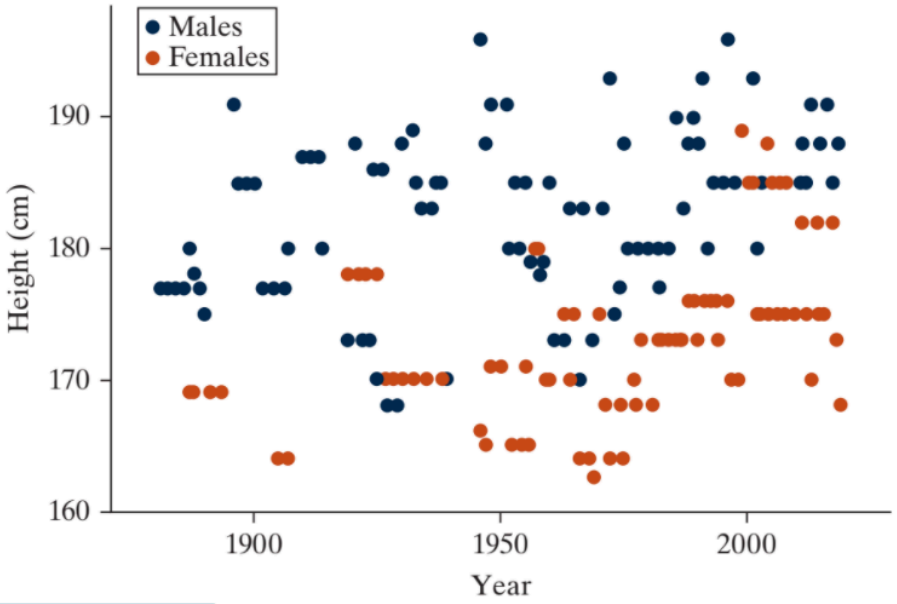



**Figure 11.1.7** Figure 1121: Scatterplot of height and year colored by event, males in blue and females in orange

**Exercise 11.1.19** How do the male heights tend to compare to the female heights?

**Exercise 11.1.20** Does the association between height and year seem to be similar between males and females?

The figure below shows the separate scatterplot of only the female winners and only the male winners.

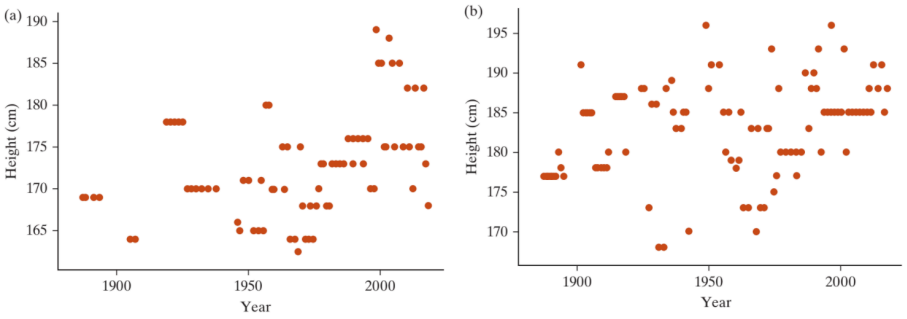



**Figure 11.1.8** Scatterplot of female champions of Wimbledon (1887-2019) and male champions of Wimbledon (1881-2019)

**Exercise 11.1.21** Describe the strength, form, and direction of the scatterplot for female Wimbledon champions. Are there any unusual observations?

**Exercise 11.1.22** Describe the strength, form, and direction of the scatterplot for male Wimbledon champions. Are there any unusual observations?

**Exercise 11.1.23** What do you predict for the value of the correlation coefficient between height and year for female winners only? Why? What do you predict for male winners?

**Exercise 11.1.24** The actual correlation coefficient for females is 0.40 and for males is 0.36. Explain to a nonstatistician why the correlation coefficients are different within the male/female groups as compared to the overall correlation of 0.19 based on the pattern of dots in the scatterplot.

Note, however, that the strength of the association is not directly impacted by the size of the sample, but by how closely the observations follow the (linear) pattern.

## Steps 4 and 5: Draw inferences and formulate conclusions

In a later section you will explore how to evaluate whether a correlation is statistically significant, but that's not necessarily needed/appropriate in this case. The data here represent the entire population of Wimbledon winners (a census).

**Exercise 11.1.25** How does the strength and direction of association compare between men and women? What does this tell us about the pattern of height of Wimbledon winners over time?

**Exercise 11.1.26** Can we say that year causes changes in the height of the Wimbledon winners? Why or why not? Can we say that being taller makes you more likely to win Wimbledon?

## Steps 6: Look back and ahead

**Exercise 11.1.27** Name one additional variable you would like measured on each Wimbledon winner and explain why.

**Exercise 11.1.28** Describe how you could do a similar study for another sport. What two variables would you collect? What would you expect to see? Why? What third variable could also be measured and be important to consider?

## 11.2 Inference for correlation coefficient

### Draft lottery

In 1970, the U.S. Selective Service conducted a lottery to decide which young men would be drafted into the armed forces (Fienberg, 1971[1]). Each of the 366 birthdays in a year (including February 29) was assigned a draft number. Young men born on days assigned low draft numbers were drafted.

We will regard the 366 dates of the year as observational units. We will consider two variables recorded on each date: *draft number* assigned to the date and *sequential date* in the year (so January 31 is sequential date 31, February 1 is sequential date 32, and so on).

**Exercise 11.2.1** In a perfectly fair, random lottery, what should be the value of the correlation coefficient between draft number and sequential date of birthday?

The figure below displays a scatterplot of the assigned draft numbers and the sequential dates. There are 366 dots, one for each day of the (leap) year.

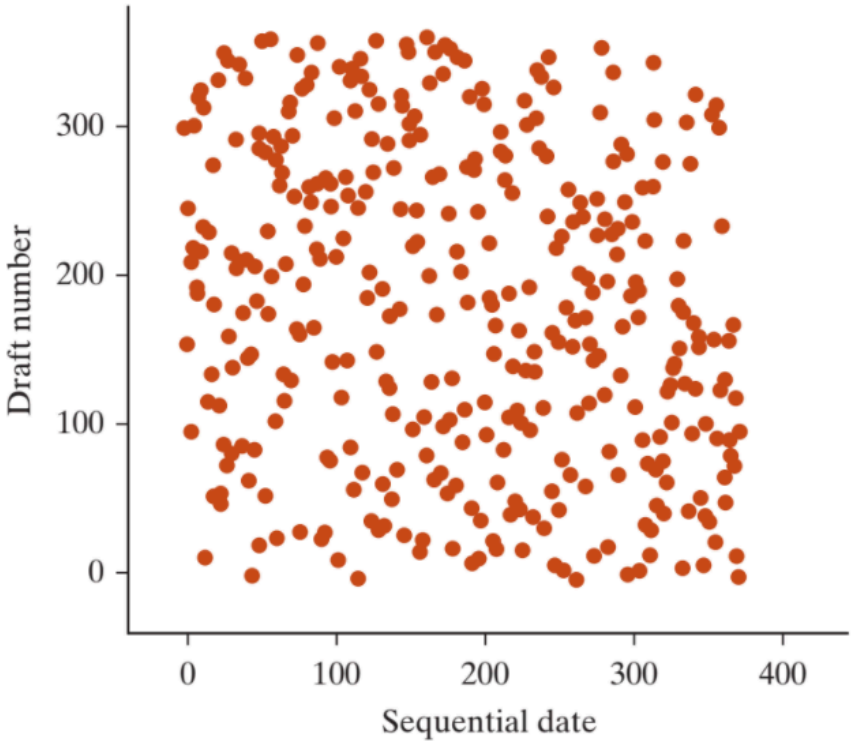



**Figure 11.2.1** A scatterplot of the draft numbers by sequential days of the year

**Exercise 11.2.2** Let's take a look at the scatterplot in the figure.

**(a)** Does the scatterplot reveal much of an association between draft number and sequential date?

**(b)** Based on the scatterplot, guess the value of the correlation coefficient.

**(c)** Does it appear that this was a fair, random lottery?

[1] doi.org/10.1126/science.171.3968.255

It's difficult to see much of a pattern or association in the scatterplot, so it seems reasonable to conclude that this was a fair, random lottery with a correlation coefficient near zero. But let's dig a little deeper:

**Exercise 11.2.3** The 1970 Draft Lottery data sheet below shows the draft number assigned to each of the 366 birthdays.

**Table 11.2.2**

| Date | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 305 | 86 | 108 | 32 | 330 | 249 | 93 | 111 | 225 | 359 | 19 | 129 |
| 2 | 159 | 144 | 29 | 271 | 298 | 228 | 350 | 45 | 161 | 125 | 34 | 328 |
| 3 | 251 | 297 | 267 | 83 | 40 | 301 | 115 | 261 | 49 | 244 | 348 | 157 |
| 4 | 215 | 210 | 275 | 81 | 276 | 20 | 279 | 145 | 232 | 202 | 266 | 165 |
| 5 | 101 | 214 | 293 | 269 | 364 | 28 | 188 | 54 | 82 | 24 | 310 | 56 |
| 6 | 224 | 347 | 139 | 253 | 155 | 110 | 327 | 114 | 6 | 87 | 76 | 10 |
| 7 | 306 | 91 | 122 | 147 | 35 | 85 | 50 | 168 | 8 | 234 | 51 | 12 |
| 8 | 199 | 181 | 213 | 312 | 321 | 366 | 13 | 48 | 184 | 283 | 97 | 105 |
| 9 | 194 | 338 | 317 | 219 | 197 | 335 | 277 | 106 | 263 | 342 | 80 | 43 |
| 10 | 325 | 216 | 323 | 218 | 65 | 206 | 284 | 21 | 71 | 220 | 282 | 41 |
| 11 | 329 | 150 | 136 | 14 | 37 | 134 | 248 | 324 | 158 | 237 | 46 | 39 |
| 12 | 221 | 68 | 300 | 346 | 133 | 272 | 15 | 142 | 242 | 72 | 66 | 314 |
| 13 | 318 | 152 | 259 | 124 | 295 | 69 | 42 | 307 | 175 | 138 | 126 | 163 |
| 14 | 238 | 4 | 354 | 231 | 178 | 356 | 331 | 198 | 1 | 294 | 127 | 26 |
| 15 | 17 | 89 | 169 | 273 | 130 | 180 | 322 | 102 | 113 | 171 | 131 | 320 |
| 16 | 121 | 212 | 166 | 148 | 55 | 274 | 120 | 44 | 207 | 254 | 107 | 96 |
| 17 | 235 | 189 | 33 | 260 | 112 | 73 | 98 | 154 | 255 | 288 | 143 | 304 |
| 18 | 140 | 292 | 332 | 90 | 278 | 341 | 190 | 141 | 246 | 5 | 146 | 128 |
| 19 | 58 | 25 | 200 | 336 | 75 | 104 | 227 | 311 | 177 | 241 | 203 | 240 |
| 20 | 280 | 302 | 239 | 345 | 183 | 360 | 187 | 344 | 63 | 192 | 185 | 135 |
| 21 | 186 | 363 | 334 | 62 | 250 | 60 | 27 | 291 | 204 | 243 | 156 | 70 |
| 22 | 337 | 290 | 265 | 316 | 326 | 247 | 153 | 339 | 160 | 117 | 9 | 53 |
| 23 | 118 | 57 | 256 | 252 | 319 | 109 | 172 | 116 | 119 | 201 | 182 | 162 |
| 24 | 59 | 236 | 258 | 2 | 31 | 358 | 23 | 36 | 195 | 196 | 230 | 95 |
| 25 | 52 | 179 | 343 | 351 | 361 | 137 | 67 | 286 | 149 | 176 | 132 | 84 |
| 26 | 92 | 365 | 170 | 340 | 357 | 22 | 303 | 245 | 18 | 7 | 309 | 173 |
| 27 | 355 | 205 | 268 | 74 | 296 | 64 | 289 | 352 | 233 | 264 | 47 | 78 |
| 28 | 77 | 299 | 223 | 262 | 308 | 222 | 88 | 167 | 257 | 94 | 281 | 123 |
| 29 | 349 | 285 | 362 | 191 | 226 | 353 | 270 | 61 | 151 | 229 | 99 | 16 |
| 30 | 164 |  | 217 | 208 | 103 | 209 | 287 | 333 | 315 | 38 | 174 | 3 |
| 31 | 211 |  | 30 |  | 313 |  | 193 | 11 |  | 79 |  | 100 |

**(a)** Find and report the draft number assigned to your birthday.

**(b)** Is your draft number in the bottom third (1-122), middle third (123-244), or top third (245-366)?

**Exercise 11.2.4** This second table has the draft numbers in order for each month.

**Table 11.2.3**

| Rank | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 17 | 4 | 29 | 2 | 31 | 20 | 13 | 11 | 1 | 5 | 9 | 3 |
| 2 | 52 | 25 | 30 | 14 | 35 | 22 | 15 | 21 | 6 | 7 | 19 | 10 |
| 3 | 58 | 57 | 33 | 32 | 37 | 28 | 23 | 36 | 8 | 24 | 34 | 12 |
| 4 | 59 | 68 | 108 | 62 | 40 | 60 | 27 | 44 | 18 | 38 | 46 | 16 |
| 5 | 77 | 86 | 122 | 74 | 55 | 64 | 42 | 45 | 49 | 72 | 47 | 26 |
| 6 | 92 | 89 | 136 | 81 | 65 | 69 | 50 | 48 | 63 | 79 | 51 | 39 |
| 7 | 101 | 91 | 139 | 83 | 75 | 73 | 67 | 54 | 71 | 87 | 66 | 41 |
| 8 | 118 | 144 | 166 | 90 | 103 | 85 | 88 | 61 | 82 | 94 | 76 | 43 |
| 9 | 121 | 150 | 169 | 124 | 112 | 104 | 93 | 102 | 113 | 117 | 80 | 53 |
| 10 | 140 | 152 | 170 | 147 | 130 | 109 | 98 | 106 | 119 | 125 | 97 | 56 |
| 11 | 159 | 179 | 200 | 148 | 133 | 110 | 115 | 111 | 149 | 138 | 99 | 70 |
| 12 | 164 | 181 | 213 | 191 | 155 | 134 | 120 | 114 | 151 | 171 | 107 | 78 |
| 13 | 186 | 189 | 217 | 208 | 178 | 137 | 153 | 116 | 158 | 176 | 126 | 84 |
| 14 | 194 | 205 | 223 | 218 | 183 | 180 | 172 | 141 | 160 | 192 | 127 | 95 |
| 15 | 199 | 210 | 239 | 219 | 197 | 206 | 187 | 142 | 161 | 196 | 131 | 96 |
| 16 | 211 | 212 | 256 | 231 | 226 | 209 | 188 | 145 | 175 | 201 | 132 | 100 |
| 17 | 215 | 214 | 258 | 252 | 250 | 222 | 190 | 154 | 177 | 202 | 143 | 105 |
| 18 | 221 | 216 | 259 | 253 | 276 | 228 | 193 | 167 | 184 | 220 | 146 | 123 |
| 19 | 224 | 236 | 265 | 260 | 278 | 247 | 227 | 168 | 195 | 229 | 156 | 128 |
| 20 | 235 | 285 | 267 | 262 | 295 | 249 | 248 | 198 | 204 | 234 | 174 | 129 |
| 21 | 238 | 290 | 268 | 269 | 296 | 272 | 270 | 245 | 207 | 237 | 182 | 135 |
| 22 | 251 | 292 | 275 | 271 | 298 | 274 | 277 | 261 | 225 | 241 | 185 | 157 |
| 23 | 280 | 297 | 293 | 273 | 308 | 301 | 279 | 286 | 232 | 243 | 203 | 162 |
| 24 | 305 | 299 | 300 | 312 | 313 | 335 | 284 | 291 | 233 | 244 | 230 | 163 |
| 25 | 306 | 302 | 317 | 316 | 319 | 341 | 287 | 307 | 242 | 254 | 266 | 165 |
| 26 | 318 | 338 | 323 | 336 | 321 | 353 | 289 | 311 | 246 | 264 | 281 | 173 |
| 27 | 325 | 347 | 332 | 340 | 326 | 356 | 303 | 324 | 255 | 283 | 282 | 240 |
| 28 | 329 | 363 | 334 | 345 | 330 | 358 | 322 | 333 | 257 | 288 | 309 | 304 |
| 29 | 337 | 365 | 343 | 346 | 357 | 360 | 327 | 339 | 263 | 294 | 310 | 314 |
| 30 | 349 | | 354 | 351 | 361 | 366 | 331 | 344 | 315 | 342 | 348 | 320 |
| 31 | 355 | | 362 | | 364 | | 350 | 352 | | 359 | | 328 |

**(a)** Use this table to determine the median draft number for your birth month.

**(b)** Collaborate with your classmates to determine and report the median draft number for all 12 months.

**(c)** Do you notice any pattern or trend in the median draft numbers over the course of the year? (Hint: If you do not see a trend, compare the six medians from January to June with the six medians from July to December.)

**Exercise 11.2.5** The correlation coefficient for the full dataset is $r = -0.226$. What does this number reveal? Is it consistent with the scatterplot?

**Exercise 11.2.6** Let's think about how we would complete a test of significance for correlation.

**(a)** Suggest two possible explanations (hypotheses) which could have generated the value of the nonzero correlation coefficient.

**(b)** In your own words, how could we go about determining whether random chance is a plausible explanation for the observed correlation value between sequential date and draft number? Explain how the 3S strategy could be applied here, in particular identifying a simulation strategy you could conduct "by hand." Note: You do not need to actually carry out a simulation analysis.

**(c)** What is the statistic? How would you simulate? How would you evaluate the strength of evidence?

The null hypothesis to be tested is that the lottery was conducted with a fair, random process. The null hypothesis would therefore imply that there is *no* association between sequential date and draft number for this process. The alternative hypothesis is that this lottery was *not* conducted with a fair, random process, so there is an association between sequential date and draft number.

How can we assess whether the observed correlation coefficient of $r = -0.226$ is far enough from zero to provide convincing evidence that the lottery process was not random? Like always, we ask how unlikely it would be for a fair, random lottery to produce a correlation value as far from zero as $-0.226$. Also like always, we answer that question by simulating a large number of fair random lotteries (a true null hypothesis), calculating the correlation coefficient for each one, and seeing how often we obtain a correlation coefficient as or more extreme (as far from zero) as -0.226.

**Exercise 11.2.7** Open the Corr/Regression applet[2]. Copy the data from DraftLottery[3] into the applet (remember to include the column titles). Check the Correlation Coefficient box and confirm that the correlation coefficient is -0.226.

**(a)** Check the Show Shuffle Options box and select Correlation from the Statistic pull-down menu to the right to keep track of that statistic. Then press Shuffle Y-values to simulate one fair, random lottery. Record the value of the correlation coefficient between the shuffled draft numbers and sequential date.

**(b)** Press Shuffle Y-values four more times to generate results of four more fair, random lotteries. Record the values of the shuffled correlation coefficients in a table.

**(c)** Did any of your simulated statistics from fair, random lotteries produce a correlation coefficient as extreme (far from zero) as the observed -0.226?

**(d)** Change the Number of Shuffles from 1 to 995 and press Shuffle Y-values to simulate 995 more fair, random lotteries. Look at the null distribution of these 1,000 correlation coefficients. Where is this distribution centered? Why does this make sense?

**(e)** Next to the Count Samples box choose Beyond from the pull-down menu. Specify the observed correlation coefficient (-0.226) and press Count. What proportion, of the 1,000 simulated random lotteries produced a correlation coefficient as extreme (as far from zero in either direction) as -0.226? Report the approximate p-value.

**(f)** Interpret this p-value: This is the probability of what, assuming what?

**(g)** What conclusion would you draw from this p-value? Do you have strong evidence that the 1970 draft lottery was not conducted with a fair, random process? Explain the reasoning behind your conclusion.

## Exploring further

Once they saw these results, statisticians were quick to point out that something fishy happened with the 1970 draft lottery. The irregularity can be attributed to improper mixing of the balls used in the lottery drawing process. (Balls with birthdays early in the year were placed in the bin first, and balls with birthdays late in the year were placed in the bin last. Without thorough mixing, balls with birthdays late in the year settled near the top of the bin and so tended to be selected earlier.) The mixing process was changed for the 1971 draft lottery (e.g., two bins, one for the draft numbers and one for the birthdays), for which the correlation coefficient turned out to be $r = 0.014$.

**Exercise 11.2.8** Use your simulation results to approximate the p-value for the 1971 draft lottery. Is there any reason to suspect that this 1971 draft lottery was not conducted with a fair, random process? Explain the reasoning behind your conclusion. Also explain why you don't need to paste in the data from the 1971 lottery first.

[2] static-content.zybooks.com/applets/2021/regshuffle/regshuffle.htm?hideExtras=2

[3] static-content2.zybooks.com/isi/data/chap10/DraftLottery.txt

## 11.3 Least squares regression

### Predicting height from footprints

Can a footprint taken at the scene of a crime help to predict the height of the criminal? In other words, is there an association between height and foot length? To investigate this question, a sample of 20 statistics students measured their height (in inches) and their foot length (in centimeters).

**Exercise 11.3.1**

**(a)** What are the observational units in this study?

**(b)** Identify the two variables recorded for each observational unit. Which is the explanatory variable, and which is the response? Also classify these variables as categorical or quantitative.

**(c)** Is this an observational study or a randomized experiment?

**Exercise 11.3.2** The sample data for the 20 students appear in the Corr/ Regression applet[1]

Open the applet and look at the scatterplot for the data already there. Describe the association between the variables as revealed in the scatterplot. (Hint: Remember to comment on direction, strength, and form of association as well as unusual observations.)

**Exercise 11.3.3** Would you say that a straight line could summarize the relationship between height and foot length reasonably well?

**Exercise 11.3.4** Check the **Show Movable Line** box to add a blue line to the scatterplot. If you place your mouse over one of the green squares at the ends of the line and drag, you can change the slope of the line. You can also use the mouse to move the green dot up and down vertically to change the intercept of the line.

**(a)** Move the line until you believe your line "best" summarizes the relationship between height and foot length for these data. Write down the resulting equation for your line (using traditional statistical notation).

**(b)** Why do you believe that your line is "best"?

**(c)** Did all students in your class obtain the same line/equation?

**(d)** How can we decide whether your line provides a better fit to the data than other students' lines? Suggest a criterion for deciding which line "best" summarizes the relationship.

One way to draw the best fit line is to minimize the distance of the points to the line (these distances are called *residuals*).

**Exercise 11.3.5** Would points above the line have a positive or negative residual, or is it impossible to tell? What about points below the line?

**Key Idea 11.3.1** A **residual** is the difference between an observed response and the corresponding prediction made by the least squares regression line (residual = observed − predicted). Thus, negative residuals occur when points are below the best fit line and positive residuals occur when points are above the best fit line.

[1] `static-content.zybooks.com/applets/2021/regshuffle/regshuffle.htm?hideExtras=2`

**Exercise 11.3.6** Check the **Show Residuals** box to visually represent these residuals for your line on the scatterplot. The applet also reports the sum of the values of the residuals (**SAE**). SAE stands for "sum of the absolute errors." The acronym indicates that we need to make residuals positive before we add them up and that sometimes people call residuals "errors."

**(a)** Record the SAE value for your line. What is the best (lowest) SAE in the class?

**Exercise 11.3.7** It turns out that a more common criterion for determining the "best" line is to instead look at the sum of the squared residuals (**SSE**). This approach is similar to simply adding up the residuals but is even more strict in not letting individual residuals get too large. Check the **Show Squared Residuals** box to visually represent the squared residual for each observation. Note that we can visually represent the squared residual as the area of a square where each side of the square has length equal to the residual.

**(a)** What is the SSE (sum of squared residuals) for your line? What is the best (lowest) SSE in the class?

**(b)** Now continue to adjust your line until you think you have minimized the sum of the squared residuals. Report your new equation and your new SSE value. What is the best SSE in the class?

**Key Idea 11.3.2** The least squares regression line minimizes the sum of squared residuals.

**Exercise 11.3.8** Now check the **Show Regression Line** box to determine and display the equation for the line that actually does minimize (as can be shown using calculus) the sum of the squared residuals.

**(a)** Record the equation of the least squares regression line by indicating the appropriate slope and intercept of the line. Note that we've used variable names in the equation, not generic x and y. And put a caret ("hat") over the y variable name to emphasize that the line gives predicted values of the y (response) variable.

**(b)** Did everyone in your class obtain the same equation?

**(c)** Is the slope positive or negative? Explain how the sign of the slope tells you about whether your data display a positive or a negative association.

**Key Idea 11.3.3** The equation of the best fit line is written as $\hat{y} = a + bx$, where

- $a$ is the y-intercept
- $b$ is the slope
- $x$ is a value of the explanatory variable
- $\hat{y}$ is the predicted value for the response variable

**Key Idea 11.3.4** For a given dataset, the signs (positive or negative) for the correlation coefficient and the slope of the regression line must be the same.

**Exercise 11.3.9** Let's investigate what the slope means in the context of height and foot length.

**(a)** Use the least squares regression line to predict the height of someone whose foot length is 28 cm. (Simply plug in the value of 28 cm for foot length in the equation of the line.)

**(b)** Use the least squares regression line to predict the height of someone whose foot length is 29 cm.

**(c)** By how much do your predictions in (a) and (b) differ? Does this number look familiar? Explain.

**(d)** Interpret the slope in context: The slope of the regression line predicting height based on foot length is _____, meaning that for every additional _____ cm increase in foot length, the predicted height increases by _____ inches.

**Key Idea 11.3.5** The **slope** coefficient of a least squares regression model is interpreted as the predicted change in the response ($y$) variable for a one-unit change in the explanatory ($x$) variable.

**Exercise 11.3.10** Let's investigate the meaning of the y-intercept in the context of height and foot length.

**(a)** Use the least squares regression line to predict the height of someone who has a foot length of 0 cm.

**(b)** Your answer to (a) should look familiar. What is this value?

**(c)** Explain how your prediction of the height of someone whose foot length is 0 cm is an example of extrapolation.

**Key Idea 11.3.6** The **y-intercept** of a regression line is interpreted as the predicted value of the response variable when the explanatory variable has a value of zero (though be wary of extrapolation (see the next Key Idea) in interpreting the intercept or other values outside the original data range).

**Key Idea 11.3.7** Predicting values for the response variable for values of the explanatory variable that are outside of the range of the original data is known as **extrapolation** and can give very misleading predictions.

**Exercise 11.3.11** Earlier, we explored the notion of a residual as the distance of a point to a line.

**(a)** Using the equation for the best fit line that you reported in question 8, find the residual for the person whose foot length is 32 centimeters and height is 74 inches. Note: Find this value by taking the actual height and subtracting the person's predicted height from the equation.

**(b)** Is this person's dot on the scatterplot above or below the line? How does the residual tell you this?

Residuals are helpful for identifying unusual observations. But not all observations of interest have large residuals.

**Exercise 11.3.12** Uncheck the **Show Movable Line** box to remove it from the display and check the **Move observations** box.

**(a)** Now click on one student's point in the scatterplot and drag the point up and down (changing the height, without changing the foot length, the original regression line will remain in grey). Does the regression line change much as you change this student's height?

**(b)** Press Revert and repeat (a) using a student with a very small x (foot size) value and then a point with an x value near the middle and then a point with a very large x value. Which of these seem(s) to have the most influence on the least squares regression line? Explain.

**Key Idea 11.3.8** An observation or set of observations is considered **influential** if removing the observation from the dataset substantially changes the values of the correlation coefficient and/or the least squares regression equation. Typically, observations that have extreme explanatory variable values (far below or far above $\bar{x}$) are potentially influential. They may not have large residuals, having pulled the line close to them.

Residuals also help us measure how accurate our predictions are from using the regression line. In particular, we can compare the "prediction errors" from the regression line to the prediction errors if we made no use of the explanatory variable.

**Exercise 11.3.13** Press the **Revert** button to reload the original dataset. Recheck the **Show Movable Line** box to redisplay the blue line. Notice that this line is flat at the mean of the $y$ (height) values.

**(a)** Check the Show Squared Residuals box (under the Movable Line information) to determine the SSE if we were to use the average height as the predicted value for every x (foot size). Record this value.

**(b)** What is the slope of this line?

**(c)** If the slope of the best fit line is zero, our data shows _____ (positive/ negative/no) association between the explanatory and response variables.

**Exercise 11.3.14** Now compare this to the SSE value from the regression line. Which is smaller? Why does that make sense?

## Coefficient of Determination

**Key Idea 11.3.9** A quantity related to the correlation coefficient is called the coefficient of determination, or R-squared ($R^2$). To measure how much the regression line decreases the "unexplained variability" in the response variable, we can calculate the percentage reduction in SSE:

$$R^2 = \frac{\text{SSE}(\bar{y}) - \text{SSE}(\text{regression line})}{\text{SSE}(\bar{y})}$$

The **coefficient of determination** ($R^2$) is the percentage of total observed variation in the response variable (height) that is accounted for by changes (variability) in the explanatory variable (foot length).

**Exercise 11.3.15**

**(a)** Find $R^2$ and report it.

**(b)** Complete the following statement: The coefficient of determination is _____ ; this means that _____ % of the variation in people's _____ is attributable to changes in their _____.

**(c)** Find the value of the correlation coefficient using the applet and confirm that when you square the correlation coefficient you get the same number as the applet reports for the coefficient of determination.

If we think of $\hat{y} = \bar{y}$ as the "worst" line (no association with $x$), then we are seeing how much better the actual line is than this worst line. The best possible line would have zeros for all of the residuals and would have an $R^2$ of 100%.

## Exploring further

**Exercise 11.3.16** What other variables could explain variation in height?

The figure below illustrates which people in the sample are men and which are women and then plots the least squares regression line separately for men and women.

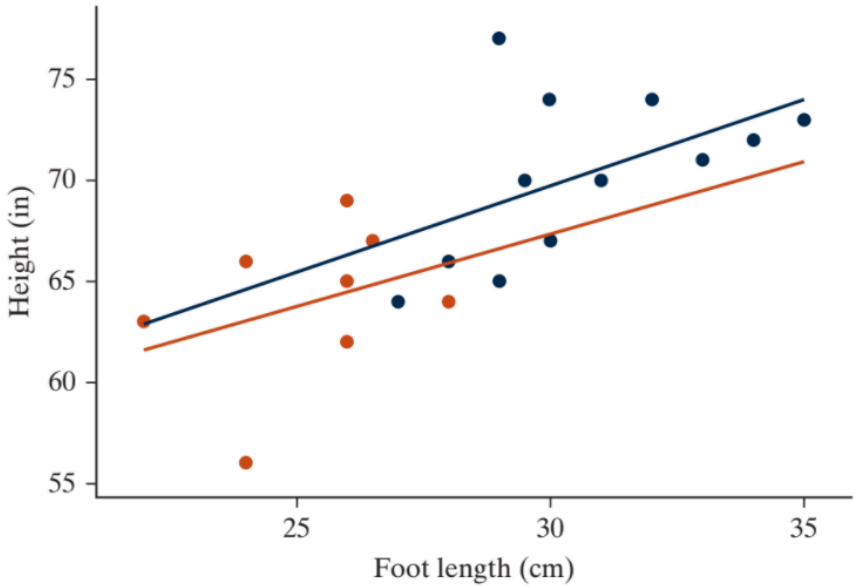



**Figure 11.3.10**

**Exercise 11.3.17** How do the female foot lengths tend to compare to the male foot lengths? How do the female heights tend to compare to the male height?

**Exercise 11.3.18** What do you notice about the slopes and intercepts for the two lines? How do you interpret these differences in this context?

The least squares regression line for men is predicted height $= 44.03 + 0.8561$(foot length), and the least squares regression line for women is predicted height $= 45.90 + 0.7152$(foot length) .

**Exercise 11.3.19** What do you think the $R^2$ will be just for the men? Just for the women? How do you think these values will compare to the overall $R^2$ of 0.506? Why?

In this case, the $R^2$ for the female-only data is 0.116 and for the male-only data is 0.267. Both groups have lower $R^2$ values than the overall $R^2$. $R^2$ measures how close the observations are to their respective lines. When we examine only the female data, the residuals are large relative to the smaller variation in female heights. Whereas the combined line (ignoring sex) has similar residuals, they are smaller compared to the larger total variation in heights.

## Colophon

This book was authored in PreTeXt.